Consumer Misperceptions and Redistribution in Financial Products: Evidence from Brazilian Consórcios*

Otávio Braga
Duke University

João Pugliese†
Stanford University

Lucas Teixeira
Central Bank of Brazil

[Job Market Paper]

October 29, 2026
Click Here for Latest Version

Abstract

Financial innovations may expand access to credit, but also introduce complexity that leads to consumer losses and redistribution across consumers. We study *consórcios*, a group savings-and-credit product accounting for 25% of new vehicle credit in Brazil. Consumers pool monthly payments and receive funding through lotteries or bids, allowing consumers with liquidity to obtain funding earlier at the expense of others. Using administrative data covering 59 million contracts and a survey of 1,296 consumers, we document four facts. First, 81% of contracts lose money relative to saving and borrowing in the market. Second, losses are regressive. Third, firms' allocation decisions favor bidding, which can improve expected returns, while consumers waiting for lotteries face long waits and large expected losses. Fourth, consumers overestimate chances of receiving funding early through lotteries. These findings challenge explanations of participation in group savings products based on commitment and suggest that misperceptions about allocation probabilities may help sustain regressive contract designs. In a randomized information experiment, disclosing true lottery probabilities increases liquidity-constrained consumers' willingness to forgo a real bonus for access to groups with more lotteries, but has no corresponding effect among consumers able to bid. Our findings show that correcting misperceptions can differentially affect demand across consumer types, with possible implications for market composition and redistribution.

*We are grateful to Arun Chandrasekhar, Matthew Gentzkow, Melanie Morten, and Marcel Fafchamps for their guidance. We also thank Nava Ashraf, Lautaro Chittaro, Liran Einav, Tishara Garg, Eva Lestant, Anirudh Sankar, Santiago Varela Seoane, Tim de Silva, Silvia Vannutelli, and Lucas Warwar for helpful comments. We owe special thanks to Leonardo Zanetti for his outstanding research assistance. This study was approved by the Stanford Institutional Review Board (Protocol #75340) and pre-registered at the AEA RCT Registry (#18745). We gratefully acknowledge financial support from the Stanford King Center on Global Development, Stanford Graduate Research Opportunities Fund, and George P. Shultz Dissertation Support Fund. João acknowledges support from the Gale and Steve Kohlhagen Fellowship in Economics, awarded by the Stanford Institute for Economic Policy Research. The views expressed here are those of the authors and do not necessarily reflect those of the Central Bank of Brazil. All errors are our own.

†Corresponding author: jfpugli@stanford.edu.

1 Introduction

Access to credit shapes major household decisions, from buying or improving homes to acquiring vehicles. Financial innovations promise to expand opportunities, particularly in developing economies where high borrowing costs constrain access to durables (Karlan and Morduch, 2010). Yet some innovations introduce complexity that makes costs and benefits difficult to assess, exposing consumers to financial losses (Beshears et al., 2018). And the same product may benefit consumers who can take advantage of contract features, often at the expense of others (Campbell and Ramadorai, 2026). Understanding when these innovations create winners and losers, and how information disclosure about product attributes can affect different consumers, is central to consumer protection policy.

This paper studies *consórcios*, a group savings-and-credit product used by millions of consumers in Latin America to finance major purchases.[1] In Brazil, the focus of our study, consórcios account for 25% of new vehicle credit and a growing share of housing credit, and one in seven households holds a consórcio contract. Consumers join groups organized by firms to pool savings and take turns being awarded funds from the pool. Each month, firms allocate awards across group members, combining lotteries with opportunities to bid for earlier access, which requires liquidity.

Consórcios are a form of rotating savings and credit associations (RoSCAs), one of the most prevalent financial institutions in the world (World Bank, 2025; Breza and Kaur, 2026). The literature on RoSCAs emphasizes demand for commitment as a key explanation for participation, alongside their role in expanding access to credit (Besley et al., 1993; Bryan et al., 2010; Basu, 2011). In typical RoSCAs, funds are allocated either through lottery or through bidding, with bids paid to other members as dividends, compensating them for waiting. In consórcios, bids are prepayments of future installments rather than transfers to other members, and lottery and bid awards can occur in the same month. This allocation mechanism allow consumers with liquidity to receive funds earlier at the expense of others, while introducing complexity that makes expected financial returns difficult to assess.

We study how the allocation mechanism interacts with differences in consumers' liquidity and imperfect information to shape who gains and loses from these group financial contracts. We combine market-wide administrative data, surveys of consumers and brokers, and a randomized information experiment to measure the distribution of financial returns, document misperceptions about the allocation mechanism and lottery probabilities, and test how correcting those beliefs affects consumers' choices according to their reported liquidity.

In consórcios, participants make monthly payments over several years but are awarded funds at an uncertain date. An earlier award means higher financial returns, all else equal. Consumers

[1] Specifically, consórcio refers to the Brazilian and Uruguayan version of this product. In Argentina and other countries, the same type of product is often called *plan de ahorro* or *grupos de ahorro* (savings plan/group).

with liquidity can bid to receive funding earlier and improve their returns, while those who cannot bid must rely on lotteries. Yet assessing lottery probabilities is difficult. Groups contain hundreds or thousands of contracts, and firms determine how many awards are allocated through lotteries and bids each month, subject to available resources. Consumers must therefore form expectations about firms' decisions about lottery and bid awards, and other participants' bidding behavior, which are not fully known when they join.

To quantify these financial consequences and investigate consumers' perceptions of allocation probabilities, we combine administrative data from the Central Bank of Brazil covering 59 million consórcio contracts from 2010 to 2025 with survey data from 1,296 consumers. We measure financial returns against a market replication that saves the same payments before the award and borrows at market rates afterward. This benchmark captures both forgone savings interest and avoided borrowing costs, and is motivated by survey evidence showing that consórcio consumers use other savings and credit products.

We document four key facts. First, 81% of consórcio contracts incur financial losses. Losses come from both awarded contracts and contracts canceled before participants receive funding. Among awarded contracts, 60% have negative returns. The median awarded contract breaks even only when the benchmark savings rate is -5% a year in real terms, holding borrowing rates fixed at market levels. Losses remain widespread even when benchmark borrowing rates are raised to the 90th percentile of market rates. Second, losses are regressive. Lower-income participants incur larger losses per BRL paid into the pool, largely because they are more likely to cancel their contracts before receiving funding.

Third, these losses are partly explained by the interaction between allocation mechanism and consumers' liquidity. Waiting for lotteries predictably leads to large losses, while bidding can lead to higher returns. Simulating bidding strategies to calculate expected returns, we find participants who wait for the lottery face large expected losses of 14 to 24 cents per BRL paid into the pool. In contrast, strategies with moderate bids can yield higher expected returns.

Relatedly, firms prioritize bids over lotteries, with bids accounting for 86% of awards in the first half of groups' durations. This can bring fee revenue forward and attract bidders, but reduces non-bidders' chances of receiving funding early. Consumers who wait for the lottery have only about an 11% chance of an award by the group's midpoint and face long waits. A lottery-only group with uniform award probabilities would raise the expected returns of those who wait for the lottery by 11 to 18 cents per BRL paid into the pool.

Fourth, despite the financial importance of award timing, consumers substantially overestimate the probability of early funding. In our surveys, respondents report believing their chance of an award by the contracts' midpoint is 31%, compared with 12% in the data, weighting asset categories by respondents' choices. These misperceptions lead consumers to understate the expected financial losses from relying exclusively on lotteries by about one-third.

Taken together, these facts raise a question: why do consumers without liquidity participate in consórcios despite large expected losses and few lotteries? The widespread misperceptions about lottery probabilities suggest that consumers may overvalue participation. But the literature on RoSCAs emphasizes commitment as a central motivation for participation, which may justify financial losses. Importantly, the value of commitment may itself depend on the allocation mechanism: the prospect of receiving funding early can sustain participation.[2]

We formalize this idea in a model of hyperbolic discounting. The model shows that the commitment value of rotating finance depends on the probability of receiving an early award, and declines convexly as lottery awards become less frequent. This convexity implies that the very low probabilities of early funding observed in consórcios substantially erode their commitment value. At the same time, high fees and large financial losses increase the commitment value needed to justify participation. Together, low lottery probabilities and high fees substantially narrow the range of parameters under which participation is rationalized by commitment alone. Groups with more lotteries would therefore provide both stronger commitment value and higher expected financial returns for non-bidders.

Misperceptions about lottery probabilities can help reconcile these findings with observed participation.[3] Overestimating the chances of early funding increases the perceived value of participation while also reducing the perceived advantage of groups with more lotteries. And if misperceptions make consumers less sensitive to differences in real lottery probabilities, firms also have weaker incentives to offer groups with more lotteries. Misperceptions may therefore help explain both why illiquid consumers participate despite large expected losses and why firms continue to offer groups with few lotteries.

These predictions motivate our information experiment. Correcting beliefs about lottery probabilities may affect consumers differently depending on their liquidity and prior beliefs. Consumers without liquidity to bid depend on lotteries and therefore stand to benefit more from groups offering more lottery awards, while consumers who initially overestimate lottery probabilities may revise their valuations more strongly.

We survey 1,296 consumers and 292 brokers to measure beliefs about allocation probabilities, and conduct a randomized information experiment with 845 consumers. To measure valuations of groups with more lotteries, we partner with a consórcio firm to develop a prospective offer for such a group. We elicit whether respondents will forgo a real survey bonus for access to this offer, providing an incentive-compatible measure of valuation.

The surveys show optimism about lottery probabilities is widespread across consumer character-

[2] A similar argument is found in Basu (2011), where the chance of early access helps participants not postpone entry into the rotating mechanism.

[3] Other misperceptions are also plausible. Consumers may underestimate their chances of canceling, overestimate their prospects of bidding, or misperceive the growth in their installments due to inflation indexation (Braga et al., 2026).

istics and stronger among prospective participants. This suggests that differences in sophistication, at least as captured by these beliefs, are secondary in explaining who benefits, while liquidity plays a central role in determining consumers' ability to take advantage of bidding. Even past and current participants, as well as brokers, still overestimate lottery probabilities, suggesting limited learning from experience. Yet brokers and past consumers hold more accurate beliefs about winning bids, which are disclosed in monthly meeting minutes. Finally, consumers report strategies that lead to large expected losses, such as relying on lotteries or delaying bids, and these are more related to consumers' liquidity than to beliefs. These patterns suggest that correcting beliefs may affect consumers' choices differently depending on their liquidity, even when misperceptions are similar.

To test the effects of correcting misperceptions, we randomize respondents into an active control group and two information treatments. The *debiasing* (D) treatment uses a short video to convey average bidding shares and lottery wait times, and to explain how more early bid awards increase lottery waits. The *debiasing + group size mechanics* (DG) treatment additionally explains variation in group size and how larger groups reduce individual lottery probabilities. The videos do not mention whether consórcios lead to financial losses or gains or compare them to savings and borrowing, focusing exclusively on the typical design of the allocation mechanism.

Both treatments partially correct misperceptions, closing about 40–50% of the gap between average beliefs and actual allocation probabilities and slightly increase consumers' confidence in their answers. Treated respondents also have stronger perceptions that consórcios favor consumers who can bid and are less likely to recommend them to consumers without liquidity to bid.

We find that disclosure of allocation probabilities can shift demand differently depending on consumers' liquidity. The debiasing (D) treatment raises liquidity-constrained consumers' valuations of the lottery-intensive offer. Using a pre-registered moderator, it increases the share choosing access to the offer over the largest bonus by 8.3 percentage points among consumers without sufficient liquidity to bid ($p = 0.07$). The treatment effect is concentrated among optimists. The DG information treatment has no effect, and the difference between the arms' effects is statistically significant.

In side-by-side choices where contract attributes are randomized and explicitly presented, consumers prefer smaller groups and more lotteries, suggesting a gap between stated preferences and market choices. The treatments do not significantly change these stated preferences, consistent with an information channel rather than updating preferences for attributes.

In hypothetical choices among consórcios, loans, and savings, treatment responses instead vary with prior beliefs: both treatments reduce consórcio choices among optimists. The DG treatment also increases choices of consórcios among non-optimists. These results suggest that disclosure can also shift demand toward outside financial alternatives among optimists.

Our findings support disclosure policies that make allocation schedules and estimates of lottery probabilities accessible to consumers. While disclosure policies have often been found to be insuf-

ficient across many settings (Campbell, 2016; Garz et al., 2021), they may be especially powerful in settings where heterogeneous responses can also have implications for market composition and redistribution. In consórcios, where consumers who do not bid help sustain groups that award funding through bids, such adjustments could affect both contract design and the distribution of returns. If disclosure increases the valuation of lottery-intensive groups, or shifts demand away from consórcios, it could weaken demand for bid-intensive contracts and increase incentives for firms to offer more lotteries. Thus, disclosure may affect not only consumers' individual decisions but also which allocation mechanisms firms can sustain in equilibrium. Establishing the equilibrium effects of disclosure, however, requires accounting for firms' responses and consumers' sorting across contracts.

This paper contributes to different strands of the literature. First, we contribute to the literature on rotating savings and credit associations, which emphasizes credit frictions and demand for commitment to explain participation and contract design (Besley et al., 1993, 1994; Calomiris and Rajaraman, 1998; Anderson et al., 2009; Anderson and Baland, 2002; Ambec and Treich, 2007; Gugerty, 2007; Basu, 2011; Afzal et al., 2017). Studying the largest formal market for rotating finance, we show how consumers' expectations about allocation timing interact with these explanations, and how liquidity differences and imperfect information shape the distribution of financial returns. We also show how consumers' imperfect information can influence the sustainability and regressiveness of rotating savings mechanisms. In this sense, we connect group-based financial institutions studied in developing economies to a broader literature on consumer protection and redistribution in financial products (Garz et al., 2021; Campbell and Ramadorai, 2026). This is especially relevant as, across many countries, rotating finance products are increasingly offered by profit-maximizing intermediaries.

Second, our information experiment connects to other experiments studying consumer protection and disclosure policies in different settings (Bertrand and Morse, 2011; Ferman, 2016; Seira et al., 2017; Alan et al., 2018; Adams et al., 2021; Balakina et al., 2026). We provide novel experimental data linking beliefs about nonprice attributes to consumers' choices. We also show the disclosure of nonprice attributes can affect consumers differently based on their prior beliefs or differential ability to interact with contract features. In particular, the specific mechanism of perceptions about probabilities involving lotteries relates to papers studying demand for prize-linked savings accounts (Kearney et al., 2010; Filiz-Ozbay et al., 2015) and state-run lotteries (Lockwood et al., 2025).

Third, a growing body of evidence studies the distributive effects of retail finance in different settings, such as life insurance, credit cards, and mortgage refinancing (Gurun et al., 2016; Egan, 2019; Gottlieb and Smetters, 2021; Byrne et al., 2023; Fisher et al., 2024; Berger et al., 2024; Egan et al., 2026; Agarwal et al., 2026). This literature empirically studies how redistribution emerges from supply-side responses to heterogeneous consumer decision frictions, as suggested by

theoretical work (Gabaix and Laibson, 2006; Heidhues and Kszegi, 2017). Similar to cancellations in our setting, Gottlieb and Smetters (2021) show how lower-income households tend to lapse more in life insurance policies, partially due to liquidity shocks. We provide experimental evidence that suggests disclosure policies may affect not only consumer decisions but also market composition, in a direction that is consistent with lower redistribution. We also provide evidence that consumers can hold similar misperceptions about contract attributes, but consumers with higher liquidity are better placed to take advantage of contract features.

Finally, we are directly related to other papers studying consórcios. To our knowledge, this paper provides the first market-wide evidence on the financial returns to consórcios, along with survey evidence of consumer misperceptions. Other studies measure the effects of durable ownership by using consórcio lotteries as an instrument, examining outcomes such as employment and earnings (Doornik et al., 2024b) and fertility (Doornik et al., 2025). Doornik et al. (2024a) study how credit contracts that mandate a savings period can expand access to credit by screening out consumers, using consórcios as an example of such a contract. Relatedly, we show cancellations, a form of dynamic screening, can imply large and regressive financial losses.

2 Setting and financial value

2.1 Institutional context: contracts, awards, and firms

When joining a consórcio, a consumer chooses the contract's face value — the funds made available upon award — and its term. Consórcios are long-term commitments, typically lasting 4 to 8 years for vehicles and 10 to 20 years for real estate. The contract is then assigned to a group for a specific asset category. Groups range from several hundred to over five thousand contracts, and consumers usually do not choose the exact group they join.[4] Appendix B describes the institutional setting in more detail.

The consumer then makes monthly payments to the firm managing the group, covering pool contributions and fees.[5] Because consórcios charge a fee instead of interest, they are often marketed as "interest-free". Fees are set at the group level, so firms do not price discriminate among members of a group, although fees vary across groups.

The key feature of the contract is that the date of award is uncertain, potentially leading some consumers to wait many years. Upon award, participants receive a letter of credit granting them the right to use funds to purchase the asset.[6]

[4]Groups may admit new contracts after formation, with terms adjusted to the group's remaining duration. Face values may vary within a group, subject to regulatory limits: the smallest must be at least half the largest.

[5]Payments and face values are indexed to reference prices or price indices and adjusted semiannually or annually.

[6]A credit assessment takes place, and credit may be withheld if deemed risky, until further payments are made.

Awards are made either by lottery or through bids, which allow participants with sufficient liquidity to obtain earlier access to funds by prepaying future installments, reducing future payments one-for-one.[7] A winning bid brings forward both contributions to the group and administration fees. Bids compete in a first-price auction, while the lottery consists of random draws among all eligible participants in a given month, including that month's bidders.

Firms choose the number of lottery and bid awards made each month, conditional on the group's budget. The Central Bank requires at least one lottery award each month. Firms include large commercial banks, carmakers, insurance companies, and independent administrators, such as Honda, Santander, and *Banco do Brasil*.

Finally, nonpayment leads to different consequences before and after award. Before award, three consecutive missed payments lead to cancellation: payments to the pool are refunded at a later date, net of a fine. After award and usage of credit, the contract becomes a collateralized loan, and missed payments constitute a default. In practice, cancellations are common (from 40% to 50%) while defaults are rare (less than 3%).[8]

2.2 Consumers' access to other financial products and financial habits

A first question is whether consórcio consumers have access to other savings and credit products. Data from a representative survey on financial literacy, conducted by the Central Bank in 2023, show that 147 out of 1,956 adult respondents were active consórcio consumers.[9] Among these, 91% had a bank account, 78% reported saving money through a financial instrument, and 54% reported having some credit product. Access to financial products is also higher when compared to other respondents, even after controlling for income and education (Appendix Table A.1).

The survey also allows us to compare the income and financial habits of consumers. Consórcios reach a wide range of consumers: 23% of consumers live in households earning up to two minimum wages, 42% in households earning two to five, and 35% in households earning more than five, although consumers are richer on average than the overall population. Even in the lowest income bracket, 82% of consumers have a bank account, against 59% of non-consumers. Both overall and within the lowest income bracket, consumers are more likely to report having a financial goal (71% vs. 45% overall), to keep money for bills apart from day-to-day money (70% vs. 61%; 85% vs. 57% in the lowest income bracket), and to agree that they set long-term goals and work toward them (67% vs. 57%). Consumers are no more likely than non-consumers to buy lottery tickets when

[7] Some groups allow bids to be partially paid with awarded funds, reducing liquidity requirements. Because this option is available to all members of a group, it does not change their relative bidding opportunities. Some groups also offer "fixed bids", with winners randomly drawn among bidders.

[8] Default rates are lower than market averages, which may be explained by selection into the contract or screening: Doornik et al. (2024a) suggest the pre-award savings period screens out participants by income risk.

[9] With about two adults per Brazilian household, and assuming no household has more than one holder, this translates into about 1 in 7 households.

short of money, suggesting that a taste for lotteries is not a main driver of demand.

2.3 Measuring financial returns compared to a market replication

We value a consórcio by comparing its cash flows with a market replication that saves the same pre-award payments and borrows any remaining amount needed to obtain the same funds at the award date. The comparison captures both the opportunity cost of saving through the consórcio and the benefit of cheaper borrowing. The procedure is detailed in Appendix F.

We define the *consórcio value* as the difference in wealth generated by the two options, holding the durable-acquisition date constant, and expressed in present-value terms:

$$V_i(t^A|\tau, r^S, r^B) := NPV(\text{Consórcio}|t^A, \tau, r^S) - NPV(\text{MarketReplication}|t^A, r^B, r^S), \quad (1)$$

where t^A is the award time, τ is the proportional fee, and r^S and r^B are benchmark savings and borrowing interest rates, with r^S being the discount rate. We refer to the value per BRL paid into the pool as a *return*. For an awarded contract, payments into the pool add up to its face value P_i, so the return is V_i/P_i. For a canceled contract, the denominator is the amount paid into the pool before cancellation, and since there is no award, we set the NPV of the market replication to zero.

We calculate V_i based on average market rates over 20112025. We set the savings interest rate to $r^S = 4\%$ (post-tax) and the market borrowing rate r^B to 4%, 19%, and 24% for real estate, cars, and motorcycles, respectively, in real terms.[10] We consider this to be a reasonable benchmark given the evidence of financial access among consórcio consumers. For each V_i, we can compute break-even borrowing or savings interest rates, holding the other rate at its benchmark.[11]

Unlike a conventional NPV discounted at the savings rate, this measure accounts for the benefit of avoiding market borrowing. It also avoids the ambiguity of internal rates of return, since consórcio cash flows can change sign twice.[12] When the borrowing and saving interest rates are equal, the measure reduces to the conventional NPV.

Importantly, when comparing across contracts, the measure does not fully account for the benefits of earlier ownership of the durable, net of depreciation. Thus, it may understate the advantages of receiving an early award.

[10] Data for average interest rates are obtained from the Central Bank. Appendix Figure A.1 presents the corresponding real interest rates alongside government bond yields.

[11] Break-even borrowing rates are undefined for late awards, which require no borrowing, while break-even savings rates are undefined for most early awards, because the saving period is short.

[12] Such cash flows may have multiple internal rates of return, which are also hard to interpret (Hirshleifer, 1958).

2.4 The central aspects of consórcios illustrated by an example

We illustrate typical contract outcomes, the measurement of returns, and the core logic of winners and losers within consórcio groups with an example of three agents. Suppose three agents, L, B, and C, join a consórcio group with three identical contracts to buy a car, with a face value of 10,000, a 60-month term, and a 20% administration fee. Each owes 12,000, initially paid in monthly installments of 200. But each has different outcomes:

- **Agent B bids and receives funding early.** Suppose Agent B has liquidity available, bids, and wins an award in the first month. He bids 5,400, equivalent to a 45% bid — bids are normalized by total payments. The bid amount is deducted from his future contributions, reducing remaining installments to about 109 per month.
- **Agent L waits for a lottery and receives funding late.** Suppose Agent L waits for the lottery until she wins in month 48. By then she has contributed 9,600, almost the full price of the car, but still owes twelve installments.
- **Agent C waits and then cancels.** Suppose Agent C pays 12 installments while waiting for an award, a total of 2,400, but then stops paying and the contract is canceled. He receives a refund, net of fines and fees. The refund date is based on a random draw, $t = 36$ in this example.

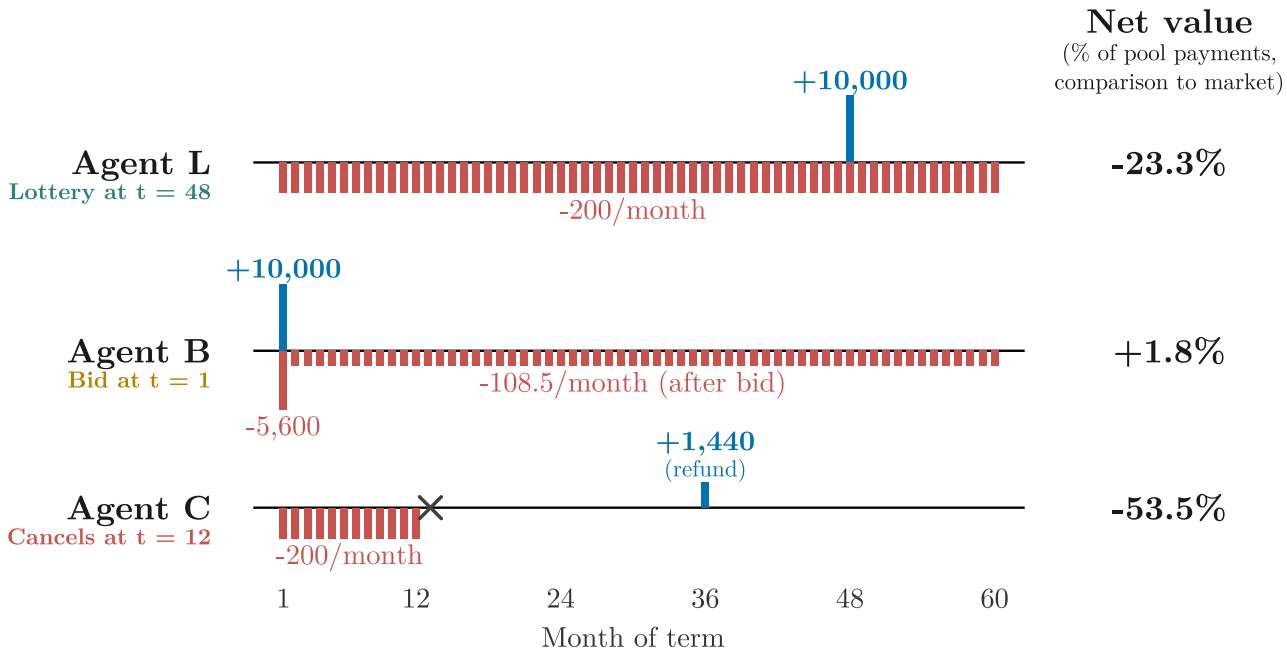


Figure 1: Three identical contracts with distinct outcomes.

Figure 1 shows the cash flows of each agent. The right column shows the returns, applying Equation (1) to the cash flows. Agent B has positive returns, which means his contract has a borrowing cost below the market rate — a subsidized loan. In contrast, the returns of Agents L and C are negative due to a combination of fees and the time value of money.

The example highlights central aspects of consórcios. First, the product can create winners and losers even though contracts are identical. The timing of awards, bids, and cancellations all contribute to substantial differences in financial returns across participants.

Second, bidding can shape the distribution of winners and losers. But bidding is gated by access to liquidity: Agent B's bid amounts to 27 months of installments paid at once. Those without liquidity must wait for a lottery and may cancel before receiving funds. How many lottery awards take place is both a decision of firms and a consequence of consumers' willingness to bid.

Third, transfers may run counter to the usual compensation for waiting. In conventional financial markets, borrowers compensate lenders for providing funds earlier. In consórcios, however, bidders may simultaneously benefit from higher financial returns and earlier access to funding to purchase an asset, while those who wait bear the cost of this advantage. Prepayment bidding thus makes this redistribution systematic rather than random, in contrast to other rotating savings and credit associations (RoSCAs), where allocations are determined by lottery or bids compensate participants who wait.

Finally, a consórcio group with bidders requires other participants who wait: without them, bidders would have no one to borrow from. Participants who cannot bid therefore have the most at stake: waiting earns no interest and still costs a fee.

A direct consequence is that lottery probabilities are a central component of the expected payoff of consumers who do not bid. For those without liquidity, the lottery is the only source of awards, and will define whether their contract resembles cheap borrowing or expensive savings.

3 Four facts about the consórcio market

In this section, we combine two datasets to document four facts about the consórcio market. We use market-wide administrative data from the Central Bank to document the financial value of consórcios as described in Section 2.3 and the observed lottery probabilities. We then turn to an original survey dataset of 1,296 prospective and past consumers to document misperceptions about lottery probabilities.

First, 81% of contracts face losses, including both awarded and canceled contracts. In particular, 60% of awarded contracts lose. Second, lower-income participants suffer larger losses, largely due to cancellations. Third, bidders can predictably obtain higher returns, while non-bidders suffer large expected losses, driven by low lottery probabilities. Fourth, consumers hold wide misperceptions

about lottery chances, and have high uncertainty about beliefs.

3.1 Data

The Central Bank data is obtained from SAG *(Sistema de Acompanhamento de Grupos de Consórcios)*. Our sampling frame considers the universe of groups and contracts that started between 2010 and 2025 and contains 48,531 groups with 59,372,033 contracts, spread across 24,955,784 unique individuals and 2,312,928 unique businesses. Appendix C describes the data extraction and sample construction in detail. The survey data has responses from 1,296 prospective and past consumers. The measurement of beliefs and sampling are detailed in Section 5.

The administrative data contains key inputs to calculate the financial value of contracts and award paths of groups. For every contract, we observe its face value, term, administration fees, bid amounts, start date, award date, and cancellation date if applicable. We deflate nominal amounts to 2025 prices. We merge income data from the national credit registry.

Because groups are long, to include ongoing groups in our analysis we predict unobserved award and cancellation outcomes and winning bids, keeping observed outcomes fixed. We restrict predictions to groups observed for at least 25% of their planned duration and estimate duration models using their observed histories. This expands the group sample by 44.7% and the contract sample by 73.5%, with outcomes predicted for about 15% of the contracts in ongoing groups. Out-of-sample validation shows that predictions closely match realized award paths and returns, with small errors that tend to understate differences between bidding and lottery awards. Prediction details are in Appendix D.

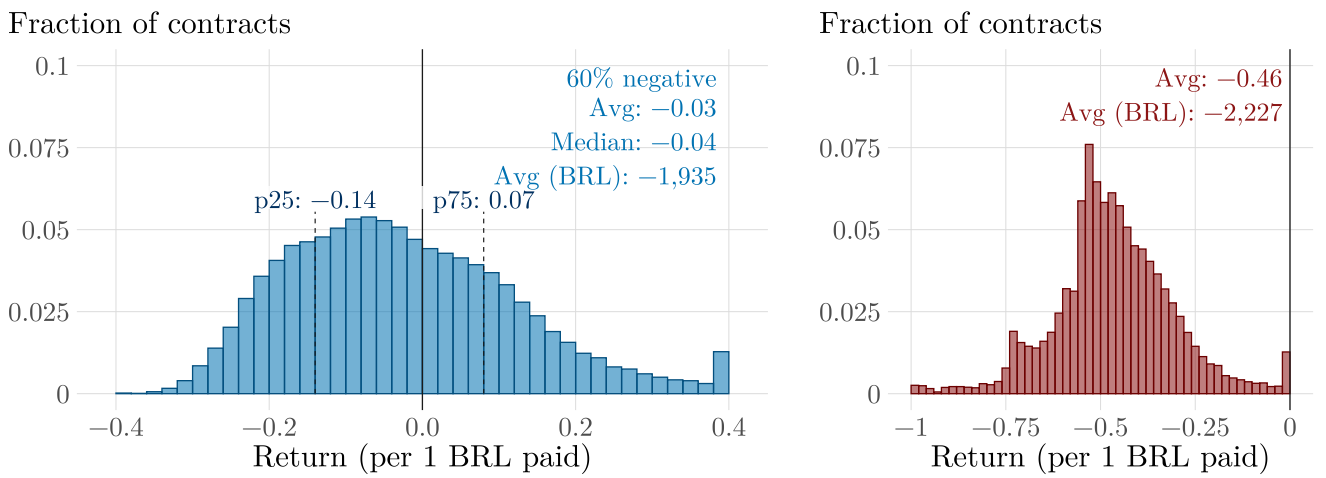
3.2 Summary statistics of individuals, contracts, and groups

Appendix Table A.2 reports summary statistics for contracts and groups by asset category. The summary statistics show how consórcios are major long-term obligations used by a wide range of consumers. First, the median contract's installment, including the administration fee, is about 17.5% of the holder's reported income. Second, the average contract term is 72 months for cars, 56 months for motorcycles, and 184 months for real estate.

Administration fees, which cover the intermediation services of the firm, are high: the average fee charged by a firm is 14% for cars, 20% for motorcycles, and 18% for real estate.[13] Firms also charge reserve fees to absorb the cost of defaults. Average reserve fees are around 3% for the three asset categories. In our calculations below, we do not include the reserve fee.

Groups are relatively large, giving firms discretion to choose between bids and lottery awards. They average between 1,000 (cars) and 2,200 (real estate) participants, which implies 12 to 19

[13]For a comparison, chit funds, a type of group-based savings in India, are capped at 7%.



Panel A. Awarded contracts *Panel B. Canceled contracts*

Figure 2: Negative returns are widespread, and cancellations are especially costly.

Notes: This figure shows financial returns per BRL paid into the pool, calculated with Equation (1) across awarded and canceled contracts. For canceled contracts, the market replication is set to zero. The sample includes all contracts in groups started between 2010 and 2025 that are completed or observed for at least 25% of their planned duration. Outcomes for contracts yet to be awarded in ongoing groups are predicted. Returns beyond the axis range fall in the end bins. Dashed lines in panel A mark the 25th and 75th percentiles.

participants for each month of duration and multiple awards per month.

3.3 Fact 1: 81% of contracts face financial losses

Returns are negative for most contracts, and widely dispersed. Among contracts awarded credit, the average return is −0.03 per BRL paid into the pool, and 60% have negative returns (Figure 2, panel A). The median return is −0.04, and the middle half of contracts yield returns between −0.14 and 0.07 per BRL paid into the pool.[14] Returns are negative for motorcycles and real estate and close to zero for cars (Appendix Table A.3). The differences reflect fees, which are smaller for cars, and the outside borrowing option, which is subsidized for real estate.

Negative returns at benchmark rates raise the question of how poor outside options must be for individual consórcios to break even with the market replication. We calculate the savings and borrowing rates that would make each contract's financial value equal to zero. The borrowing break-even applies to early awards, while the savings break-even describes contracts with later awards. Break-even rates can be non-monotonic in the contract term and award time, so they serve primarily to illustrate returns.

[14] Appendix Figure A.2 shows the same distribution varying the sample. Panel A shows groups observed past 50% of their planned duration, which is a stricter criterion for predictions, and panel B shows completed groups, which require no imputation. Completed groups have less negative returns, reflecting composition: those groups are older, with smaller fees and bids, and are almost entirely motorcycles and cars. Appendix Figure A.3, panel A, shows that average fees have increased by 3.6 percentage points since 2010. Panel B shows that, for the same award month, average bids have increased by 2.4 percentage points.

For contracts around the median return of −0.04, the median break-even savings rate is −4.9% a year, roughly what cash held at home would lose to inflation over the period (Appendix Table A.4). At the 25th percentile return of −0.14, the break-even rate is −7.9%, worse than holding cash. By comparison, a standard savings account pays about 1 to 2% in real terms, while a certificate of deposit or government bond can yield 4%.

For contracts around the median return, the break-even borrowing rate is 10.1 and 5.6 percentage points above the market average for motorcycles and real estate, respectively, and 0.3 percentage points below it for cars. Motorcycle and real estate contracts therefore break even only for participants who would otherwise borrow well above market rates.

We also assess how sensitive these results are to the benchmark rates (Appendix Figure A.4 and Appendix Table A.5). The savings rate matters little: setting it to 0% or 8% moves the average over all contracts, including cancellations, by less than 0.01, and the share of awarded contracts with negative returns stays between 54% and 66%. Borrowing rates 50% above market averages, roughly at the 90th percentile of the distribution of rates, raise the average return of awarded contracts to +0.04 per BRL, but also spread returns further apart, and 47% still lose. Thus, widespread losses are robust to a broad range of plausible outside options.

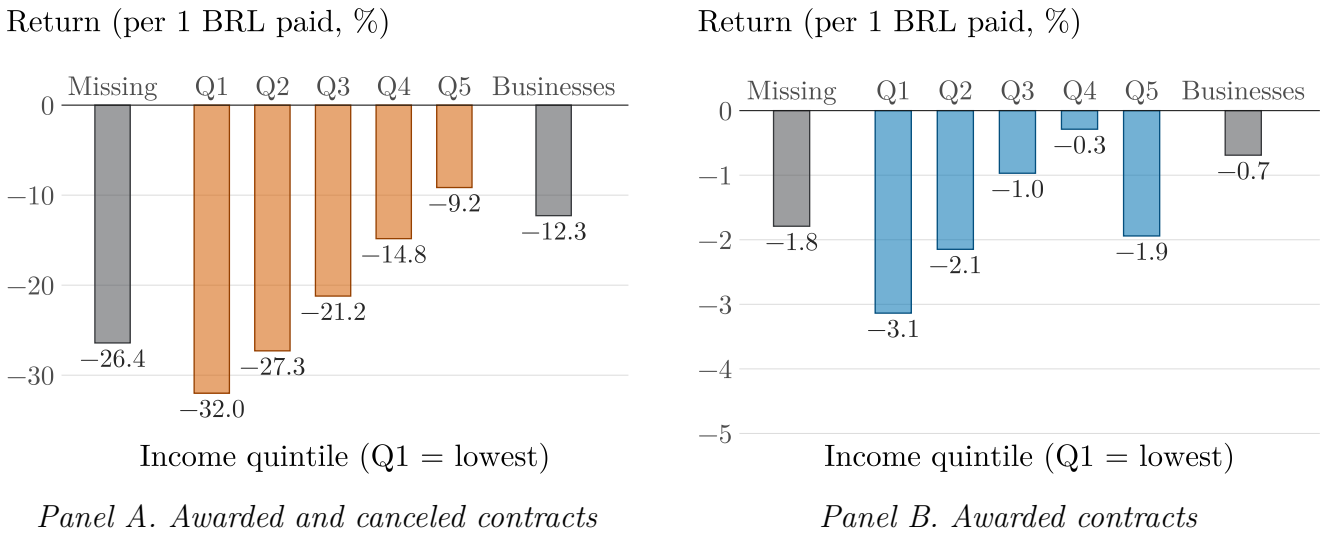
Figure 2, panel B turns to canceled contracts, which receive a refund rather than an award. These contracts lose, on average, 0.46 per BRL paid into the pool. Fees are not refunded, there are cancellation fines, and refunds take time to be issued. Furthermore, in absolute value, the average loss on a canceled contract is close to that on an awarded contract (BRL 2,227 against BRL 1,935), with the caveat that an awarded contract resulted in the purchase of an asset.

3.4 Fact 2: Lower-income participants suffer larger losses, largely due to cancellations

To study returns by income, we restrict the sample to completed groups. We include both awarded and canceled contracts, measuring returns per BRL paid into the pool as in Section 2.3.

Lower-income participants experience substantially larger financial losses (Figure 3, panel A). Considering cancellations, participants in the bottom income quintile lose 32% of their contributions, compared with 9% in the top quintile. This gradient is largely driven by cancellations: among awarded contracts, returns show an inverse U-shaped relationship with income (panel B). Businesses also appear to have larger returns relative to consumers. Cancellation rates fall from 60% in the bottom income quintile to 29% in the top quintile (Appendix Figure A.5). Although we do not causally link the allocation mechanism with cancellations, it is reasonable to assume cancellations are more likely among those who do not intend to bid.[15]

[15] In a 2026 industry survey, participants cited long waits or unclear rules (36%), financial difficulties (35%), and changes in purchase plans (25%) as reasons for cancellation.



Panel A. Awarded and canceled contracts *Panel B. Awarded contracts*

Figure 3: The income gradient in returns is largely driven by cancellations.

Notes: This figure shows financial returns per BRL paid into the pool, calculated with Equation (1) using the subsample of completed groups from Central Bank data. Income quintiles refer to quintiles taken over the distribution of self-reported incomes from the national credit registry, adjusted to 2025 prices. "Missing" refers to missing income. "Businesses" refer to contracts owned by businesses.

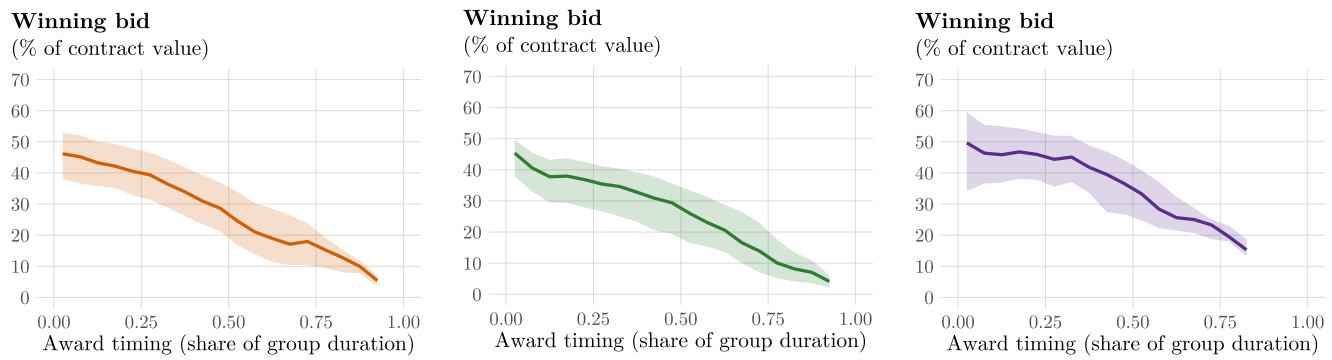
Higher-income participants are also more likely to receive early awards, defined as awards within the first 25% of group duration (Appendix Figure A.6). This is especially true for real estate groups: the top income quintile is almost three times as likely to receive an early award as the bottom quintile. Conditional on not canceling, the relationship varies across asset categories: the probability of an early award increases with income in real estate but not in cars or motorcycles. Within-group comparisons yield similar results (Appendix Figure A.7, panels A and B). We also find a strong relationship between a contract's face value relative to the group average and the probability of early awards, despite bids scaling with the contract's face value (Appendix Figure A.7, panels C and D).

Bidding imposes a much larger liquidity burden on lower-income participants: within each asset category, a winning bid costs about 15 times as many months of reported income in the bottom income quintile as in the top quintile (Appendix Figure A.8).

3.5 Fact 3: Non-bidders predictably lose, and bidders can predictably obtain higher returns

Bidding allows participants to obtain funds earlier, which improves returns, but costs liquidity and reduces the net amount borrowed from the group, reducing the gains from cheaper borrowing.[16] Bids often range from 30% to 60% of a contract's value in early periods (Figure 4). Despite this

[16] At a given award timing, bid awards earn less than lottery awards, most of all early in the term, when winning bids are highest (Appendix Figure A.9, panel A).



Panel A. Cars *Panel B. Motorcycles* *Panel C. Real estate*

Figure 4: Winning bids are highest early in the group's life.

Notes: This figure shows winning bids as a share of total contractual payments (face value plus administration fee), the base on which bid percentages accrue, against the share of the group's duration elapsed at the award. Award timing follows the group's clock, from the group's start. Lines are medians and bands the 25th to 75th percentiles within twentieths of the group's duration, in all groups of the three asset categories. Bids include contracts with fixed-bid clauses. Cells with fewer than 30 winning bids, all in the last fifth of the duration, are omitted.

trade-off, contracts with bid awards obtain higher financial returns. Bid-awarded contracts have average returns 0.04 higher and median returns 0.10 higher than lottery-awarded contracts (Figure 5). The gap widens as outside rates rise, because early funding is worth more when saving pays more or borrowing costs more (Appendix Figure A.4, right column).

To evaluate expected returns, we retrospectively compare a strategy that never bids with strategies that submit the same bid, as a share of the contract's total payments, every month until awarded, holding contract characteristics and other participants' bids fixed (for details, see Appendix E). A contract joins a group at its start, pays the category's average fee, and is awarded at the earlier of a lottery draw, which follows the group's estimated lottery hazard, and a bid award, which occurs once its bid exceeds its group's lowest winning bid that month, among bids of at least 30% of face value paid partly with own resources and excluding fixed bids. Table 1 reports, averaged over contracts, the probability of an award within 12 and 24 months, the median award month, and the expected return.[17]

Never bidding produces expected returns between −0.14 and −0.24 per BRL paid into the pool, with long expected waits. For the median never-bidding contract, the loss is equivalent to saving at real annual rates of −4.9% for cars, −8.5% for motorcycles, and −2.7% for real estate, against our 4% benchmark. By contrast, bidding substantially advances award timing and improves expected returns (Table 1). For some bidding strategies, expected returns become positive, including for cars at average market fees and borrowing rates.

How these differences vary with different rate benchmarks? Larger borrowing rates increase the benefit of early awards and therefore the return of bids. At same time, lower interest rates increase

[17] We also simulate strategies sampling from contracts' observed date of entry into the group. Both expected wait times, relative to the contract term, and returns remain roughly unchanged.

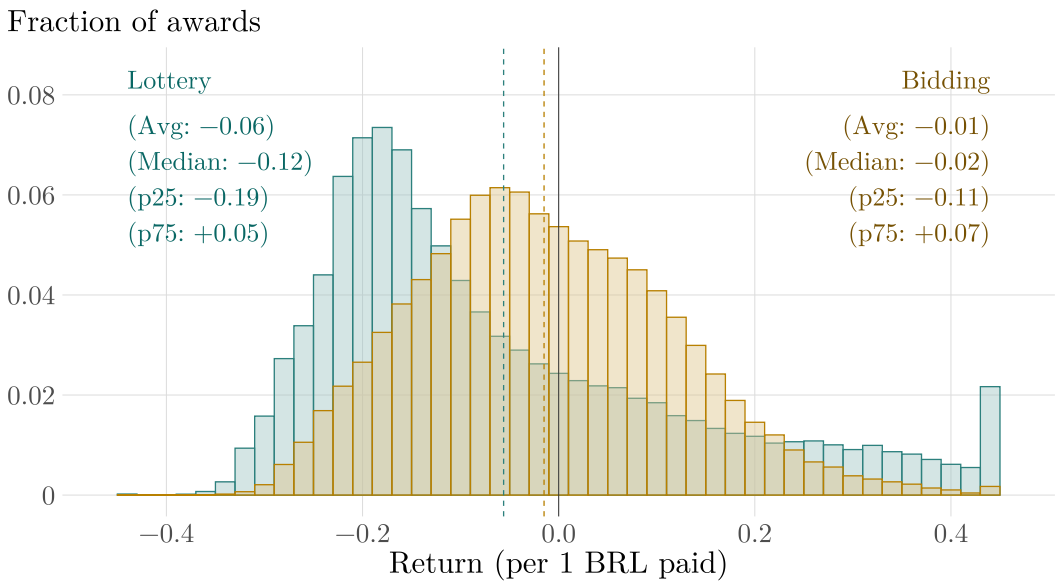


Figure 5: Bid-awarded contracts have higher unconditional returns.

Notes: This figure shows financial returns per BRL paid into the pool, calculated with Equation (1) across bid-awarded and lottery-awarded contracts. The sample includes awarded contracts in groups started between 2010 and 2025 that are completed or observed for at least 25% of their planned duration. Outcomes for contracts yet to be awarded in ongoing groups are predicted. Returns beyond the axis range fall in the end bins. Dashed vertical lines mark each mode's average return.

the value of late awards and the return of lotteries (Appendix Figure A.2, panels B, C and F).

The large expected losses from never bidding are a direct consequence of firms' allocation decisions. Firms choose the mix of lottery and bid awards, subject to a regulatory minimum of one lottery award per month. Lottery awards are back-loaded (Figure 6, panel A): bids account for 86% of awards in the first half of the term, and for a majority of first-half awards in 89% of groups, covering 97% of contracts, as large groups mute the minimum requirement.[18] Firms could favor bid awards because they bring fee payments forward, or because bidders carry lower credit risk. The data do not support the second explanation: bid awards default more often, even at the same award time (Appendix Figure A.10).[19]

We can use the data to reconstruct the cumulative probability of an award conditional on not bidding. The resulting distribution's median wait is 82% of group duration, and the probability of receiving credit by the group's midpoint is only 11% (Figure 6, panel B). Waits vary across groups, but the expected wait for a lottery award exceeds 70% of the term in groups covering 91% of contracts.

[18]Between groups started in 2010–2014 and in 2023–2024, the share of early awards made by bid rose from 80% to 90% for cars and from 69% to 82% for real estate, and stayed at 89% for motorcycles (Appendix Table A.6).

[19]Default rates fall steeply with award time, possibly because the pre-award period screens participants (Doornik et al., 2024a). Bidding lowers the outstanding balance but brings awards forward, when defaults are more common. Within three years of the award, bid awards are 1.16 percentage points more likely to default than lottery awards, and 0.38 points more likely at the same award month. A large share of bid awards therefore does not lower overall default rates.

Table 1: Award timing and returns of never bidding versus bidding monthly.

Strategy	Award probability (%)		Median award time (months)	Expected return
	by 12 mo.	by 24 mo.		
	(1)	(2)	(3)	(4)
Panel A. Cars (14% fee)				
Never bid	3	6	65	−0.14
Bid 40%	61	76	8	0.11
Bid 50%	85	94	3	0.12
Bid 60%	96	99	3	0.09
Panel B. Motorcycles (21% fee)				
Never bid	3	9	54	−0.21
Bid 40%	94	98	3	0.12
Bid 50%	98	99	2	0.09
Bid 60%	99	100	2	0.03
Panel C. Real estate (18% fee)				
Never bid	2	3	158	−0.24
Bid 40%	22	37	45	−0.11
Bid 50%	58	71	10	−0.07
Bid 60%	86	93	6	−0.07

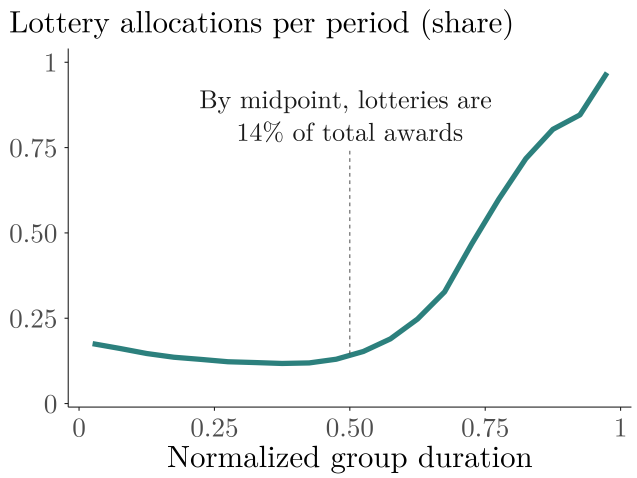
Notes. This table shows simulated outcomes for a contract that joins a group at its start, under different bidding strategies. Bids are assumed to be constant every month until awarded. The contract characteristics are the asset category's average fee, reported in the panel heading, and each group's planned term. Bidding strategies use the observed minimum bids as winning thresholds, with other participants' bids assumed fixed. See Appendix E for details. Non-bidding strategies are calculated using the group's estimated lottery hazard. Returns are calculated with Equation (1) at the benchmark rates. Rows average over contracts, weighting each group by its size and sampling weight.

To quantify the importance of the allocation mechanism, we simulate a counterfactual with lottery awards spread uniformly over the group's duration and average contract characteristics. Expected returns of never bidding increase by 0.17, 0.18, and 0.11 for cars, motorcycles, and real estate, respectively, becoming slightly positive for cars and closing 87% and 46% of the loss for motorcycles and real estate.

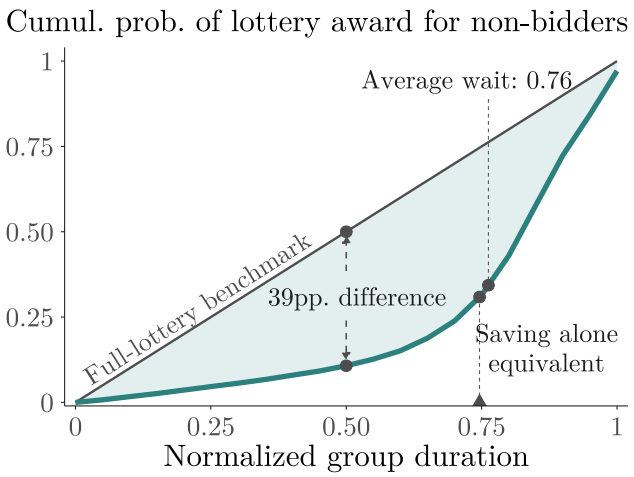
The takeaway is that participants who do not plan to bid, or who plan to bid only late, are buying a contract with large expected losses. Furthermore, those who do not bid face a long wait for a lottery award and a very low chance of early award. A natural question is whether consumers appear to be aware of these statistics.

3.6 Fact 4: Consumers hold large misperceptions about lottery chances

Lastly, through our survey data, we document evidence consistent with optimism about lottery chances, alongside considerable uncertainty about these beliefs, consistent with weakly informed



Panel A. Flow of lottery awards



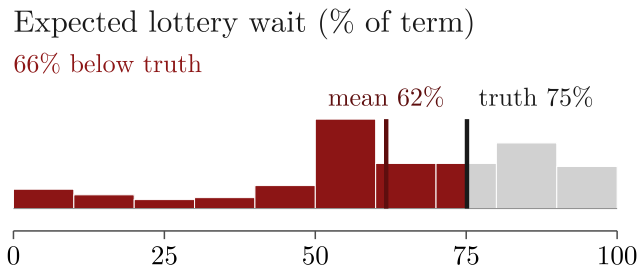
Panel B. Cumulative probability of lottery award

Figure 6: Lottery awards are back-loaded, creating long waits for participants who never bid.

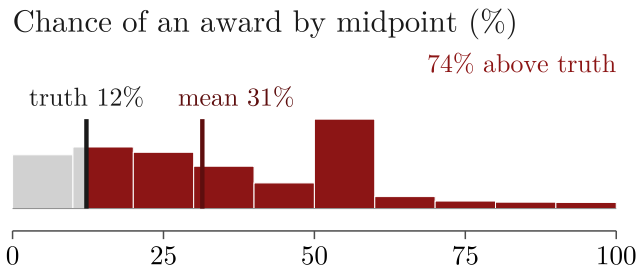
Notes: This figure shows how lottery awards are distributed over a group's duration. Panel A shows the share of lottery awards, pooling the observed awards of all groups started between 2010 and 2025. Panel B shows the cumulative probability of a lottery award for a participant who never bids, from each group's estimated lottery probabilities (Appendix D), weighted by group size. The triangle marks how long a participant who deposited the same monthly installment in a savings account at a 4% real rate would take to accumulate the face value, as a share of the term, averaged across asset categories. Estimates by asset category are displayed in Appendix Figure A.11.

priors. We frame questions under a strategy of never bidding and relying exclusively on lotteries, using respondents' simulated contract terms. We report beliefs and true values re-weighting asset categories to reflect respondents' choices.

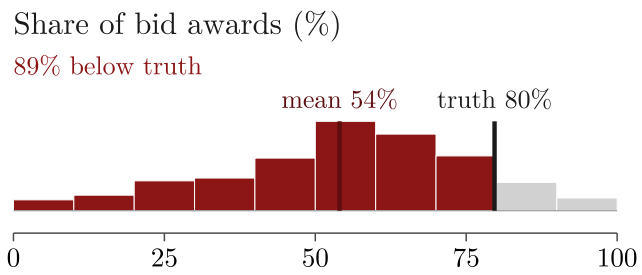
On average, respondents expect an award after 62% of the contract duration, compared with 75% in the administrative data (Figure 7, panel A). They also estimate a 31% chance of receiving an award by the group's midpoint, compared with 12% in the data (panel B). A majority of respondents overestimate their lottery prospects. The average beliefs sit roughly at the 10th percentile of the distribution of lottery wait times across groups in the data.



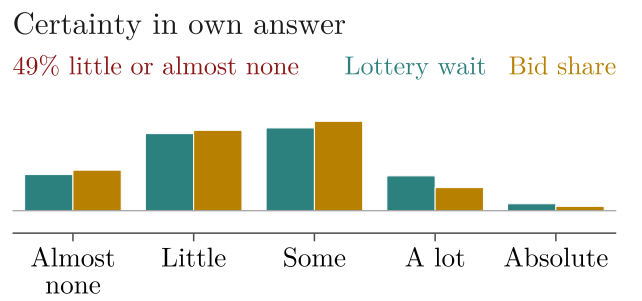
Panel A. Expected wait for a lottery award



Panel B. Chance of a lottery award by the midpoint



Panel C. Share of awards won by bid by midpoint



Panel D. Certainty in own answers

Figure 7: Consumers hold large misperceptions about lottery chances.

Notes: This figure shows consumers' beliefs about lottery wait times and the share of bid awards. True values are calculated from Central Bank administrative data and averaged across asset categories weighted by respondents' asset choices in the contract simulation. The share of respondents whose estimates are optimistic, with respect to lotteries, is calculated separately for each asset category. See Section 5.1.1 for a description of belief measurement. Panel A reports respondents' expected wait for a lottery award as a fraction of the contract's duration. Panel B reports their perceived probability of a lottery award by the group's midpoint. Panel C reports the perceived share of awards won through bidding. Panel D reports respondents' confidence in their answers.

Respondents also report substantial uncertainty. Almost half of the consumers report having little or no confidence in their answers, while only 16% report high or complete certainty (panel D). We also see bunching at 50% across questions, including questions that were quantitative or used visual aids, consistent with uncertainty or poorly formed expectations.

Misperceptions also extend to the allocation mechanism: respondents believe that bid awards account for 54% of awards in the first half of a contract, compared with 80% in the data. Nearly 90% underestimate this share, suggesting consumers are unaware of a key input to understand their lottery chances. Qualitative questions yield similar patterns (Appendix Figure A.12).

How much do these beliefs matter for financial returns? We value a never-bidding contract, at each respondent's chosen asset category and term, under two award-timing distributions: the one in the administrative data, and one that matches respondents' average expected wait and chance of an award by the midpoint. At true lottery odds, never bidding loses 0.20 per BRL paid into the

pool. At average beliefs, it loses 0.13. Average beliefs therefore hide about a third of the expected loss.[20]

In Section 5.3, we further examine heterogeneity by financial literacy, education, and experience, as well as beliefs among brokers, and show misperceptions are widespread.

3.7 Takeaways

The facts illustrate how, for a large share of participants, participating in a consórcio group is likely to lead to losses relative to saving and borrowing at market rates. Even under significantly more expensive counterfactual borrowing rates, or lower savings interest rates, a large share of participants would still suffer losses. Furthermore, participants with lower income also tend to cancel more and face relatively larger losses.

Returns are also predictable by bidding strategies. Strategies consisting of bidding from the start of a contract can lead to higher and sometimes positive returns relative to the market. However, these strategies are mostly available for consumers with sufficient liquidity to bid. Not bidding and waiting for a lottery leads to long waits and predictable losses. Furthermore, consumers do not appear to be aware of just how long the wait for a lottery award might be, how low the chance of an early lottery award is, or that bids are much more frequent than lotteries.

4 Participation in group savings: existing explanations and misperceptions

The facts documented in Section 3 raise a question: why do non-bidders join consórcios despite large expected losses and a low share of lotteries? The low share of lotteries, in particular, implies low chances of early access and long wait times. In this section, we draw on a large literature in development economics that studies participation in RoSCAs, considering the standard explanations and their limitations in this setting. We then show how misperceptions about lottery probabilities can interact with these explanations and help rationalize participation.

We write a model to examine individual decisions. Connecting to our survey and information experiment, we focus our discussion on non-bidders, who suffer large expected losses. Their participation is pivotal for the sustainability of groups that pool both types; otherwise, bidders would have no one to borrow from. In Appendix H, we extend the model to explicitly consider bidders, group mechanics, and an equilibrium with competitive firms.

[20] The calculation covers financial returns only, not any value of owning the asset early beyond the benchmark rates.

4.1 Setup: indivisible consumption

Consider a mass of agents indexed by i who want to acquire a lumpy indivisible good with normalized price 1. Let $t \in \{1, 2, 3\}$ denote time periods. Let $\frac{1}{2} > y \geq \frac{1}{3}$ denote per-period income, so the agent can only buy the good alone at $t = 3$. Let c_t denote nondurable consumption, and $d_t \in \{0, 1\}$ denote ownership of the indivisible good at t. The indivisible good yields a flow of ξ in $t \in \{1, 2\}$ and $\Delta \geq \xi$ at $t = 3$, possibly capturing value beyond the model. Let $v(\cdot)$ denote concave utility from nondurables with $v(0) = 0$ and $v'(0) = \infty$. Consumers are quasi-hyperbolic discounters with present-bias parameter $\beta \leq 1$. Period utility is $u_t = v(c_t) + \xi d_t$ for $t = 1, 2$ and $u_3 = v(c_3) + \Delta d_3$. The consumer may save at $r^S = 0$, a normalization.

A rotating finance contract (τ, ℓ) charges a fee τ, leading to installments $q_t(\tau) = (1 + \tau)/3$ and provides a probability $p_t(\ell)$ of obtaining the good at t, which depends on the share of lottery awards ℓ. The random RoSCA benchmark is given by $(0, 1)$. We assume a member who has been awarded pays her remaining installments. A member not yet awarded may cancel at $t = 2$ and forfeits what she paid. To simplify exposition, we omit details of group operation, budget, and the bidding procedure when $\ell < 1$.

The self at t maximizes $u_t + \beta \sum_{s>t} u_s$, taking her later selves' choices as given. The decisions boil down to two options: save alone or join the rotating finance group. In general, an option to borrow at $r^B > r^S$ will reduce the parameter space in which individuals join the group.

4.2 How does the literature understand rotating finance instruments?

The literature broadly discusses the demand and sustainability of rotating finance groups as a result of poor outside options to save and borrow.

First, following Besley et al. (1993), when credit is unavailable, pooling savings offers a chance of obtaining an indivisible good earlier than saving alone. However, limited access to credit alone seems insufficient to explain participation: as Section 3.5 shows, non-bidders would, on average, obtain the good earlier by saving the installment amount. While consumers may still value the small chance of an early award, difficulty saving may be more important than credit constraints in explaining their participation.[21]

A second interpretation is more prominent in the literature: rotating finance groups serve as commitment devices to help people save (Bryan et al., 2010; Breza and Kaur, 2026). Demand for commitment can emerge from variants of time inconsistency or other microfoundations, such as household conflict (Anderson and Baland, 2002; Ambec and Treich, 2007; Gugerty, 2007). The

[21] The credit rationalization also seems insufficient to explain the rigid allocation mechanism structure and sustainability of groups. Besley et al. (1994) shows that a credit market composed of the same group resources is often more efficient, therefore raising the question of why a system of peer-to-peer lending would not replace the rotating structure if access to credit was the key motivation.

proposition below captures this idea, similar to the results in Basu (2011).

Proposition 1. *Fix a contract (τ, ℓ), with $q(\tau) < y$ affordable, and consider consumers who would not buy the good by saving alone. Then:*

(i) A random RoSCA, $(0, 1)$, is joined iff $\underline{\beta}^R \leq \beta \leq \bar{\beta}^R$.

(ii) A consórcio, (τ, ℓ), is joined iff $\underline{\beta}^C(\tau, \ell) \leq \beta \leq \bar{\beta}^C(\tau, \ell)$. Furthermore, $\underline{\beta}^C(\tau, \ell)$ is decreasing in ℓ, and increasing in τ. Moreover, $[\underline{\beta}^C(.), \bar{\beta}^C(.)] \subseteq [\underline{\beta}^R, \bar{\beta}^R]$.

(iii) The ratio $\underline{\beta}^C(\tau, \ell)/\underline{\beta}^R \geq 1$ is decreasing and convex in ℓ.

All proofs are in Appendix H.8. The lower bound of the interval in β, expressed in (i) and (ii), captures the effectiveness of commitment: the lower it is, the higher is the scope for present-bias to drive demand for rotating finance groups. The upper bound captures the agent who is sufficiently patient and may prefer to save alone.

As Proposition 1(ii) shows, consórcios are less effective as commitment devices because of the lower share of lotteries $\ell < 1$, and to some extent, of the high fees charged. In particular, the probability of obtaining the good at each period is part of the commitment force of rotating finance, as it incentivizes the participant to make payments to the group while waiting for the award.[22] Furthermore, as (iii) shows, the rate at which consórcios depart from RoSCAs is convex in ℓ, suggesting that greater reductions from $\ell = 1$ increasingly weaken the strength of consórcios as commitment devices. The very low ℓ observed in the market therefore complicates the explanation of consórcios as commitment devices under correctly perceived odds. Furthermore, Basu (2011) shows how sophisticated agents who join should not renege on the contract. Contrary to most other RoSCAs, we see high cancellation rates, suggesting the need for additional mechanisms such as stochastic present-bias (e.g., Duflo et al., 2011). Additionally, partial naivety can also explain cancellations, but would weaken demand for commitment (John, 2020).

4.3 Wrong beliefs about lottery probabilities

As shown in Section 3.6, consumers in our sample hold substantial misperceptions about lottery chances, particularly overoptimism. These beliefs may partly reflect how lotteries are presented and discussed in the consórcio market. Firms are not mandated to disclose data about expected chances over time. Moreover, the randomness of lotteries may lead consumers to focus on the possibility of receiving an award at any point, rather than on how their probability of receiving one evolves over time.[23]

[22] At $p = 0$, demand for illiquidity is the sole commitment force. With impersonal groups, the consórcio turns into a product that is indistinguishable from many other illiquid savings instruments, such as certificates of deposit.

[23] Indeed, Time Magazine (1967) describes consórcios as "something that is both safe and lucky", suggesting that the appeal of lottery-based allocation was already prominent in early descriptions of consórcios.

Part of the difficulty learning about the true lottery probability over many time periods is that it is an equilibrium outcome: it depends on the composition of bidders and non-bidders, their bidding decisions, and firms' decisions. Evaluating these probabilities requires consumers to anticipate the behavior of other participants under uncertainty. Our reduced-form parameter captures potential failures to form these equilibrium expectations correctly. Misperceptions may also reflect how brokers and firms communicate, advertise, or shroud information about the allocation schedule, such as the number of lottery or bid awards (Gabaix and Laibson, 2006). In this sense, the contract has two prices: the fee charged, reflected in the installment, is the base price, while the lottery odds are the "hidden" price. Under this interpretation, avoiding the hidden price — the "sophisticated" action — does not necessarily rely on information about lotteries, but rather on liquidity.

To capture the role of misperceptions in the model, define the perceived probability of a lottery award as $\hat{p}(\ell) = \frac{1}{3} + \lambda\left(p(\ell) - \frac{1}{3}\right)$, where $\lambda \in [0,1]$ is a reduced-form parameter governing the accuracy of consumers' beliefs. When $\lambda = 1$, consumers correctly perceive lottery probabilities. As λ decreases, they become overoptimistic about early awards and less sensitive to the true odds.

Proposition 2. *Fix a contract (τ, ℓ) with $\ell < 1$, and a consumer who would join a pure lottery contract at fee τ.*

(i) there is $\bar{\lambda}(\tau,\ell) > 0$ such that she takes up the contract (τ,ℓ) whenever $\lambda < \bar{\lambda}(\tau,\ell)$;

(ii) the band $[\underline{\beta}^C(\tau,\ell), \bar{\beta}^C]$ widens as λ falls;

(iii) the highest fee she pays for a contract with lottery share ℓ, $\bar{\tau}(\ell;\lambda)$, rises as λ falls.

As Proposition 2(i) shows, the perception of probabilities can rationalize participation even when there is no demand for commitment, and consórcios are viable products for consumers who would reject them if they knew the right odds. Therefore, misperceptions about lottery odds can rationalize entry when ℓ is low.[24]

The perception of probabilities interacts with both commitment and early access. First, as Proposition 2(ii) shows, misperceived odds widen the band of sophisticated present-biased consumers who join. Equivalently, $\bar{\lambda}$ is increasing in β: the more present-biased the consumer, the more misperception a contract needs to retain her, since more of its value must arrive through the lottery. Second, as Proposition 2(iii) shows, perceived odds raise the willingness to pay for the chance of early access.

Furthermore, the perception of probabilities can also interact with supply side responses. If consumers are insensitive to variation in true probabilities, as predicted by models of misperceptions, firms' incentives to compete by providing groups with more lotteries are weakened.

[24]There are other possible rationalizations outside the scope of this paper. Consumers may experience anticipatory utility, hope or aspirations. Consumers may also misperceive their bidding prospects, expecting to receive future windfalls or to save and be able to bid. Or consumers may demand the option value of bidding even if they end up rarely exercising it.

5 Experimental and survey evidence on misperceptions

We next ask whether correcting these misperceptions changes consumers' beliefs and decisions. We survey consumers and brokers and randomly assign consumers to short videos that disclose how often awards go to bids and how long lottery waits are, with one version additionally explaining how group size affects lottery chances. We find the videos partially correct beliefs, raise certainty in answers, and change perceptions of bidders' advantage in the product. For the basic treatment arm, correcting beliefs shifts demand differently depending on liquidity: consumers who cannot bid increase their valuation of a prospective lottery-intensive offer more.

5.1 Survey and experimental design

We design surveys to capture beliefs about relevant attributes of consórcio contracts. We then design an information provision experiment that leverages the Central Bank data to disclose the true share of bid awards and how they relate to the wait time for a lottery.

5.1.1 Belief measurement: lottery wait times and bid award shares

First, we elicit beliefs about the payoff-relevant object for a consumer who never bids: the estimated lottery wait time and probability of early award, defined as an award in the first half of a contract.[25] Second, we elicit beliefs about the shares of bid awards. We tailor these questions to each respondent's desired asset category and contract duration and incentivize quantitative answers with accuracy bonuses worth up to 50% of the base survey payment. We measure each belief using complementary qualitative, quantitative, and probabilistic formulations. The three measures are intended to capture the same belief under different formulations (Haaland et al., 2023). For the bid share, respondents report the same object using natural language (e.g., "more than half"), a visual share, and the probability that a randomly selected award during the first half of the contract was made through bidding. Appendix Figures G.2 and G.3 provide screenshots of the three types of belief questions.

5.1.2 Information treatments

We designed our information provision experiment with a focus on the allocation mechanism, the share of lottery and bid awards, and the expected wait time for a lottery. The primary goal was to correct misperceptions about consumers' lottery award chances. In addition, we also explain

[25] For the market, a $Beta(\alpha, \beta)$ distribution fitted to these two moments tracks the cumulative distribution of lottery-only wait times within 7 percentage points (Figure 6, panel B). Because contracts are long, eliciting the full distribution was impractical, and piloting also showed that more granular measures performed poorly.

mechanics: more bid awards means fewer lottery awards, and therefore longer lottery wait times.[26]

Importantly, we provided no information about prospective financial returns, how to compare consórcios to savings or borrowing, optimal bid strategies, or any rhetorical arguments focusing on the financial advantages of consórcios.

We design two information arms, labeled D for "debiasing" and DG for "debiasing + group size mechanics". Both information arms present the true bid share at the group midpoint for the respondent's asset category, computed from the Central Bank data as in Figure 6; we refer to this number as the signal. DG additionally highlights that there is variation in group sizes, that group size matters for lottery chances because of the one-lottery-per-month requirement, and that group sizes have doubled in recent years. The intention of arm DG was to provide the link between a useful observable proxy (group size) and lottery wait times.[27] Videos were tailored for each asset category. Script details are available in Appendix G.

To avoid differential attention or attrition due to the video treatment, we make an active control group. We use a public Central Bank video containing very basic information about consórcios.

Appendix Figure A.15 reports the distribution of video completion by arm: median completion is essentially 100% and mean completion is about 95% in each arm, with roughly 91% of respondents watching at least 90% of their video.

5.1.3 Incentivized valuation of a lottery-intensive offer

If consumers misperceive lottery probabilities, it is possible they are also less sensitive to changes in the number of lottery awards. This can provide a connection between a demand-side bias and a supply-side response: firms have weaker incentives to offer lottery-intensive groups, consistent with the prevalence of low-lottery groups observed in the market data.

To test whether information changes demand for groups with more lotteries, we partner with a consórcio firm to create a prospective offer for such a group. We then measure consumers' valuation of the offer through an incentive-compatible mechanism.

The proposed group would have only lottery awards in its first 12 months, at least two lotteries per month thereafter, and a smaller group size. Simulations suggest that 40 to 60% of awards would be allocated by lottery (95th percentile of observed groups), with a 30 to 40% chance of an award by the midpoint, compared with 10 to 15% in observed groups.

Respondents were told that choosing access to this real offer could authorize the administrator to contact them about joining. We did not disclose the administrator's identity or fees, keeping

[26] In piloting interviews with consumers and brokers, they sometimes expressed that bid awards do not delay lotteries since bids add resources to the pool. This is a misperception: bid awards always remove net resources from lottery awards. Appendix Figure A.13 shows a 0.74 correlation between early bid awards and lottery wait times.

[27] Appendix Figure A.14 shows the size gradient in finished groups: by the group midpoint, a never-bidder has a 30% chance of an award in the smallest fifth of groups and 7% in the largest.

the offer centered on lottery allocations.

We elicited valuations using a five-option multiple price list, in which respondents chose between access to the offer and a bonus of up to BRL 50 or BRL 100 (two to three times the base survey payment). Participants first answered an example question and were then instructed to treat each choice as final. The multiple price list is simple but can lead to censoring, as participants could have valuations well above BRL 50 or 100, and others could be uninterested. Indeed, valuations are heavily censored: 75% are either the minimum (zero) or maximum possible value.

5.1.4 Other outcomes

To measure perceptions of bidders' advantage, we asked whether they agreed that consórcios are generally only worth it for people who can bid, whether administrators tend to favor bidders, and whether they would recommend a consórcio to a friend or relative who cannot bid.

To measure beliefs and intended bidding behavior, we elicited expected winning bids and the likelihood and maximum size of respondents' own bids, together with that bid's perceived chance of winning and expected time to award. These outcomes capture whether the information provision changed beliefs about the bidding competition and willingness to participate.

We also measured hypothetical choices using two randomized exercises. First, respondents chose between consórcio groups whose fees, monthly lottery awards, group sizes, and expected winning bids varied independently (Appendix Figure A.16), identifying each attribute's causal effect on the stated choice (Hainmueller et al., 2014). Second, respondents chose between a consórcio with average market attributes and either financing or saving, with borrowing or savings rates randomized. To facilitate comparison, we reported the downpayment and time required to accumulate it under financing, and the time required to purchase the good by saving an amount equal to the consórcio installment.

5.2 Recruitment, consumer characteristics, and balance and attrition

We recruit consumers and brokers through social media ads explicitly mentioning consórcios, targeting potential customers who have expressed interest in the product ("leads"). This approach reflects the Brazilian market, where social media use is widespread and consórcios are commonly advertised by brokers and firms. To better approximate actual potential customers, the ads did not mention monetary incentives, and we excluded respondents who reported being unfamiliar with consórcios. Recruitment ran from June 3 to August 19, 2026.

For consumers, the study consists of two survey waves (Figure 8). Wave 1 elicits beliefs, consórcio experience, financial literacy, savings behavior, and demographic characteristics. Wave 2 delivers

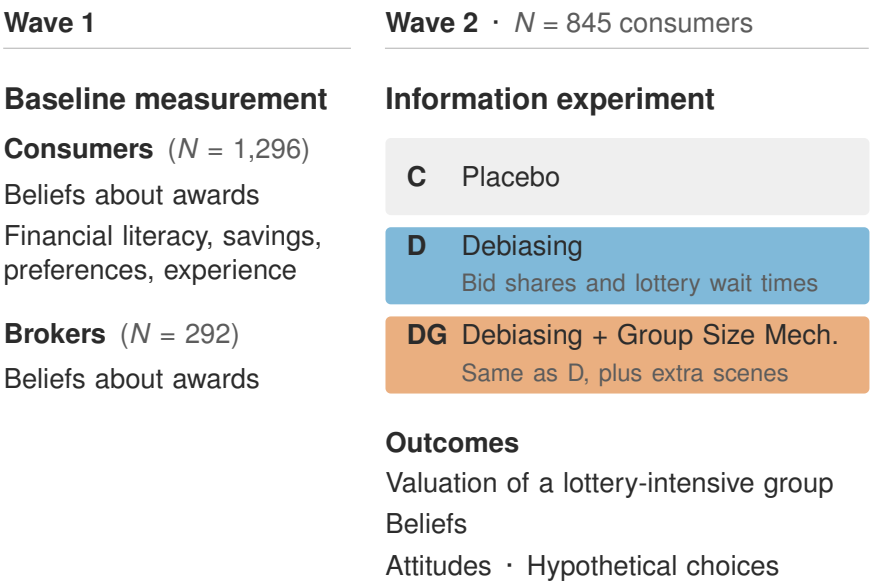


Figure 8: Experimental design.

the randomized information provision, repeats the belief elicitation, and measures outcomes.[28] Brokers are recruited alongside consumers but respond only survey.

5.2.1 Sample characteristics

Table 2 describes the consumer sample. Respondents are, on average, 47 years old and 59% male, compared with the representative consumer population, which averages 41 years and is 65% male. Simulated interest is heavily concentrated in real estate (48%) and cars (41%) and away from motorcycles (12%), whereas new entrants who actually purchase consórcios in the administrative data split more evenly across real estate (28%), cars (37%), and motorcycles (35%). Most respondents are employed (87%) and are the primary financial decision-maker in their household (83%). The Wave 2 subsample is similar (Appendix Table A.10).

The sample provides a mix of experienced and new consumers: 57% have participated in a consórcio before and a quarter report currently paying installments.

Importantly, consumers report purchase intent: over half say they are likely to join a consórcio within three months, and 45% report already considering a specific administrator. About a third have discussed a consórcio with a broker. Consumers also seem to be positive about the product: 70% report they would recommend consórcios to friends and family.

Relevant to our model and mechanism, about 71% of respondents report that they would not have their own resources to bid about 50% of the simulated contract's face value during its first years. We call these respondents liquidity-constrained and the remainder unconstrained. The liquidity constraint correlates with household income, savings behavior, the ability to cope with

[28] Contract choices remain largely stable across waves: 88% of consumers retain the same asset category.

Table 2: Consumer sample description.

	Wave 1 Sample		Population	
	Mean (SD)	Median	Mean (SD)	Median
Panel A. Demographics				
Age	47.2 (12.5)	47.0	40.6 (13.5)	39.0
Male (%)	59.1		65.4	
Employed (%)	87.2			
College complete or higher (%)	69.7			
Primary financial decision maker (%)	82.7			
Panel B. Financial profile				
Financial literacy score (0–4)	3.2 (1.0)	3.0		
Savings buffer (months)	5.1 (4.0)	4.0		
Implied monthly installment (BRL)	1,792 (2,176)	1,250	783 (710)	556
Installment as % of monthly income	27.6 (61.3)	14.8	23.6 (34.7)	15.1
Panel C. Consórcio profile				
Past consórcio participation (%)	56.9			
Currently paying a consórcio (%)	25.5			
Discussed consórcio with vendor (%)	33.9			
Likely to join (≤3 mo.) (%)	56.0			
Would recommend consórcio (%)	69.4			
Already considering an administrator (%)	44.8			
Panel D. Asset of interest				
Real estate (%)	47.7		28.3	
Cars / vehicles (%)	40.8		36.9	
Motorcycles (%)	11.5		34.8	
N	1,296		–	

Notes. This table compares the Wave 1 consumer sample, after the screening quality filter, with a representative sample of Central Bank records for consumers who joined a consórcio in 2025–2026 (Population). Binary variables report the share in percent. Continuous variables report the mean, standard deviation in parentheses, and median.

emergencies, and access to credit (Appendix Table A.7).

Brokers. We survey 292 brokers (Appendix Table A.8). Three-quarters work at businesses dedicated to consórcios: 46% at specialized consórcio firms and a further 31% at consórcio administrators, with the remainder at financial-advisory firms, banks, and car dealerships. At specialized consórcio firms and financial-advisory firms, 60% of brokers sell for two or more administrators, against 9% of brokers employed by an administrator. They are active sellers, with a median of five sales a month, although half have been in the business for three years or less. About 78% advertise on social media, almost all of them on Instagram, and 61% post at least once a week. Consumers and brokers come from the same market (Appendix Table A.9): about half of each sample is in the Southeast, both lean toward real estate and cars, and 84% of consumers who name an administrator they are considering name one that a broker in our sample sells for.

5.2.2 Balance and attrition

Appendix Table A.11 summarizes balance across covariates. Treatment groups are balanced: a joint test of orthogonality has p-values of 0.63 and 0.39. Appendix Table A.12 summarizes completion rates and attrition. For each treatment arm, we report the number assigned, the number who did not complete the survey, the number excluded by the prespecified quality filters, and the resulting main analysis sample. We distinguish survey attrition from quality-based exclusions, because the latter have observed outcomes. Non-completion is small, at about 8%, and completion rates are balanced across treatment arms.

5.3 Survey evidence on misperceptions, learning, and strategies

5.3.1 Misperceptions about lottery probabilities are widespread across subgroups

In Section 3.6, we presented evidence that consumers are, on average, optimistic about lottery chances and wait times. When there are winners and losers in identical contracts, one natural hypothesis is that consumers may be heterogeneous in sophistication: some understand hidden attributes, while others do not (Campbell and Ramadorai, 2026). Proxies for sophistication can include financial literacy, education, past experience, or the outcome itself.

To measure heterogeneity across subgroups, we combine the lottery wait and bid share questions into two different standardized indices and verify how much closer to the truth each subgroup is. For each consumer characteristic, we regress each belief index on an indicator for that characteristic and report robust 95% confidence intervals. Figure 9 combines the results: higher point estimates represent more precise beliefs.

The data does not support a straightforward sophistication story. Consumers who report they plan to bid early or who have liquidity to bid do not hold substantially more accurate beliefs about lottery probabilities or bid shares. Financial literacy, education, and prior consórcio experience all predict more accurate beliefs, but none closes the information gap by more than 25%. Financial literacy presents the strongest correlation: respondents score 0.46 standard deviations higher on the bidding-share index and 0.23 standard deviations higher on the lottery-wait index. Respondents with prior consórcio experience also score higher on both indices. While these patterns are consistent with expected directions, the magnitudes are modest.

Brokers' answers provide a comparison point to evaluate consumers' beliefs. Brokers are substantially more accurate about bid shares, sitting at about 50% of the gap between average consumers and true values. However, they do not hold more accurate beliefs about lottery wait times, and inexperienced brokers are more optimistic about lottery chances than the average consumer.

This result is consistent with how different pieces of information feed into different strategies and outside options. By revealed preference, a bidder wants an early award, and to understand

whether they can get it, the sufficient statistic is the bid cut-off in the group.

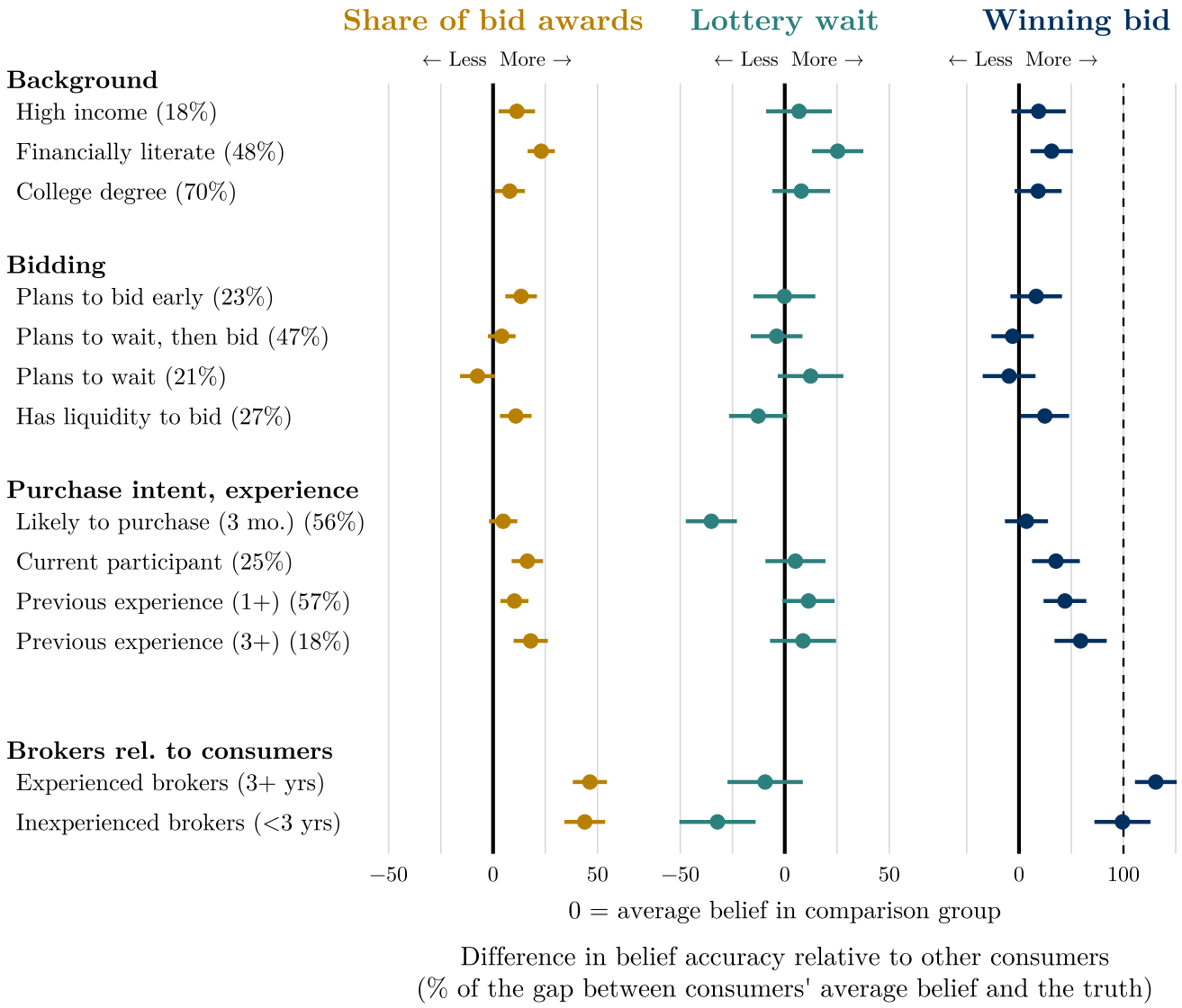


Figure 9: Heterogeneity in beliefs about award allocation across consumer subgroups.

Notes: This figure shows how much more accurate each subgroup's beliefs are than those of other consumers, as a percentage of the gap between consumers' average belief and the truth. Beliefs about the share of bid awards and about the lottery wait are each combined into an index of three standardized measures. Consumer rows compare consumers with and without each characteristic, within asset category; percentages in parentheses are the share of consumers with the characteristic. Broker rows compare brokers with all consumers. Bars show 95% confidence intervals from robust standard errors.

5.3.2 Broker and consumer responses are consistent with limited learning

As displayed in Figure 9, experience correlates only mildly with more accurate beliefs about the share of bid awards and lottery wait times. Brokers are more accurate than consumers about the share of bid awards, closing almost half of the gap. They are not more accurate about the lottery wait, and inexperienced brokers are less accurate.

Learning from repeated experience. Because contracts last many years, consumers have few own draws to update their beliefs about lottery wait times. In the fastest-learning case, a consumer starts from equal odds between the full-lottery benchmark and the true distribution. This consumer would need about six non-bidding award experiences to be 95% confident that the benchmark is wrong. A second issue is that would-be bidders also compete in lotteries until they bid. The average award time of lotteries is therefore not the same as the expected award time of a non-bidder.[29] In contrast, we do see evidence of stronger learning about winning bids, closing about 50% of the gap.

Learning from firm disclosure. Before purchase, firms disclose little information. We simulated purchases on the websites of the top 22 administrators, covering 90% of sales, and documented the information available. In only one case are the planned allocation rules available. After purchase, consumers can learn from monthly meeting minutes, which are required to report winning bids and the breakdown of awards. In principle, a consumer could use the breakdown of awards, along with the number of participants, to assess the past lottery chances in the group and obtain a forward-looking estimate. Past winning bids are therefore observable to inform future bidding decisions. Consistent with the available information, brokers and experienced consumers are substantially more accurate about winning bids, as reported in Figure 9.

Learning from brokers. In practice, most sales happen through brokers (73% of past purchases in the survey data), who report earning high sales commissions of about 3% of a contract's face value. But the incentives of commission-based agents can be distorted (Anagol et al., 2017; Egan, 2019). These incentives may also differ between clients who bid and clients who wait for the lottery. In the broker survey, we asked whether a client who waits longer for the credit is less likely to do business with the broker in the future. We randomly varied whether the client wants the credit as soon as possible through a bid or through the lottery. Brokers are 11.8 pp less likely to agree when the client relies on the lottery (54% versus 66%, $p = 0.04$). This suggests brokers lose less when a lottery client waits than when a bidding client waits. Their incentive to correct optimistic expectations about lotteries is therefore weaker.

Even absent distorted incentives, learning about lottery chances from brokers can be limited because they also report optimistic beliefs. Only 10% agree with the statement that "consórcios are only advantageous to bidders". Brokers could be answering strategically. However, 81% report owning a consórcio themselves, similar to other settings where agents tend to buy the products they sell (Linnainmaa et al., 2021).

[29] Social learning may also be limited by factors such as noise from different strategies and biased sharing.

5.3.3 Consumers report strategies that would lead to expected losses

Do consumers plan to follow strategies that lead to expected losses? Only 23% of respondents plan to bid early. Most plan to wait for the lottery, either bidding if the wait becomes too long (47%) or never bidding (21%), both of which are strategies likely to generate losses, as shown in Section 3.5.

Plans vary with reported liquidity to bid 50%, which captures a reasonable early bid threshold. Among consumers with liquidity to bid, 33% plan to bid early, compared with 19% among those without (a 13.7 pp difference, $p < 0.01$). Consumers without liquidity are instead twice as likely to plan to wait for the lottery alone (24% versus 12%, $p < 0.01$). Half of consumers with liquidity plan to wait and then bid, a share similar to those without liquidity (50% versus 46%, $p = 0.23$).

Beliefs about lottery chances are more weakly related to these plans. Within each liquidity group, consumers who are optimistic about an early lottery award, those reporting a chance above the median of 30%, are 4 to 5 pp less likely to plan to bid early. The difference is significant only among consumers without liquidity ($p = 0.04$; $p = 0.41$ among those with liquidity), and optimism is not associated with planning to wait. The difference by liquidity is about three times as large.

5.4 Effects of information experiment on survey outcomes

We next turn to estimate the effects of the randomized information provision on beliefs, choices, and attitudes. If consumers hold mistaken and overoptimistic beliefs about lottery chances, the information treatment should on average correct these beliefs. Our key hypothesis is that it will also change liquidity-constrained consumers' valuation of the prospective offer for a group with more lotteries by making it more attractive relative to the options in the market.

5.4.1 Empirical specification

We pre-registered the estimation of average treatment effects $\tau_D = E[Y|D] - E[Y|C]$ and $\tau_{DG} = E[Y|DG] - E[Y|C]$ using the augmented inverse-probability-weighted (AIPW) estimator, which adjusts for pre-treatment characteristics to improve precision (Athey et al., 2019). The outcome model is a causal forest fit on the randomization strata and on pre-treatment characteristics from the screening survey, including baseline measures of the outcome. For heterogeneity by a binary characteristic M, we project the conditional effect on M, which gives the effect at $M = 0$ and its difference at $M = 1$. We report heteroskedasticity-robust standard errors throughout.

5.4.2 Information corrects beliefs and changes product evaluations

Beliefs are partially corrected. Both treatments shifted beliefs in the expected direction, correcting average misperception gaps by about 40–50% across most measurements (Figure 10). Corrections are similar as a percentage of the prior gap for both bidding shares and lottery wait times. Only bidding shares were numerically displayed in the video, while lottery wait times were only visually suggested. Qualitative responses reveal a similar pattern (Appendix Figure A.12). We cannot reject the equality of treatment effects on beliefs across the two arms.

We also measured beliefs about average group sizes, which can serve as a key input to understand lottery chances. Both treatments increased perceived group sizes, likely due to the visual depiction of multiple awards per month. Only treatment arm *DG* was explicit about group size numbers and the existence of heterogeneity in group sizes. We find that although both treatment arms had similar average effects, responses can be large, and *DG* reduced the mean absolute error relative to the true group size.

Changes in average beliefs are accompanied by slightly greater confidence, consistent with uncertainty reduction in Bayesian updating (Augenblick and Rabin, 2021). Both treatments increase respondents' stated certainty in their beliefs by about 0.2 points on a five-point scale ($p < 0.01$).

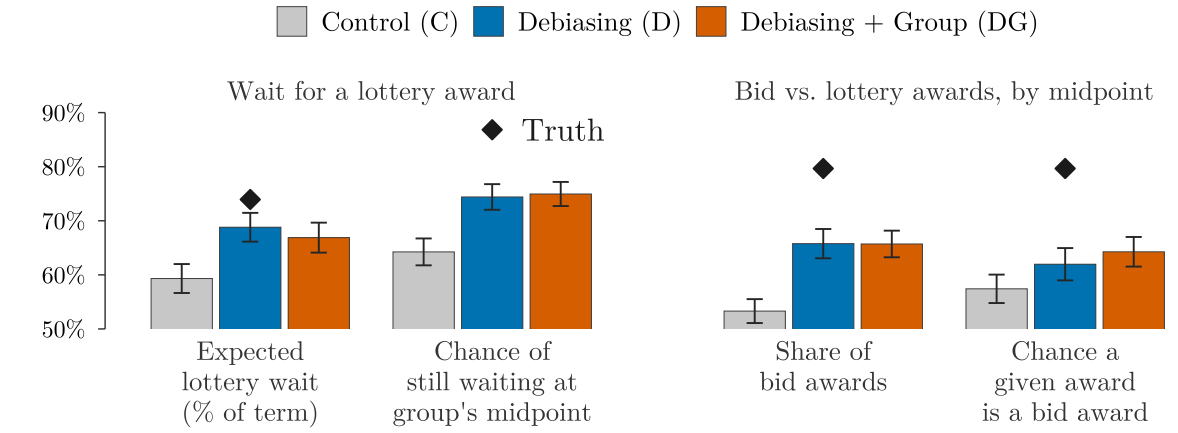
We next test whether respondents with larger initial misperceptions revise their beliefs more in response to treatment. We estimate learning rates using the following pre-registered regression,

$$Updating_i = \beta_0 + \beta_1 T_i + \beta_2 Gap_i + \beta_3 Gap_i \times T_i + \varepsilon_i, \tag{2}$$

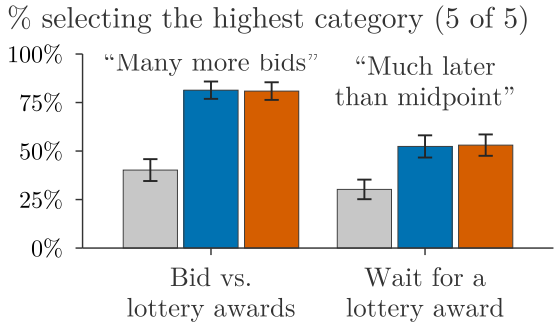
where $Updating = Posterior - Prior$ and $Gap = Signal - Prior$. As shown by Cullen and Perez-Truglia (2022), β_3 can be interpreted as the learning rate of the treatment. We estimate Equation (2) using the obviously-related instrumental variables (ORIV) estimator to account for classical measurement error in priors (Gillen et al., 2019).[30] Appendix Table A.14 displays the results under ORIV and OLS specifications for both treatments. For bidding shares, correcting for measurement error yields an estimated learning rate of 0.49, with no detectable level shift: respondents with larger initial misperceptions revise their beliefs more in response to treatment. For lottery wait times, treatment shifts reported beliefs, but we do not detect a differential response by initial belief gaps.

Information strengthens perceptions of bidder advantage. Both treatments make respondents more likely to view consórcios as favoring those who can bid and make participants less likely to recommend them to those who cannot bid (Appendix Table A.15).

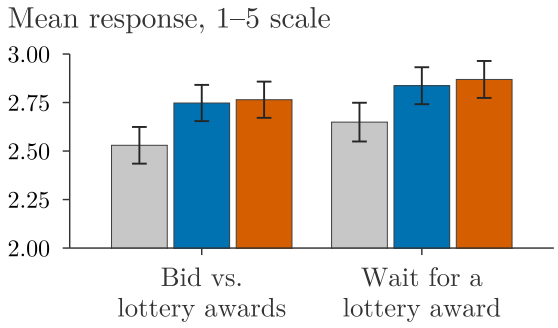
[30] As priors enter both sides of Equation (2), noisy priors can bias the measurement of β_2 and β_3. To see why, imagine a model where $Prior_i = \mu_i + \eta_{0i}$ and $Posterior_i = \mu_i + \eta_{1i}$, where μ_i is a true latent belief and η_i is random noise, independent of the latent belief and survey waves. In this model, there is no learning, but a regression based on Equation (2) recovers $\beta_2 = \frac{\sigma^2_\eta}{\sigma^2_\eta + \sigma^2_\mu}$



Panel A. Quantitative beliefs



Panel B. Qualitative beliefs



Panel C. Certainty in beliefs

Figure 10: Treatment videos partially correct misperceptions.

Notes: This figure shows Wave 2 beliefs by treatment arm. Bars are arm means estimated with augmented inverse-probability weighting (AIPW), with 95% confidence intervals. Diamonds mark the true values from Central Bank data, averaged across respondents' chosen asset categories. Appendix Table A.13 reports the treatment effects for panels A and C.

Agreement with the statement that consórcios are "only a good deal for those who can bid" increases by 10.6 pp under D ($p = 0.01$) and 8.1 pp under DG ($p < 0.05$), against a control mean of 48%. Both arms also increased agreement with a statement that "firms favor bidders" by 9.3 pp under D ($p = 0.02$), with a similar but smaller and noisier effect under DG (5.0 pp, $p = 0.20$), against a control mean of 65%.

On the recommendation margin, the share who would *not* recommend consórcios rose by 6.5 pp under DG and 6.2 pp under D (both $p < 0.05$, on a control mean of 15%), while the "do not know enough to recommend" share fell by 5.4 pp under DG ($p < 0.1$, from 23%); we cannot reject equality of the two arms on any of the five outcomes.

Information spillovers to beliefs about bids. Respondents also updated their beliefs about winning bids. Relative to the control group, whose beliefs did not change between waves, D and DG raise the expected average winning bid by 5.5 and 6.3 pp (Appendix Table A.18, column 1).

This suggests some sophistication: holding everything else fixed, learning that more awards go to bids should, if anything, lower the bid needed to win. The increase closes the gap between consumers with and without liquidity to bid: treated consumers without liquidity expect winning bids of 50 to 52% of the credit, against 43% in the control group and 48% among liquid consumers in the control group. Effects on bidding intentions are positive but small and imprecise, with the stated likelihood of bidding rising by 0.16 and 0.12 points on a five-point scale and the reported own bid by 1.7 and 2.2 pp, none statistically significant.

5.4.3 Information and demand for alternative contract designs with more lotteries

Information increases valuation of lottery-intensive contracts. In our setting, disclosure can inform all consumers, but more fundamentally, it can change how heterogeneous types of consumers sort across contracts. A key prediction is that correcting misperceptions should increase valuation of the lottery-intensive group offer among consumers who are liquidity-constrained. We test this prediction using our pre-registered liquidity variable and other variables capturing intention to bid. Because of bunching at maximum and minimum values, we define the dependent variable as choosing the prospective offer over the maximum price at the multiple price list (MPL) elicitation.

Consistent with our prediction, the Debiasing (D) treatment increases choices of the offer at the maximum valuation by 8.3 pp ($p = 0.07$), a 25% increase relative to the control average (Table 3, column 2). Results are similar and stronger when using other direct measures of intention to bid (column 3). In column 4, we restrict to participants with monotonic answers in the multiple price list procedure, and the effect is slightly larger (0.104, $p = 0.03$). The main results are similar when the multiple price list choices are converted to BRL (Appendix Table A.17).

The treatment effects of D are concentrated among individuals who are initially optimistic about early award chances. With the caveat that cells are relatively small, we define optimism as reporting a belief about the chance of an early award above the median of 30%. The treatment effect among optimists is 16.2 pp ($p = 0.04$, column 5). When interacting optimism and liquidity or intention to bid, we see effects are again concentrated in the optimistic, while non-optimistic consumers have zero or negative estimated coefficients (columns 6 and 7).

In contrast to the effects of arm D among constrained respondents, the results for arm DG are essentially zero among both constrained and unconstrained respondents. We generally reject the equality of the D and DG estimates in the main treatment effects at the 10% level, suggesting results are not due to noise. This difference is puzzling because both treatment arms led to similar corrections in beliefs. One hypothesis is that adding extra scenes to the video may have led to a change in salience or attention, whereas in D the sole focus is lottery probabilities. We see that the DG

Table 3: Treatment effects on valuation of lottery-intensive groups.

	WTP = maximum (chose offer at the top price)						
		Heterogeneity by liquidity			Heterogeneity by beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Debiasing (D)	0.059	0.083*	0.095**	0.104**	-0.013	0.025	0.024
	(0.039)	(0.046)	(0.045)	(0.046)	(0.053)	(0.059)	(0.061)
× liquidity to bid 50%		-0.095		-0.100		-0.190	
		(0.087)		(0.088)		(0.133)	
× strategy = bid early			-0.155*				-0.156
			(0.091)				(0.120)
× optimist about early award					0.162**	0.140	0.158*
					(0.078)	(0.093)	(0.089)
× optimist × liquidity						0.138	
						(0.177)	
× optimist × bids early							0.010
							(0.185)
Debiasing + Group (DG)	-0.010	-0.024	-0.010	-0.032	-0.049	-0.027	-0.041
	(0.038)	(0.044)	(0.043)	(0.045)	(0.051)	(0.057)	(0.059)
× liquidity to bid 50%		0.054		0.086		-0.111	
		(0.085)		(0.086)		(0.132)	
× strategy = bid early			-0.001				-0.037
			(0.090)				(0.121)
× optimist about early award					0.087	0.008	0.070
					(0.076)	(0.090)	(0.086)
× optimist × liquidity						0.289*	
						(0.174)	
× optimist × bids early							0.078
							(0.182)
$D = DG$ (p-value)	0.06	0.01	0.01	0.00	0.38	0.25	0.19
Sample	Full	Full	Full	WTP-consistent	Full	Full	Full
Control mean	0.33	0.33	0.33	0.31	0.33	0.33	0.33
Observations	845	845	845	810	845	845	845

Notes. The outcome is an indicator that the respondent chose the consórcio over the gift card at the top of the price list (Section 5.1.3). Estimates are augmented inverse-probability-weighted (AIPW) effects from causal forests that compare each arm with the control group. In columns 2–7, we present effects when the moderator equals zero and the difference when the moderator equals one. Optimists are respondents whose Wave 1 belief about the chance of a lottery award in the first half of the group is above the sample median of 30%. Column 4 restricts to respondents with a consistent (monotone) price-list choice. The $D = DG$ row tests equality of the two arms' effects, in the moderator-zero group. Standard errors in parentheses. Control means are control-group averages of the dependent variable. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Is the valuation effect of D consistent with mediation by beliefs? We find evidence that the treatment effects are larger among respondents who were initially optimistic about the chance of an early award (Table 3, column 5). Using the moderator test proposed by Kwon and Roth (2026), we cannot reject full mediation through beliefs ($p = 0.4$).

5.4.4 Information does not change stated choices in conjoint analysis

In choices using side-by-side comparisons of groups with randomized attributes (fees, group size, minimum number of lotteries, and average winning bids), consumers consistently prefer smaller groups and more lotteries. In the control arm, consumers are 15 percentage points more likely to choose a group offering at least two lotteries per month rather than one ($p < 0.01$), and 15 and 23 percentage points less likely to choose medium and large groups over small groups, respectively ($p < 0.01$). These preferences contrast with observed market allocations, where groups are typically large and offer few lotteries. This discrepancy suggests that decision frictions, including how products may be presented and compared, may drive a wedge between consumers' stated preferences and market choices.

Information provision does not significantly change how consumers choose between attributes. None of the treatment effects on preferences for the four attributes is statistically distinguishable from zero at the 5% level (Appendix Figure A.17). This suggests that the treatment operates through changes in information rather than underlying preferences (Handel and Kolstad, 2015), consistent with its focus on allocation mechanisms, bidding shares, and lottery wait times rather than value judgments about attributes.

The null effects also provide evidence against increased attention to attributes as the primary mechanism of treatment effects (Conlon, 2026). Both treatments emphasized lotteries, while the DG treatment additionally highlighted group size. Yet neither treatment changed average choices of these attributes.

5.4.5 Information changes stated choices between financial products

In side-by-side comparisons of a consórcio with a loan or with a saving strategy to purchase the asset, the effects of information depend strongly on prior beliefs about the chance of an early lottery award. Beliefs about early lottery awards are a strong predictor of choices in the control group: optimists are 15 pp less likely to choose the loan and 24 pp less likely to choose saving over the consórcio ($p < 0.01$, Appendix Table A.16, columns 2 and 5).

Debiasing (D) raises the choice of the loan among optimists. On average, D raises the share choosing the loan by 6.2 pp ($p = 0.06$) from a control share of 18% (column 1), and the effect is 16.5 pp larger among optimists ($p = 0.01$; column 2). The effect is stronger for consumers with liquidity to bid, contrary to our expectations. But it is in line with the fact that liquid consumers

can pay a downpayment, and that the information treatment never stated that, depending on the bid and fees, consórcios can be a subsidized loan. On the saving margin, the same pattern is weaker (the optimist difference is 14.1 pp, $p = 0.09$; column 5).

Unlike D, the DG arm reinforces the choice of consórcios among non-optimists. Among respondents who are not optimists, DG increases the choice of consórcios over saving by 16.3 pp ($p < 0.01$; column 5). This effect is concentrated among those with liquidity to bid, whose share choosing saving falls by 33.6 pp ($p < 0.01$; column 6), from 74%, compared with no effect under D (difference: $p < 0.01$). Among optimists, DG has no significant effect on either margin. The additional group-size information appears to reinforce non-optimists' preference for consórcios. Thus, despite similar average changes in beliefs, the two treatments shift choices among different consumers and different margins (the treatment arms are statistically different, with $p = 0.01$ and $p = 0.02$; columns 1 and 4).

5.5 Discussion and implications for disclosure policy

Taken together, the experimental results suggest that misperceptions about lottery chances influence both participation in consórcios and the types of contracts consumers value. Correcting these beliefs increased the perceived advantage of bidders and shifted some initially optimistic consumers toward conventional loans. At the same time, liquidity-constrained consumers increased their valuation of a lottery-intensive alternative. And in side-by-side comparisons, consumers already prefer smaller groups and more lotteries, and information does not significantly change these preferences. Thus, information appears to affect consumers' evaluations of the options available in the market, rather than changing their stated preferences over attributes. These findings suggest that imperfect information helps sustain demand for consórcios, and in particular for groups with few lotteries, even when consumers appear to prefer alternative designs when attributes are transparent.

Our results have two implications for disclosure policies. First, as in other settings, disclosure may improve consumers' decision-making. More importantly, disclosure of relevant product attributes can have different effects depending on consumers' characteristics, potentially changing the composition of consumers in the market. In our setting, the relevant characteristic is liquidity. Disclosure may be especially effective because the redistribution generated by bidding depends on the participation of both bidders and non-bidders. When non-bidders leave the market or move to lottery-intensive groups, they reduce the scope for bidding and the associated redistribution. Thus, disclosure may induce consumer sorting that, in equilibrium, reduces both bidding and redistribution.

Policymakers may also consider directly regulating the extent of bid awards. Currently, regulations require at least one lottery award. But this leaves room for firms to adjust their contract

designs and reduce the effectiveness of this requirement (Campbell, 2016). The first consórcio regulation, introduced in 1972, required at least half of the funds disbursed by a group to be allocated through lotteries. In 1991, this requirement was replaced by the current minimum of one lottery award per month. Because group sizes are unrestricted, firms can increase the number of participants while maintaining the required minimum number of lotteries. We have evidence that the share of awards allocated through bidding rose from 54–66% in 2008 to approximately 80% in 2026 (Appendix Figure A.18). Therefore, policymakers should consider regulations that mandate lottery awards as a share of total awards or total funds disbursed.

6 Conclusion

We study Brazilian consórcios, the world's largest formal market for rotating finance. We quantify the financial gains and losses from participation and examine how allocation mechanisms chosen by firms distribute value across consumers. Using consumer surveys and an information experiment, we investigate whether misperceptions about award probabilities weaken demand for greater lottery access.

Market-wide administrative data shows most contracts have negative returns relative to a market replication and losses can be sizable. We then show how bidding can improve expected returns by advancing credit receipt, while relying exclusively on lotteries leads to long waits and negative returns. Late recipients effectively finance early recipients without receiving interest, generating redistribution across participants.

Survey evidence shows that consumers' misperceptions are large and systematic. An information disclosure experiment, providing accurate information about award allocations and lottery wait times, increases liquidity-constrained respondents' valuation of access to the offer of a lottery-intensive group.

Studying consórcios brings firms' contract design choices into the analysis of rotating finance. The findings also contribute to the literature on redistribution through household financial products. While differences in financial sophistication can generate cross-subsidies across consumers, access to higher expected returns in our setting does not require better information about lottery probabilities. Consumers with liquidity to bid can obtain better returns even when they hold substantial misperceptions. Widespread information frictions can therefore have unequal consequences for consumers with different access to liquidity. This finding can inform other financial products where access to liquidity can shape contract values.

Our findings also raise questions about the role of firms and brokers in shaping consumer information. We document limited disclosure of award allocations by firms. Brokers, meanwhile, hold relatively accurate beliefs about winning bids and the share of bid awards, but remain poorly

informed about lottery probabilities. Information about bidding is more readily observable and useful in advising consumers who can bid, while information about lottery probabilities is harder to verify. Brokers may therefore transmit information selectively, even without deliberately misleading consumers. Moreover, the difficulty of verifying lottery probabilities may weaken incentives to communicate them accurately. Understanding whether supply-side communication sustains differences in consumer information, and thereby reinforces unequal financial outcomes and potentially sustain regressive contracts, is an important direction for future research.

References

Adams, P., Hunt, S., Palmer, C., and Zaliauskas, R. (2021). Testing the effectiveness of consumer financial disclosure: Experimental evidence from savings accounts. *Journal of Financial Economics*, 141(1):122–147.

Afzal, U., d'Adda, G., Fafchamps, M., and Said, F. (2017). Two Sides of the Same Rupee? Comparing Demand for Microcredit and Microsaving in a Framed Field Experiment in Rural Pakistan. *The Economic Journal*.

Agarwal, S., Presbitero, A., Silva, A. F., and Wix, C. (2026). Who Pays For Your Rewards? Redistribution in the Credit Card Market. *Finance and Economics Discussion Series*, (2023-007):1–70.

Alan, S., Cemalcilar, M., Karlan, D., and Zinman, J. (2018). Unshrouding: Evidence from Bank Overdrafts in Turkey. *The Journal of Finance*, 73(2):481–522.

Ambec, S. and Treich, N. (2007). Roscas as financial agreements to cope with self-control problems. *Journal of Development Economics*, 82(1):120–137.

Anagol, S., Cole, S., and Sarkar, S. (2017). Understanding the Advice of Commissions-Motivated Agents: Evidence from the Indian Life Insurance Market. *Review of Economics and Statistics*, 99(1):1–15.

Anderson, S. and Baland, J.-M. (2002). The Economics of Roscas and Intrahousehold Resource Allocation. *Quarterly Journal of Economics*, 117(3).

Anderson, S., Baland, J.-M., and Moene, K. O. (2009). Enforcement in informal saving groups. *Journal of Development Economics*, 90(1):14–23.

Athey, S., Tibshirani, J., and Wager, S. (2019). Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178.

Augenblick, N. and Rabin, M. (2021). Belief Movement, Uncertainty Reduction, and Rational Updating. *The Quarterly Journal of Economics*, 136(2):933–985.

Balakina, O., Balasubramaniam, V., Dimri, A., and Sane, R. (2026). Unshrouding product-specific attributes through financial education. *Journal of Development Economics*, 178:103604.

Basu, K. (2011). Hyperbolic Discounting and the Sustainability of Rotational Savings Arrangements. *American Economic Journal: Microeconomics*, 3(4):143–171.

Berger, D., Milbradt, K., Vavra, J., and Tourre, F. (2024). Refinancing Frictions, Mortgage Pricing and Redistribution. *NBER Working Paper*.

Bertrand, M. and Morse, A. (2011). Information Disclosure, Cognitive Biases, and Payday Borrowing. *The Journal of Finance*, 66(6):1865–1893.

Beshears, J., Choi, J. J., Laibson, D., and Madrian, B. C. (2018). Behavioral Household Finance. In *Handbook of Behavioral Economics: Applications and Foundations 1*, volume 1, pages 177–276. Elsevier.

Besley, T., Coate, S., and Loury, G. (1993). The Economics of Rotating Savings and Credit Associations. *American Economic Review*, 83(4):792–810.

Besley, T., Coate, S., and Loury, G. (1994). Rotating Savings and Credit Associations, Credit Markets and Efficiency. *The Review of Economic Studies*, 61(4):701–719.

Braga, O., Teixeira, L. I., and Pugliese, J. F. (2026). Inflation Beliefs and the Indexation of Household Debt.

Breza, E. and Kaur, S. (2026). Psychology and Development: Applications from Cognitive and Social Psychology. *NBER Working Paper*.

Bryan, G., Karlan, D., and Nelson, S. (2010). Commitment Devices. *Annual Review of Economics*, 2(1):671–698.

Byrne, S., Devine, K., King, M., McCarthy, Y., and Palmer, C. (2023). The Last Mile of Monetary Policy: Inattention, Reminders, and the Refinancing Channel. *NBER Working Paper*.

Calomiris, C. W. and Rajaraman, I. (1998). The role of ROSCAs: lumpy durables or event insurance? *Journal of Development Economics*, 56(1):207–216.

Campbell, J. Y. (2016). Restoring Rational Choice: The Challenge of Consumer Financial Regulation. *American Economic Review*, 106(5):1–30.

Campbell, J. Y. and Ramadorai, T. (2026). Household Finance in Retrospect and Prospect. *NBER Working Paper*.

Conlon, J. J. (2026). Inattention, Information Provision, and (Failures of) Revealed Preference. *Journal of the European Economic Association*, page jvag059.

Cullen, Z. and Perez-Truglia, R. (2022). How Much Does Your Boss Make? The Effects of Salary Comparisons. *Journal of Political Economy*, 130(3):766–822.

Doornik, B. F. N. V., Gomes, A. R., Skrastins, J., and Schoenherr, D. (2024a). Savings-and-Credit Contracts.

Doornik, B. V., Fazio, D., Ramadorai, T., and Skrastins, J. (2025). Housing and Fertility.

Doornik, B. V., Gomes, A., Schoenherr, D., and Skrastins, J. (2024b). Financial Access and Labor Market Outcomes: Evidence from Credit Lotteries. *American Economic Review*, 114(6).

Duflo, E., Kremer, M., and Robinson, J. (2011). Nudging Farmers to Use Fertilizer: Theory and Experimental Evidence from Kenya. *American Economic Review*, 101(6):2350–2390.

Egan, M. (2019). Brokers versus Retail Investors: Conflicting Interests and Dominated Products. *The Journal of Finance*, 74(3):1217–1260.

Egan, M., Matvos, G., Seru, A., Wang, L., and Yao, V. (2026). Who Pays for Payments? *NBER Working Paper Series*.

Ferman, B. (2016). Reading the Fine Print: Information Disclosure in the Brazilian Credit Card Market. *Management Science*, 62(12):3534–3548.

Filiz-Ozbay, E., Guryan, J., Hyndman, K., Kearney, M., and Ozbay, E. Y. (2015). Do lottery payments induce savings behavior? Evidence from the lab. *Journal of Public Economics*, 126:1–24.

Fisher, J., Gavazza, A., Liu, L., Ramadorai, T., and Tripathy, J. (2024). Refinancing cross-subsidies in the mortgage market. *Journal of Financial Economics*, 158:103876.

Gabaix, X. and Laibson, D. (2006). Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets. *Quarterly Journal of Economics*.

Garz, S., Giné, X., Karlan, D., Mazer, R., Sanford, C., and Zinman, J. (2021). Consumer Protection for Financial Inclusion in Low- and Middle-Income Countries: Bridging Regulator and Academic Perspectives. *Annual Review of Financial Economics*, 13(1):219–246.

Gillen, B., Snowberg, E., and Yariv, L. (2019). Experimenting with Measurement Error: Techniques with Applications to the Caltech Cohort Study. *Journal of Political Economy*, 127(4):1826–1863.

Gottlieb, D. and Smetters, K. (2021). Lapse-Based Insurance. *American Economic Review*, 111(8):2377–2416.

Gugerty, M. (2007). You Cant Save Alone: Commitment in Rotating Savings and Credit Associations in Kenya. *Economic Development and Cultural Change*, 55(2):251–282.

Gurun, U. G., Matvos, G., and Seru, A. (2016). Advertising Expensive Mortgages. *The Journal of Finance*, 71(5):2371–2416.

Haaland, I., Roth, C., and Wohlfart, J. (2023). Designing Information Provision Experiments. *Journal of Economic Literature*, 61(1):3–40.

Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2014). Causal Inference in Conjoint Analysis: Understanding Multidimensional Choices via Stated Preference Experiments. *Political Analysis*, 22(1):1–30.

Handel, B. R. and Kolstad, J. T. (2015). Health Insurance for Humans: Information Frictions, Plan Choice, and Consumer Welfare. *American Economic Review*, 105(8):2449–2500.

Heidhues, P. and Kszegi, B. (2017). Naïveté-Based Discrimination*. *The Quarterly Journal of Economics*, 132(2):1019–1054.

Hirshleifer, J. (1958). On the Theory of Optimal Investment Decision. *Journal of Political Economy*, 66(4):329–352.

John, A. (2020). When Commitment Fails: Evidence from a Field Experiment. *Management Science*, 66(2):503–529.

Karlan, D. and Morduch, J. (2010). Access to Finance. In *Handbooks in Economics*, volume 5 of *Handbook of Development Economics*, pages 4703–4784. Elsevier.

Kearney, M. S., Tufano, P., Guryan, J., and Hurst, E. (2010). Making Savers Winners: An Overview of Prize-Linked Savings Products. Technical Report w16433, National Bureau of Economic Research, Cambridge, MA.

Kwon, S. and Roth, J. (2026). Testing Mechanisms. *Review of Economic Studies*, page rdag028.

Linnainmaa, J. T., Melzer, B. T., and Previtero, A. (2021). The Misguided Beliefs of Financial Advisors. *The Journal of Finance*, 76(2):587–621.

Lockwood, B. B., Allcott, H., Taubinsky, D., and Sial, A. (2025). What Drives Demand for State-Run Lotteries? Evidence and Welfare Implications. *Review of Economic Studies*, 92(4):2578–2623.

Seira, E., Elizondo, A., and Laguna-Müggenburg, E. (2017). Are Information Disclosures Effective? Evidence from the Credit Card Market. *American Economic Journal: Economic Policy*, 9(1):277–307.

Time Magazine (1967). Brazil: A Lot of Car Buying by Lot. *Time Magazine*.

World Bank (2025). The Global Findex Database 2025: Connectivity and Financial Inclusion in the Digital Economy.

A Additional figures and tables

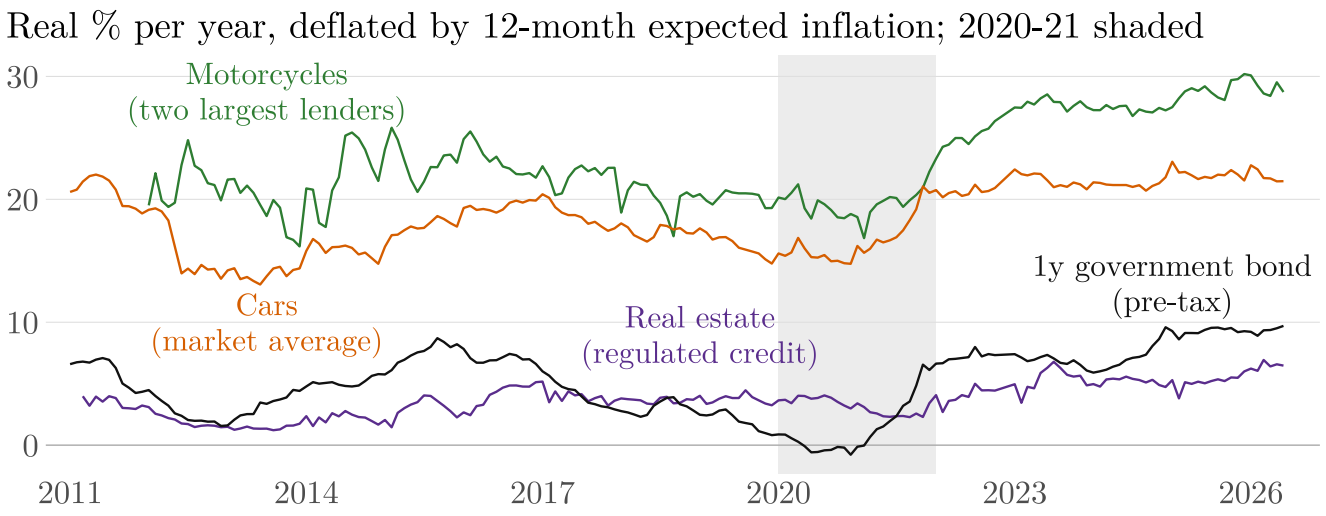
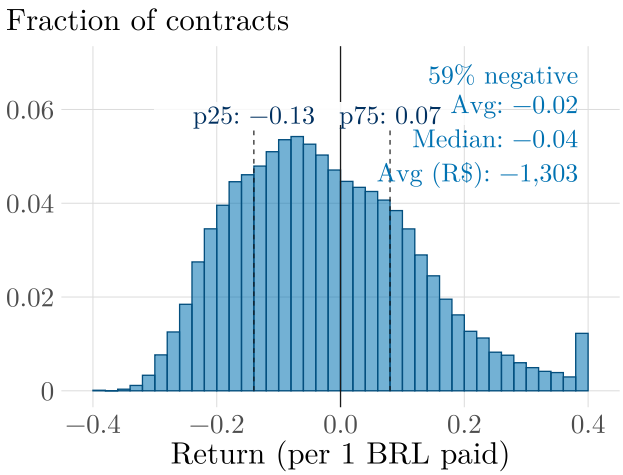
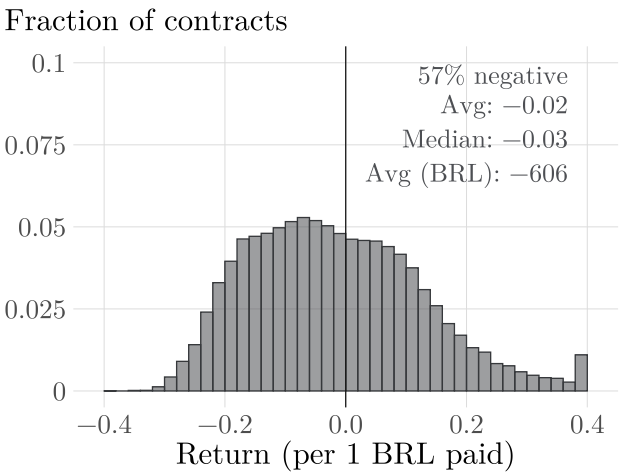


Figure A.1: Market interest rates.

Notes: This figure shows monthly real interest rates on one-year government bonds and on loans for cars, motorcycles, and real estate, from 2011 to June 2026. The government bond rate is the yield on fixed-rate Treasury bonds (*Tesouro Prefixado*), interpolated to one year to maturity and before income tax, from the Treasury's *Tesouro Direto* price history. The car rate is the Central Bank's average rate on new vehicle loans to individuals (SGS series 20749), which car loans dominate. The real estate rate is the Central Bank's average rate on housing loans to individuals at regulated rates (SGS series 20773). No motorcycle series exists, so the motorcycle rate averages the fixed-rate vehicle-loan rates of Banco Honda and Banco Yamaha Motor, the two largest motorcycle lenders, from the Central Bank's interest rates by financial institution. These two lenders also represent 80% of the consórcio motorcycle market. All rates are deflated by the median twelve-month-ahead inflation expectation in the Central Bank's Focus survey. The shaded band marks 2020–2021, when pandemic-era policy pushed short-term real rates below zero; we exclude these years when averaging rates for the benchmarks in Section 2.3.



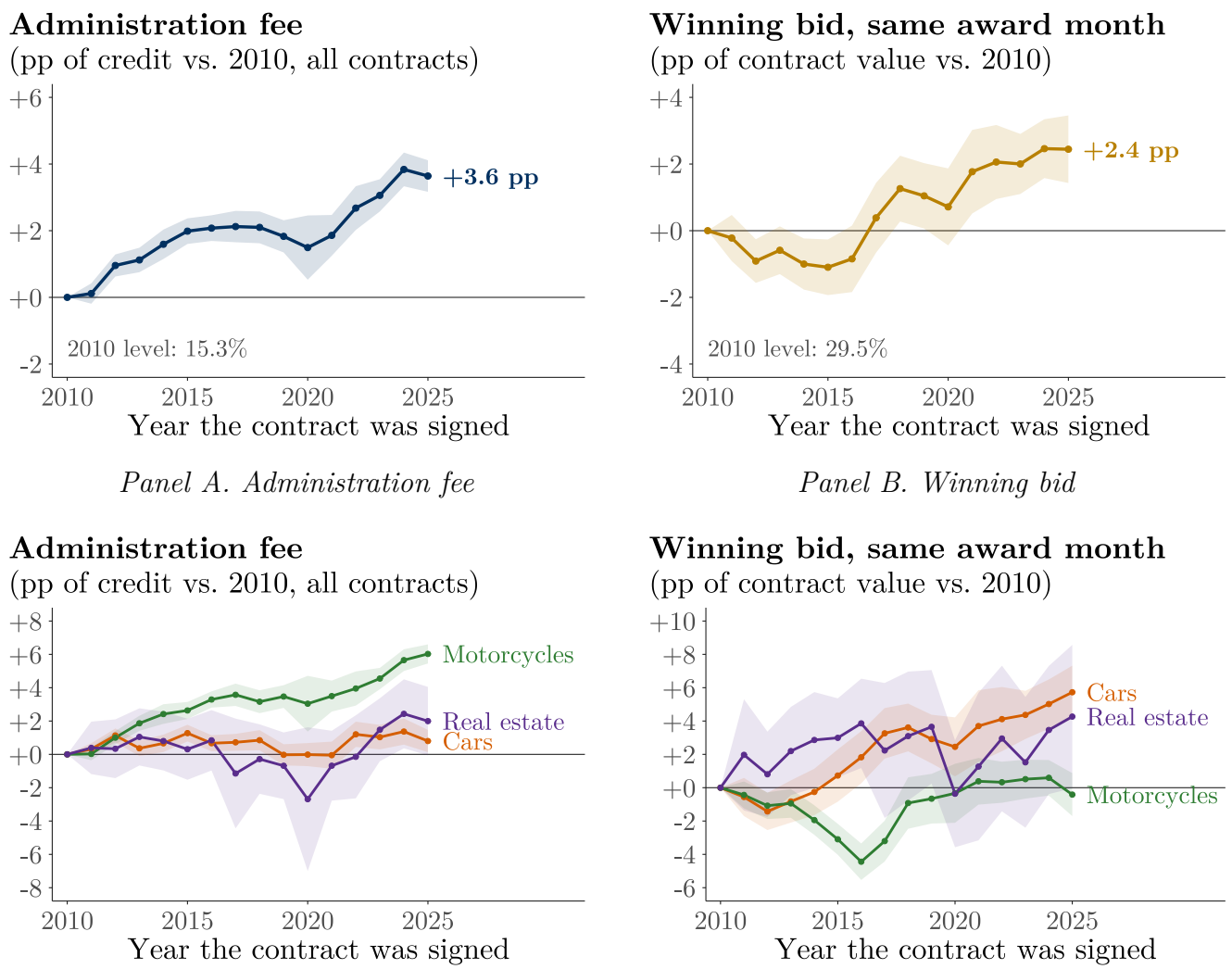
Panel A. Ongoing groups with 50% observed duration



Panel B. Completed groups

Figure A.2: Returns under a stricter observation gate and in completed groups.

Notes: This figure shows net value calculated with Equation (1) across awarded contracts, as in Figure 2, panel A, in two other samples. In panel A, the sample includes awarded contracts in groups started between 2010 and 2025 that are completed or observed for at least 50% of their planned duration. Outcomes for contracts yet to be awarded in ongoing groups are predicted. In panel B, the sample includes only completed groups, which require no prediction. Returns beyond the axis range fall in the end bins. Dashed lines in panel A mark the 25th and 75th percentiles.



Panel A. Administration fee

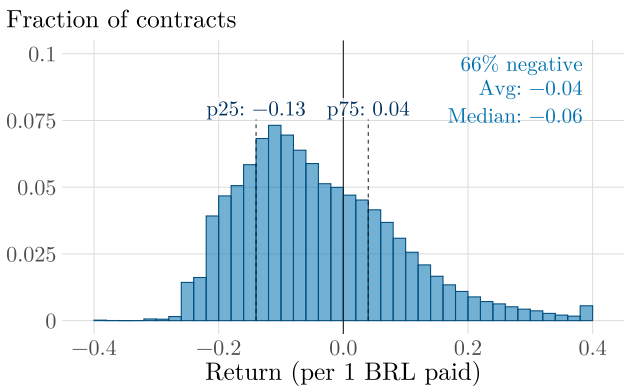
Panel B. Winning bid

Panel C. Administration fee, by asset category

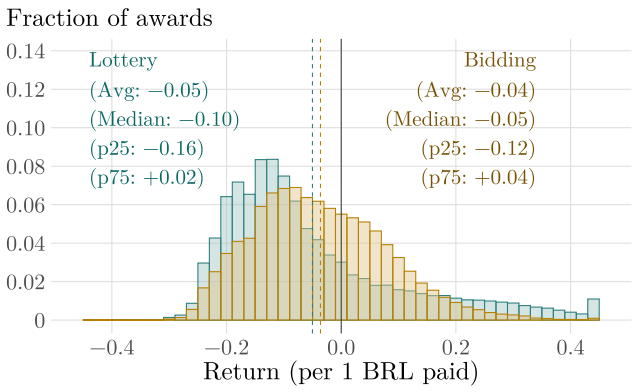
Panel D. Winning bid, by asset category

Figure A.3: Fees and bids by year the contract was signed.

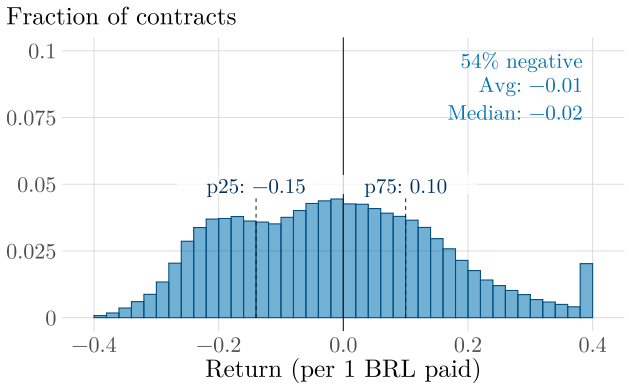
Notes: This figure shows the average change in administration fee and winning bids from 2010 until 2025. Each point is the coefficient on a year-of-signature indicator, in percentage points relative to contracts signed in 2010, from a regression with asset category dummies. Bands are 95% intervals with standard errors clustered by group. Panel A is the administration fee on all contracts. Panel B is the winning bid among contracts awarded by bidding, including fixed-menu bids, holding fixed the month of the contract at which the award happened, so that later cohorts are not compared with earlier awards. Panels C and D run the same regressions separately for each asset category.



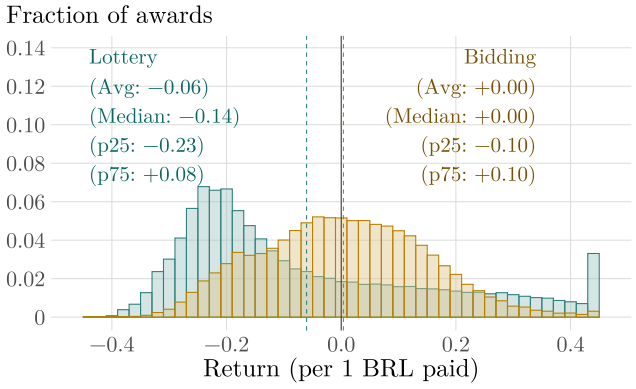
Panel A. Savings 0%, awarded contracts



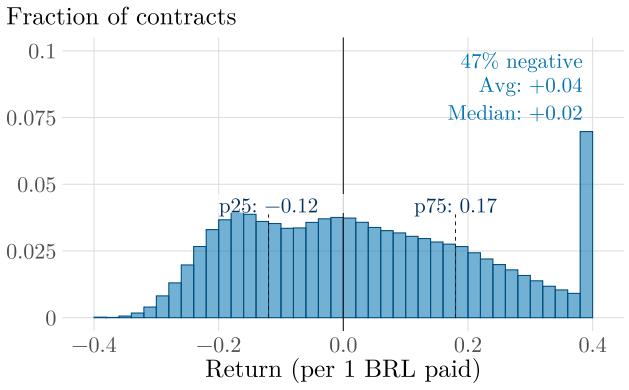
Panel B. Savings 0%, bid vs. lottery



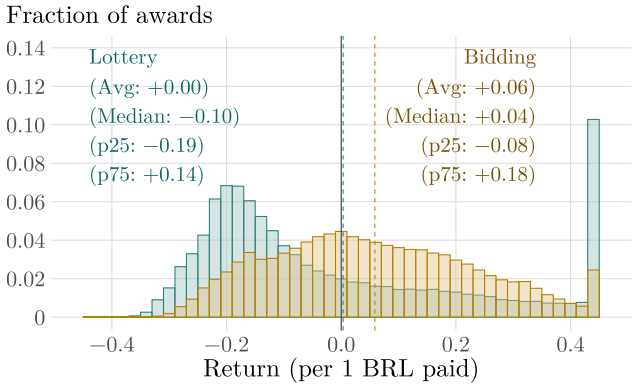
Panel C. Savings 8%, awarded contracts



Panel D. Savings 8%, bid vs. lottery



Panel E. Borrowing ×1.5, awarded contracts



Panel F. Borrowing ×1.5, bid vs. lottery

Figure A.4: Returns under alternative benchmark rates.

Notes: This figure shows the returns of awarded contracts in Figure 2, panel A (left), and of bid and lottery awards in Figure 5 (right), under alternative benchmark rates. In panels A to D, the savings rate is 0% or 8%, and borrowing rates move with it, keeping their spreads over the savings rate. In panels E and F, the savings rate stays at 4% and borrowing rates are 50% higher: 6%, 28.5%, and 36% for real estate, cars, and motorcycles. The sample is the same as in Figure 2. In the right column, the distribution within each award mode in completed groups is approximated from its quantiles under each scenario; the averages are exact. Returns beyond the axis range fall in the end bins.

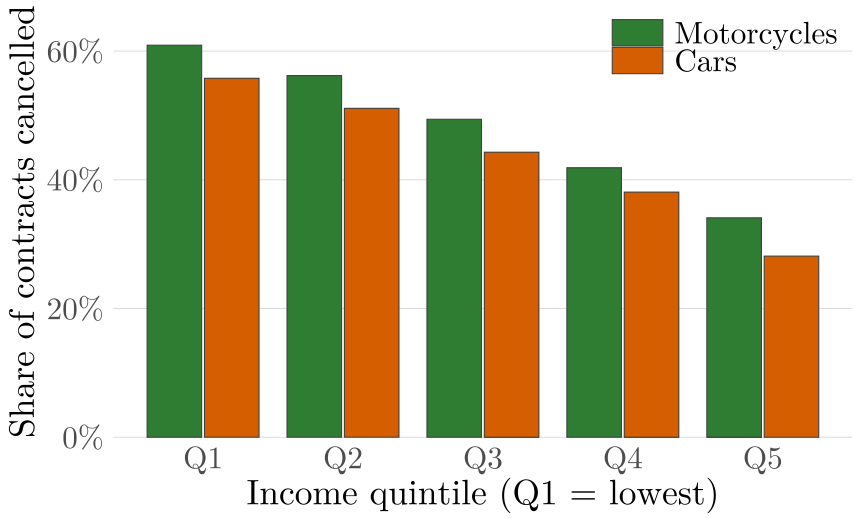
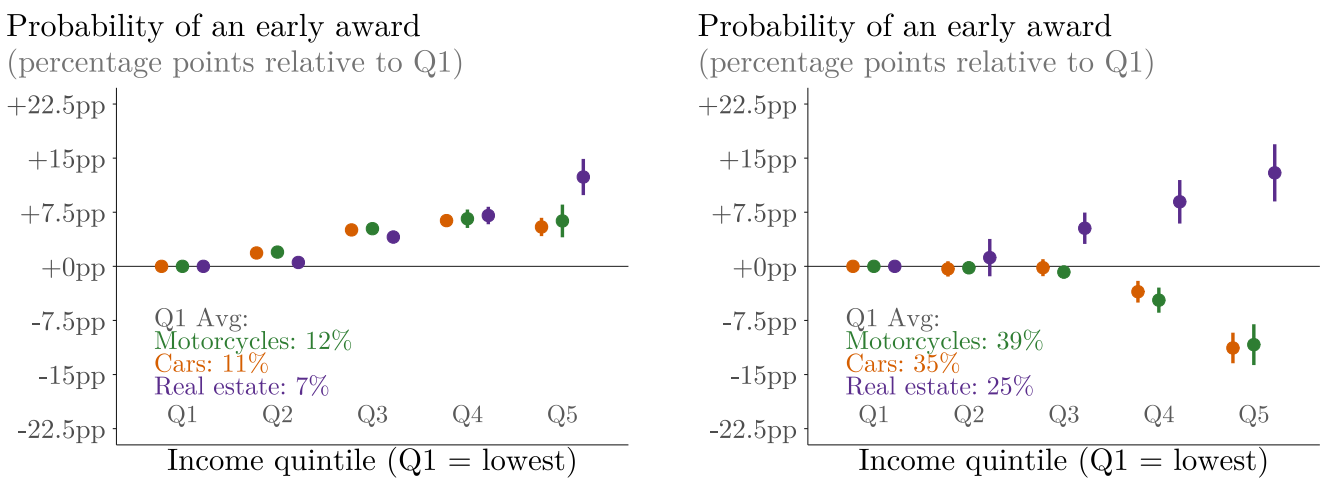


Figure A.5: Share of contracts canceled by income quintile, completed groups.

Notes: This figure shows the share of contracts canceled in each income quintile, in completed groups. Income quintiles are taken over self-reported incomes from the national credit registry. Real estate is omitted because few members of its completed groups have an income record.

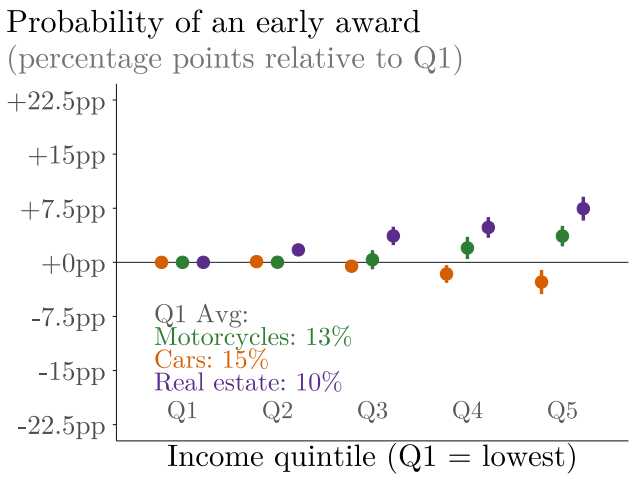


Panel A. All eligible contracts

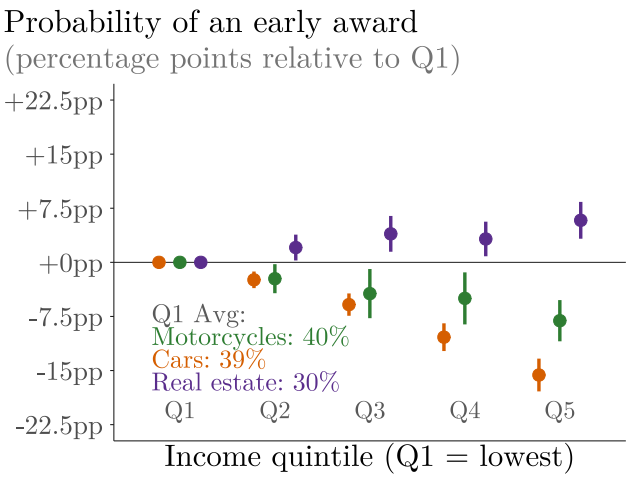
Panel B. Excluding canceled contracts

Figure A.6: Higher-income participants receive more early awards in the aggregate, but not uniformly among contracts that do not cancel.

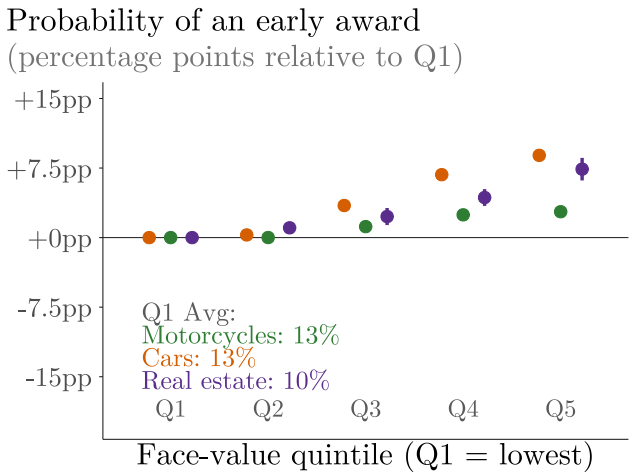
Notes: This figure shows the difference in the probability of an early award between each income quintile and the lowest, by asset category. An early award is an award within the first 25% of the group's planned duration. Panel A includes all contracts, and panel B excludes canceled contracts. Quintiles rank contracts by self-reported income at joining, from the national credit registry. The sample includes groups started between 2010 and 2025 that are completed or observed for at least 25% of their planned duration. The key reports the probability of an early award in the lowest quintile. Vertical lines are 95% confidence intervals.



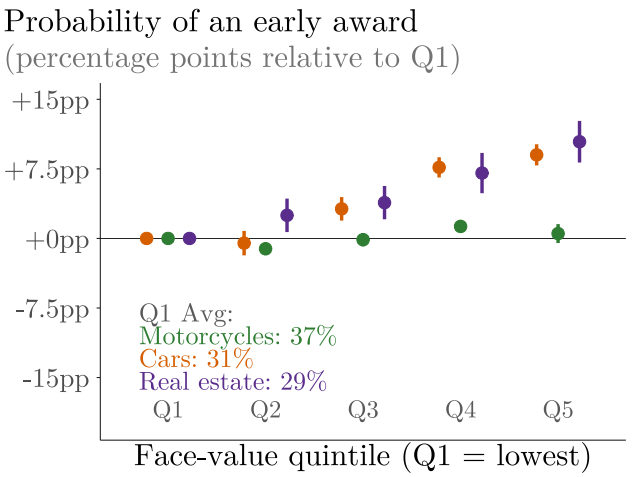
Panel A. Income, all eligible contracts



Panel B. Income, excluding canceled contracts



Panel C. Face value, all eligible contracts



Panel D. Face value, excluding canceled contracts

Figure A.7: Access to early awards by within-group quintiles of income and face value.

Notes: This figure shows the difference in the probability of an early award between each within-group quintile and the lowest, by asset category. An early award is an award within the first 25% of the group's planned duration. Panels A and C include all contracts, and panels B and D exclude canceled contracts. Quintiles rank contracts within their group, by income in panels A and B and by face value in panels C and D. The sample includes groups started between 2010 and 2025 that are completed or observed for at least 25% of their planned duration. The key reports the probability of an early award in the lowest quintile. Vertical lines are 95% confidence intervals.

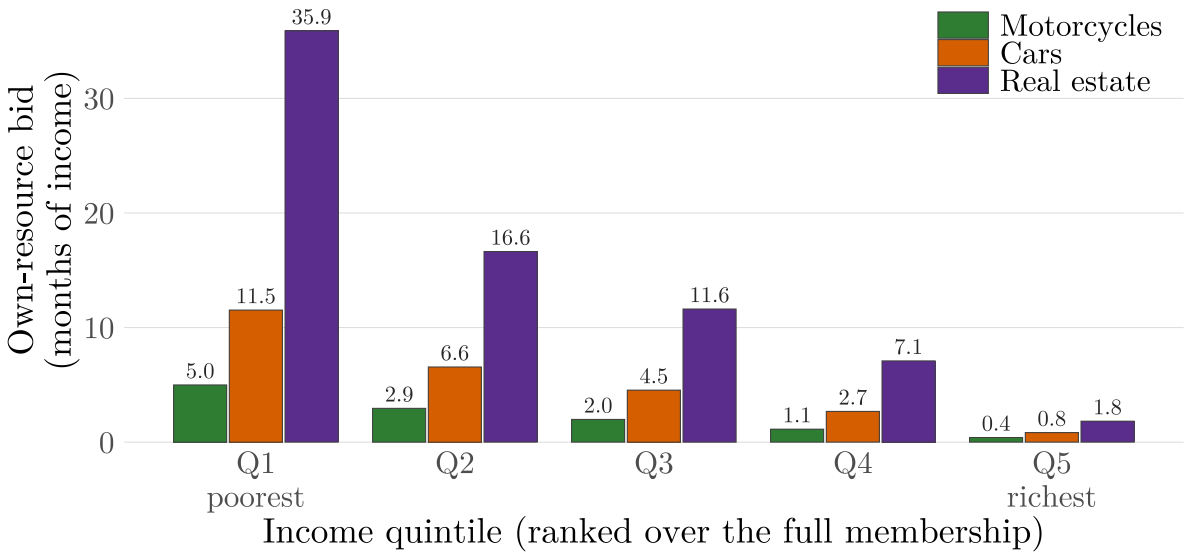
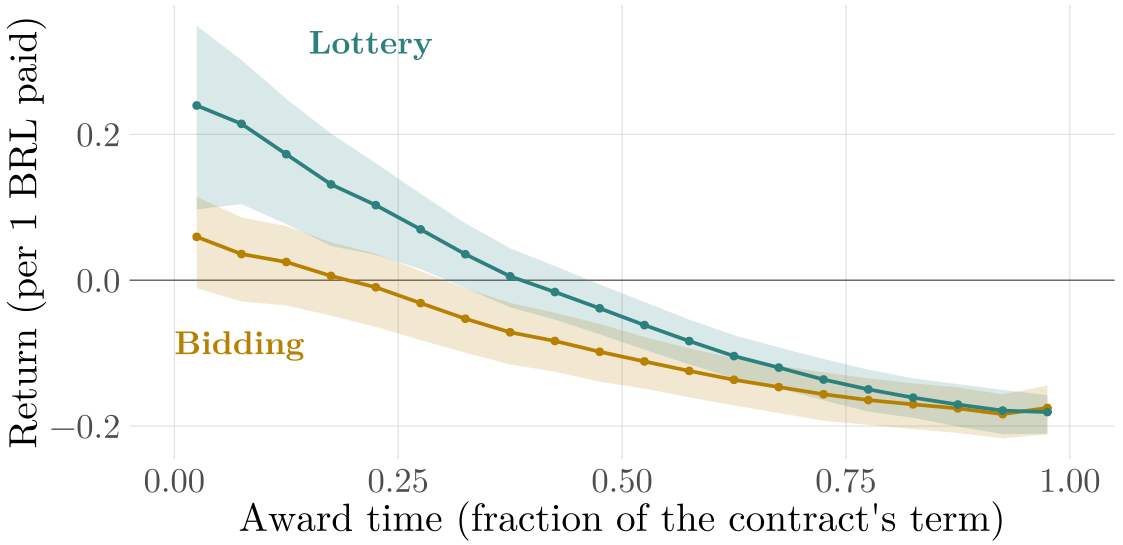
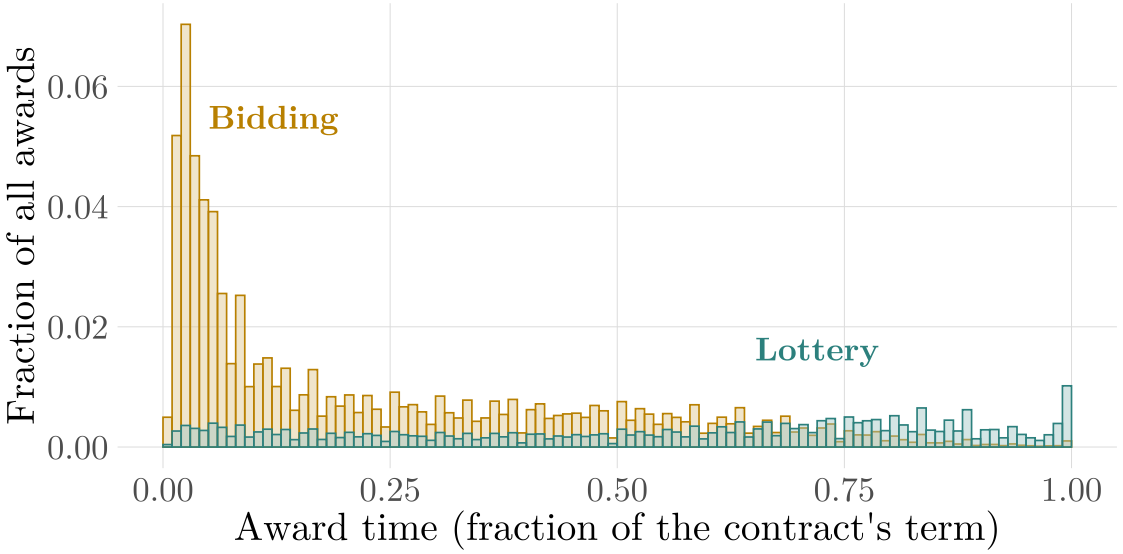


Figure A.8: Median own-resource cost of winning cash bids, in months of income, by income quintile and asset category.

Notes: This figure shows the median cost of a winning bid paid from the member's own resources, in months of income, among contracts awarded by bidding, by income quintile and asset category. The own-resource bid excludes the part of the bid paid out of the credit itself (embedded bid). Income is the real monthly income recorded in the national credit registry at joining. Quintiles are cut over all members of the three categories pooled, so the first quintile holds the same incomes in every category. The sample includes groups started between 2010 and 2025. Medians are weighted by group sampling weights.



Panel A. Returns fall steeply with award timing, and are lower for bid awards than for lottery awards at any given timing



Panel B. Bid awards are concentrated early in the contract term

Figure A.9: Returns and award timing by award mode.

Notes: This figure shows the returns and the timing of awards by bidding and by lottery. Award time is the month of the award as a fraction of the contract's own term. In panel A, lines are the average return in each 5% interval of the term, and bands span the 25th to 75th percentiles. Panel B shows the share of all awards falling in each 1% interval of the term. The sample includes awarded contracts in groups started between 2010 and 2025 that are completed or observed for at least 25% of their planned duration. Outcomes for contracts yet to be awarded in ongoing groups are predicted.

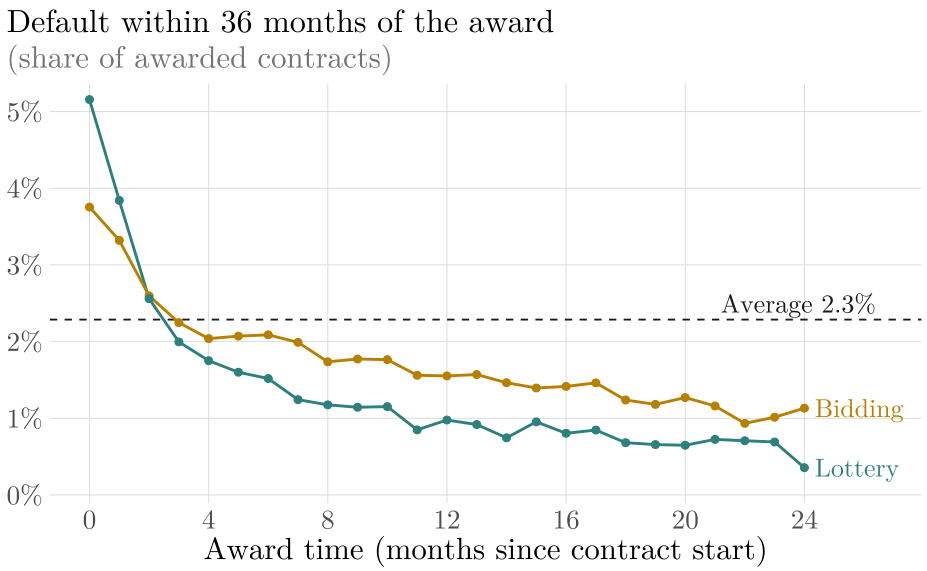


Figure A.10: Default within 36 months of the award, by month of award and allocation mechanism.

Notes: This figure shows the share of awarded contracts that default within 36 months of the award, by the month of the contract in which the award happened (counted from signing, up to 24) and by allocation mechanism. A default is a legal collection action filed against the contract. Only contracts observed for at least 36 months after the award are included, so every contract has the same time at risk. The dashed line is the average over both mechanisms. In a linear regression of default on a bid-award indicator, bid awards are 1.16 percentage points more likely to default (s.e. 0.03), and 0.38 points (s.e. 0.03) with award-month fixed effects; standard errors are heteroskedasticity-robust.

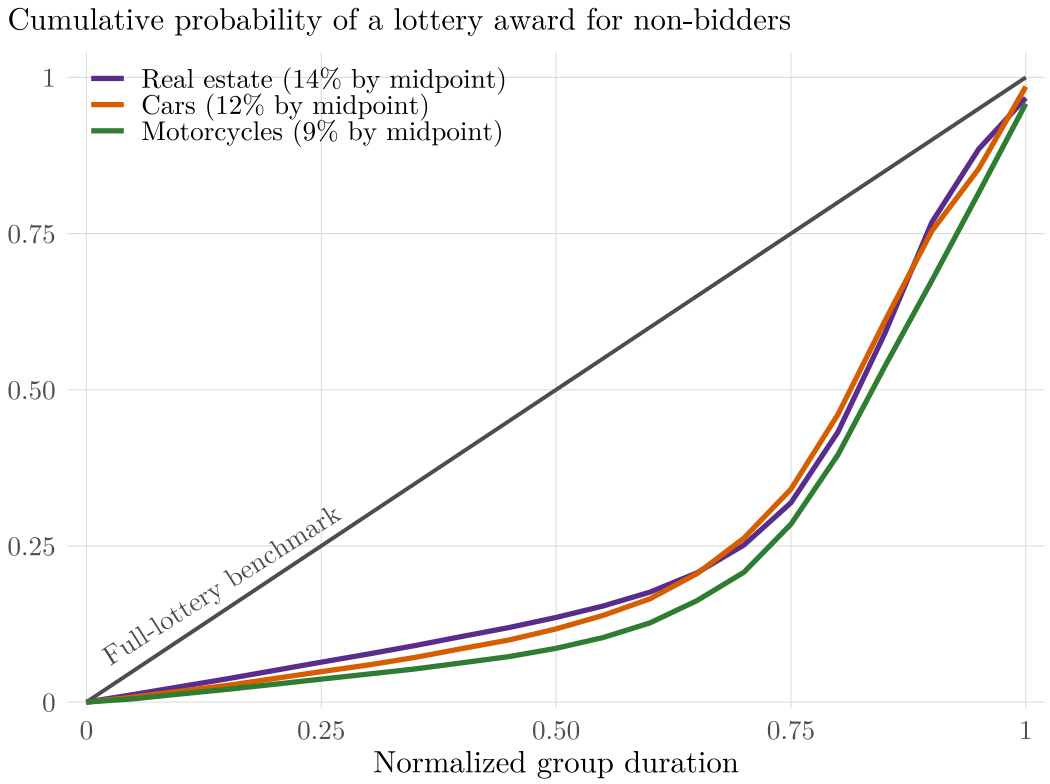


Figure A.11: Cumulative probability of a lottery award for non-bidders, by asset category.

Notes: This figure shows the cumulative probability that a contract that never bids receives a lottery award, by the share of the group's duration elapsed, for each asset category, as in Figure 6, panel B. Curves average the groups' estimated lottery-award probabilities, weighted by group size. The legend reports each category's probability of a lottery award by the group's midpoint. The 45-degree line is the benchmark in which every award is made by lottery.

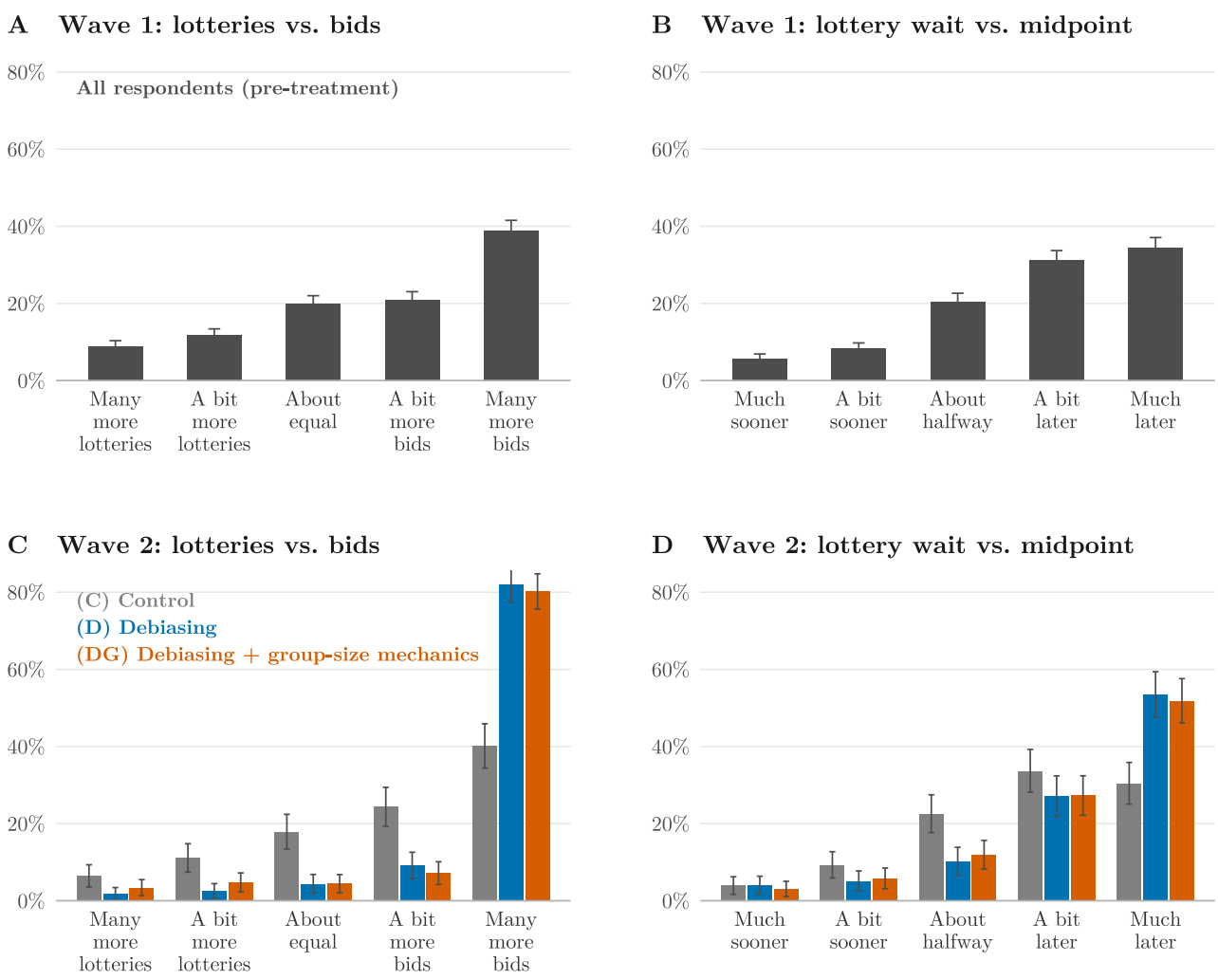


Figure A.12: Qualitative beliefs about award shares and lottery wait times.

Notes: This figure shows answers to the qualitative questions on whether lotteries or bids make up more of the awards in the first half of the group, and on when a lottery award arrives relative to the group's midpoint. Panels A and B show Wave 1, before treatment, for all respondents. Panels C and D show Wave 2, by treatment arm. Bars are shares of respondents, and vertical lines are 95% confidence intervals.

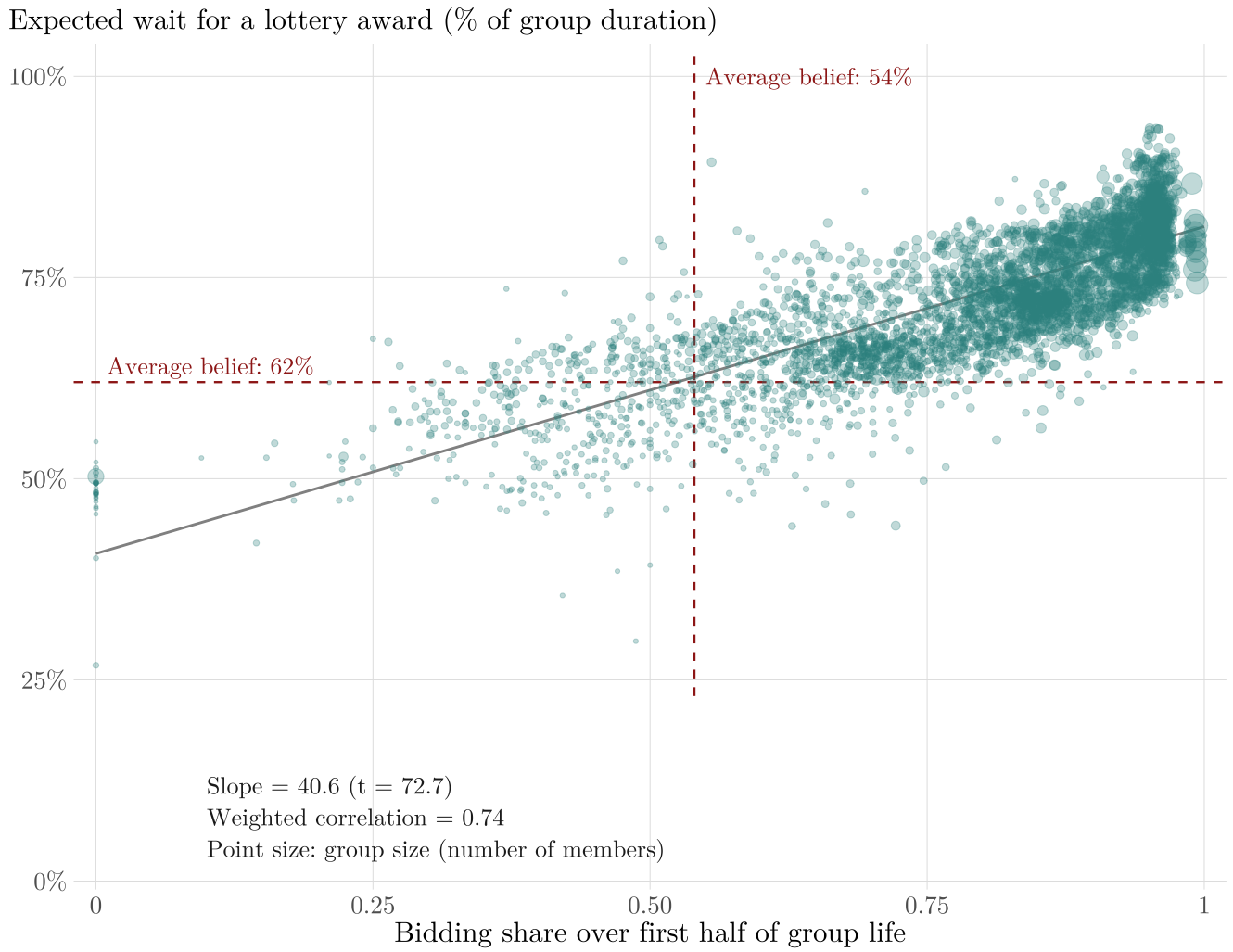


Figure A.13: Bidding intensity and lottery waits across completed groups.

Notes: This figure shows, for each completed group, the share of awards made by bid in the first half of the group and the expected wait for a lottery award, conditional on an award, as a share of the group's duration. Point size is group size, and the solid line is the group-size-weighted least-squares fit. Dashed lines mark respondents' average beliefs about the same two quantities (Section 3.6). Groups whose cumulative probability of a lottery award by the end of the term is below 0.8 (3.7% of groups) are excluded.

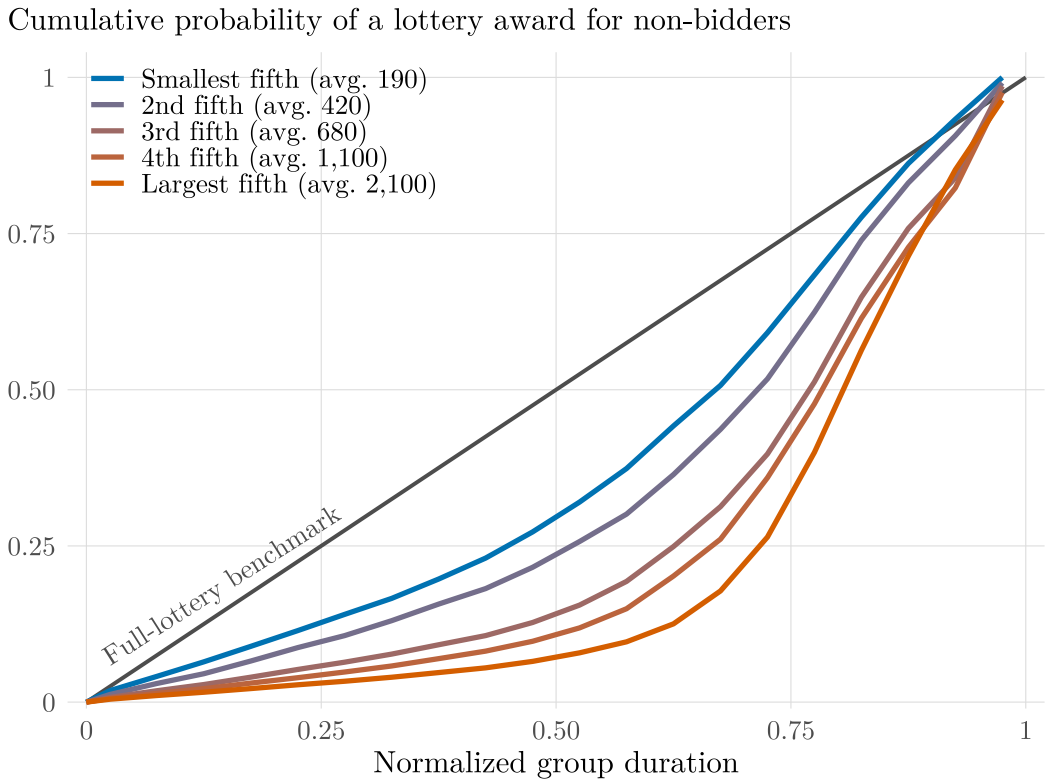


Figure A.14: Cumulative probability of a lottery award for non-bidders, by group size.

Notes: This figure shows the cumulative probability that a contract that never bids receives a lottery award, by the share of the group's duration elapsed, for each fifth of completed groups ranked by size. Curves average the groups' lottery-award probabilities, weighted by group size. The legend reports each fifth's average number of members. The 45-degree line is the benchmark in which every award is made by lottery. The sample includes 4,537 completed groups.

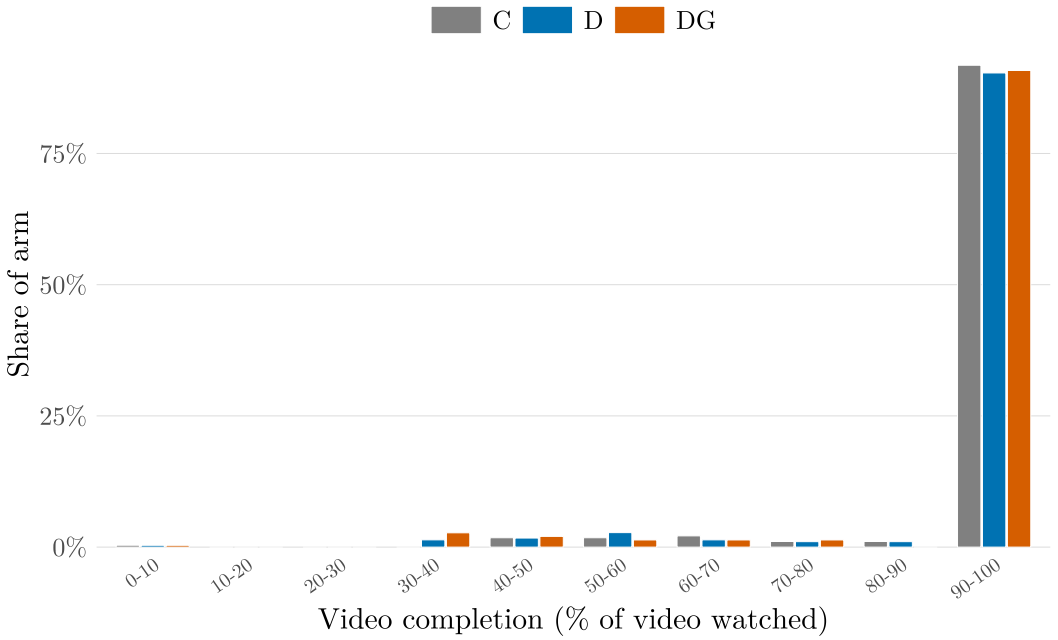


Figure A.15: Distribution of information-treatment video completion, by arm.

Notes: This figure shows how much of their video respondents watched in Wave 2, by treatment arm. Completion is the share of the video watched, so arms whose videos differ in length are comparable. Bars are the shares of each arm's respondents in each 10-point completion bin. The sample includes Wave 2 respondents who passed the quality filters.

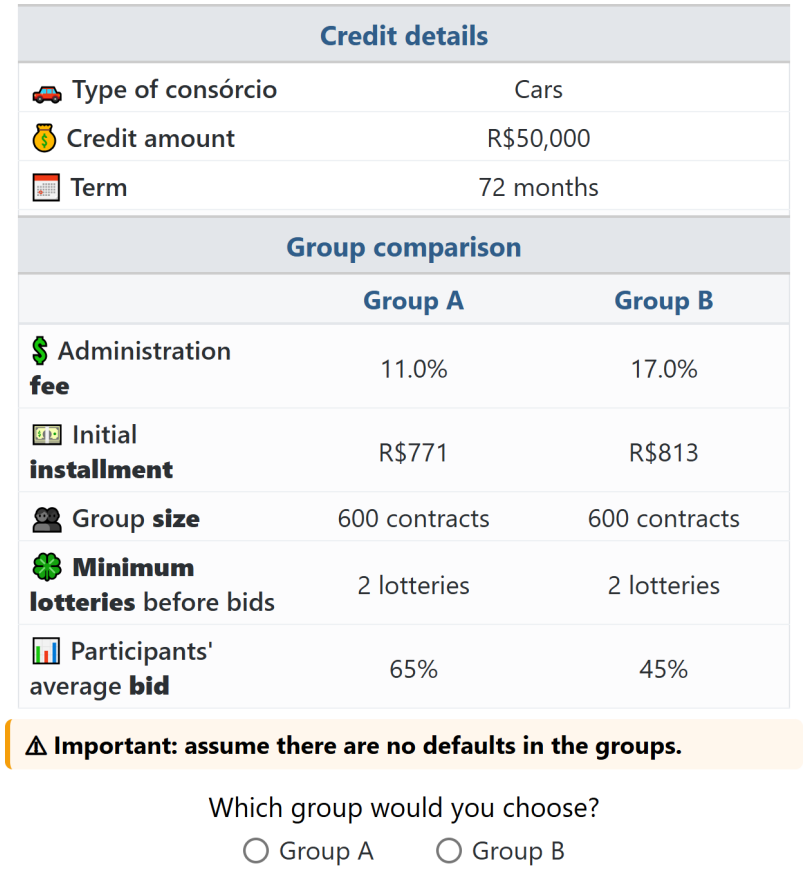


Figure A.16: Hypothetical group-choice task.

Notes: English translation of one of the two group choices each respondent made in Portuguese. The asset, credit amount, and term are those of the contract the respondent simulated earlier; the administration fee, group size, minimum lotteries before bids, and participants' average bid are randomized independently, and the initial installment follows from the fee. In this example, the two groups differ only in the fee and the average bid.

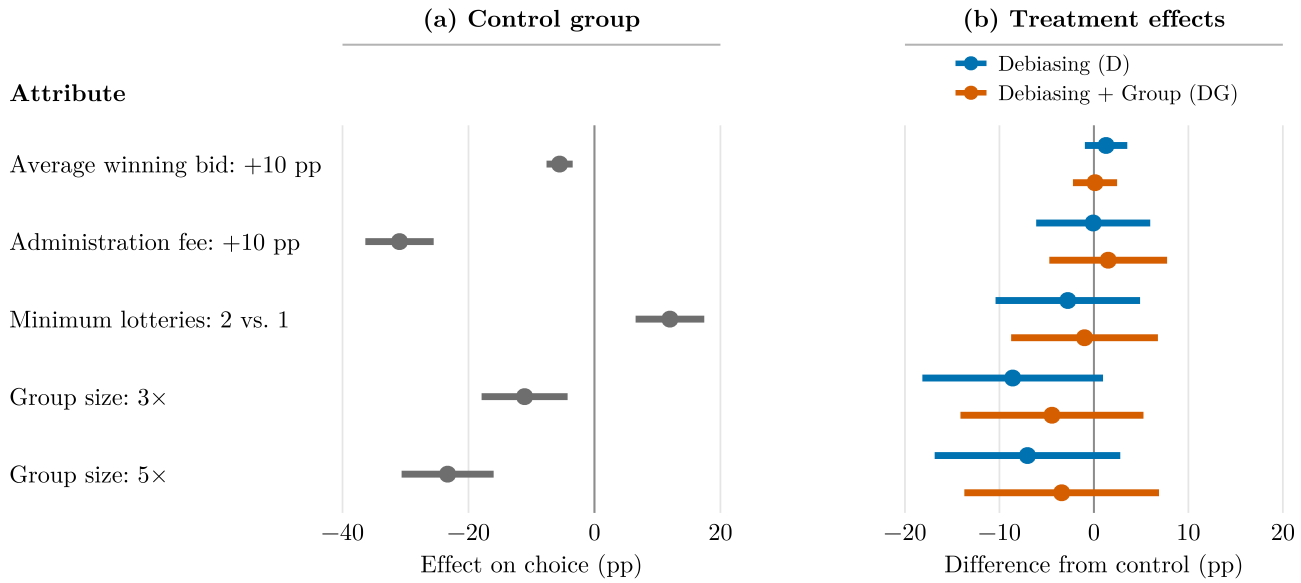


Figure A.17: Attribute valuation in hypothetical group choices.

Notes: Panel A shows the effect of each attribute on the probability that the control group chooses a group; panel B shows how much each treatment arm changes that effect. The average winning bid and the administration fee were randomized as discrete levels; we enter them linearly and report the effect of a 10 pp increase. Minimum lotteries and group size enter as indicators relative to the baseline levels, and group sizes are shown as the ratio to the smallest group, rounded. Linear probability model with PDS Lasso controls; bars are 95% confidence intervals clustered by respondent.

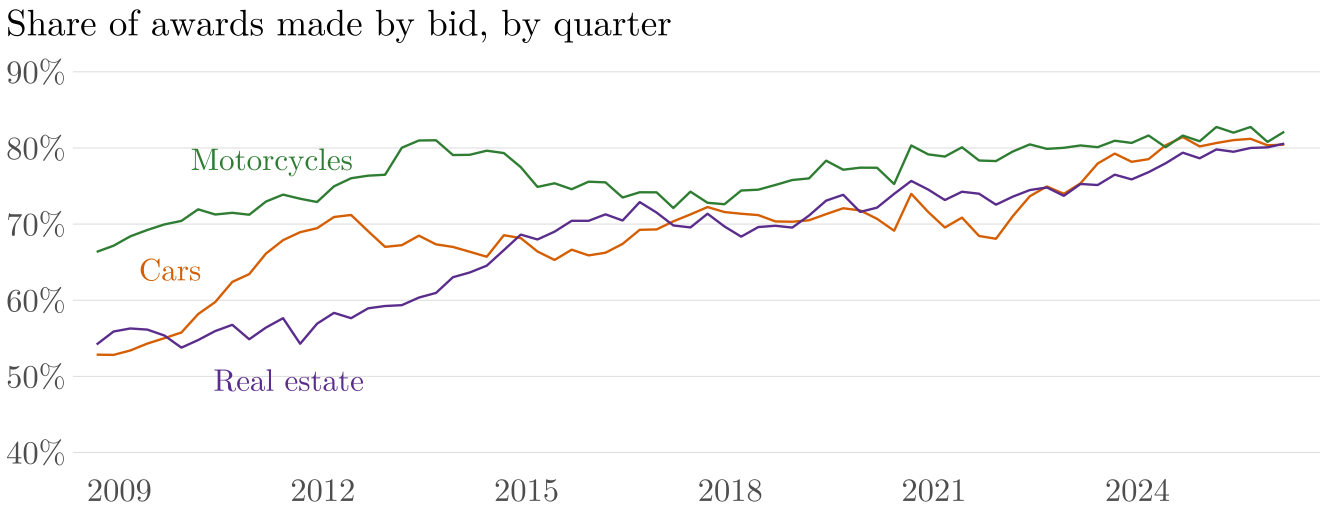


Figure A.18: Share of awards made by bid, 2008–2026.

Notes: This figure shows the share of each quarter's awards made by bid, by asset category, from the Central Bank's public consórcio data. Each quarter pools the awards of all active groups, regardless of how far each group is into its duration. Earlier data do not separate bid and lottery awards.

Table A.1: Financial products held by consórcio consumers and non-consumers.

			Difference	
	Consórcio consumers	**Non-consórcio consumers**	**No controls**	**Income+Educ controls**
	(1)	(2)	(3)	(4)
Panel A: Accounts and cards (currently holds)				
Bank account	91.2	68.2	22.9***	14.7***
			(2.6)	(2.6)
Savings account	56.5	33.6	22.9***	16.7***
			(4.2)	(4.1)
Credit card	77.6	49.5	28.0***	18.7***
			(3.6)	(3.7)
Panel B: Savings (previous 12 months)				
At least one savings product	77.6	52.5	25.1***	15.6***
			(3.6)	(3.5)
At least one, excluding bank accounts	55.2	28.6	26.6***	20.6***
			(4.3)	(4.1)
Savings or checking account	63.9	41.4	22.5***	12.9***
			(4.1)	(4.0)
Treasury bonds or CDs	17.1	6.2	10.9***	7.9**
			(3.2)	(3.1)
Stocks	18.5	6.0	12.5***	10.7***
			(3.3)	(3.3)
Private pension	10.2	2.8	7.4***	5.8**
			(2.5)	(2.5)
Panel C: Credit products (currently holds)				
At least one of the products below	53.7	16.7	37.1***	34.0***
			(4.2)	(4.2)
Vehicle financing	31.3	3.9	27.4***	26.0***
			(3.9)	(3.8)
Home-equity loan	6.8	0.7	6.1***	5.9***
			(2.1)	(2.1)
Mortgage	16.3	3.2	13.2***	11.2***
			(3.1)	(2.9)
Unsecured bank loan	15.0	4.5	10.4***	9.4***
			(3.0)	(3.0)
Payroll loan	23.8	8.1	15.7***	15.3***
			(3.6)	(3.6)
Respondents (N)	147	1,809	1,956	1,956

Notes: This table shows the financial products held by consórcio consumers and non-consumers in the Central Bank of Brazil Financial Literacy Survey (*Pesquisa de Letramento Financeiro*, https://dadosabertos.bcb.gov.br/dataset/pesquisa-de-letramento-financeiro), among adults aged 18 and over. Consórcio consumers are respondents who hold a consórcio or acquired one in the previous two years. Columns 1–2 report the share (%) holding each product. Columns 3–4 report OLS coefficients on a consórcio-consumer indicator, in percentage points; column 4 adds income-bracket and schooling indicators. Robust standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.2: Contract and group characteristics by asset category.

	Cars	**Motorcycles**	**Real Estate**
	Mean (SD)	Mean (SD)	Mean (SD)
Panel A: Contract level			
Credit value at joining (USD thousand)	8.2 (5.9)	2.3 (1.4)	34.4 (30.2)
Admin fee (%)	13.5 (4.7)	20.4 (4.5)	18.3 (5.4)
Term (months)	72 (18)	56 (17)	184 (37)
Installment incl. fee (% of income), median	17.9	17.2	17.3
Individual holder (vs. firm) (%)	71.2	97.3	77.4
Year of joining	2018.2 (4.3)	2017.7 (4.7)	2020.3 (4.3)
Awarded, not canceled (%)	39.3	54.6	15.9
Canceled (%)	43.3	32.2	50.4
Active (no award, no cancellation) (%)	17.4	13.2	33.7
Contracts (N)	1,039,222	748,342	333,650
Contracts (share, %)	49.0	35.3	15.7
Panel B: Group level			
Group size (quotas)	999 (1,280)	1,080 (1,026)	2,214 (1,746)
Planned duration (months)	72 (13)	57 (15)	186 (30)
Formation year	2016.3 (4.4)	2016.1 (4.9)	2018.6 (4.9)
Quotas awarded (%)	64.1 (30.0)	72.0 (32.1)	25.0 (24.9)
Quotas canceled (%)	40.0 (18.7)	46.6 (16.2)	46.3 (22.5)
Groups (N)	26,750	17,809	3,972
Groups (share, %)	55.1	36.7	8.2

Notes: This table shows means of contract and group characteristics by asset category, with standard deviations in parentheses. Panel A uses a random sample of about 4% of contracts in groups started between 2010 and 2025. Credit value is the face value at joining, in thousands of USD at 5.5 BRL per USD. The installment row reports the median monthly installment, including the administration fee, as a share of the holder's income in the national credit registry, among contracts with reported income. Panel B uses all groups started between 2010 and 2025. Group size counts every quota reported over the group's life, including canceled quotas that are resold, so it exceeds the number of members at any one time. Shares of quotas awarded and canceled are measured at the group's last filing.

Table A.3: Average returns by asset category

		Decomposition of fee-free return		
	Return	Transfer	Borrowing advantage	Fee
	(1)	(2)	(3)	(4)
Panel A. Awarded contracts				
Cars	0.008	0.011	0.123	0.111
Motorcycles	−0.043	0.016	0.151	0.185
Real estate	−0.095	0.031	−0.000	0.126
All segments	−0.025	0.015	0.124	0.146
Panel B. Canceled contracts				
Cars	−0.045	−0.029	–	0.017
Motorcycles	−0.060	−0.034	–	0.024
Real estate	−0.034	−0.027	–	0.019
All segments	−0.051	−0.032	–	0.021

Notes: This table shows average returns per unit of face value by asset category, for awarded and canceled contracts. Canceled contracts are also normalized by their face value, not by the amount paid in as in Figure 2, panel B, so their returns and transfers are small relative to that figure. The sample includes contracts in groups that are completed or observed for at least 25% of their duration; outcomes of contracts not yet resolved in ongoing groups are predicted (Appendix D). Column (1) is the return from Equation (1). Columns (2)–(4) decompose the fee-free return, which equals (2) + (3) − (4). Transfer is the present value of the installment plan at the savings rate. Borrowing advantage is the additional value of receiving the funds at the award date, at the asset category's market borrowing rate. Fee is the present value of the administration fee. Canceled contracts still awaiting their refund draw enter column (1) with a predicted refund and are excluded from columns (2)–(4).

Table A.4: Break-even outside rates across the return distribution

Return percentile	Return	Median award timing	Break-even borrowing		Break-even savings	
			Rate	Spread over market (pp)	Rate	Spread over market (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All segments						
10th	−0.21	0.73	–	–	−10.0%	−14.0
25th	−0.14	0.55	103.5%	+82.3	−7.9%	−11.9
50th	−0.04	0.17	29.7%	+7.0	−4.9%	−8.9
75th	+0.07	0.06	12.3%	−7.0	–	–
90th	+0.17	0.05	8.0%	−11.6	–	–
Panel B. Cars						
10th	−0.18	0.77	–	–	−5.9%	−9.9
25th	−0.12	0.60	109.8%	+90.8	−5.9%	−9.9
50th	+0.00	0.24	18.7%	−0.3	4.5%	+0.5
75th	+0.10	0.05	9.7%	−9.3	–	–
90th	+0.20	0.05	6.5%	−12.5	–	–
Panel C. Motorcycles						
10th	−0.23	0.72	–	–	−13.3%	−17.3
25th	−0.16	0.52	132.3%	+108.3	−18.1%	−22.1
50th	−0.06	0.10	34.1%	+10.1	–	–
75th	+0.04	0.07	19.3%	−4.7	–	–
90th	+0.16	0.04	12.8%	−11.2	–	–
Panel D. Real estate						
10th	−0.24	0.68	–	–	−5.2%	−9.2
25th	−0.17	0.48	–	–	−8.7%	−12.7
50th	−0.09	0.12	9.6%	+5.6	–	–
75th	−0.02	0.06	5.2%	+1.2	–	–
90th	+0.03	0.07	2.9%	−1.1	32.6%	+28.6

Notes: This table shows break-even interest rates at percentiles of the return per unit of face value, for awarded contracts. The sample includes contracts in groups that are completed or observed for at least 25% of their duration; outcomes of contracts not yet awarded in ongoing groups are predicted. The break-even borrowing rate is the borrowing rate at which the return is zero, holding the savings rate at 4%. The break-even savings rate is the savings rate at which the return is zero, holding the borrowing rate at its benchmark (19% for cars, 24% for motorcycles, and 4% for real estate). Spreads are the break-even rate minus the benchmark rate. A positive borrowing spread means the contract breaks even only for a participant who would borrow at that many points above the market rate. Award timing is the median award month as a share of the term. Each row takes the contracts with returns within 0.01 of the percentile and reports their median break-even rate; a dash marks a cell where the rate is undefined for more than half of these contracts. The same loss spread over more periods is offset by a smaller rate difference, so break-even rates can move closer to market rates as the term increases and need not be monotonic in the return.

Table A.5: Average returns under alternative outside rates

	Market borrowing rates			Borrowing rates 50% higher		
Savings rate	0%	4%	8%	0%	4%	8%
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Finished and ongoing groups (at least 25% of term observed)						
Cars	−0.015	−0.015	−0.013	+0.030	+0.024	+0.020
Motorcycles	−0.049	−0.048	−0.041	−0.010	−0.010	−0.010
Real estate	−0.089	−0.056	−0.030	−0.076	−0.043	−0.021
All segments	−0.041	−0.036	−0.029	−0.003	−0.001	+0.000
Panel B. Finished groups only						
Cars	−0.010	−0.010	−0.009	+0.039	+0.033	+0.028
Motorcycles	−0.044	−0.040	−0.036	−0.004	−0.004	−0.004
Real estate	−0.084	−0.066	−0.052	−0.069	−0.054	−0.042
All segments	−0.031	−0.028	−0.025	+0.012	+0.010	+0.009
Panel C. Awarded contracts, finished and ongoing groups, all segments						
Mean	−0.040	−0.025	−0.013	+0.039	+0.044	+0.047
Median	−0.061	−0.040	−0.020	−0.001	+0.016	+0.032
Share negative	66%	60%	54%	50%	47%	45%

Notes: This table shows returns per unit of face value when the same contracts are repriced under alternative benchmark rates. Panels A and B report the average over all contracts, awarded and canceled; panel C reports the mean, median, and share negative of awarded contracts. The baseline is column (2): a 4% savings rate and borrowing rates of 4%, 19%, and 24% for real estate, cars, and motorcycles. Columns (1) and (3) change the savings rate and move borrowing rates with it, keeping their spreads over the savings rate. Columns (4) to (6) raise borrowing rates by 50% at a 4% savings rate, to 6%, 28.5%, and 36%, and keep those spreads when the savings rate changes. Canceled contracts still awaiting their restitution draw are valued at a predicted restitution. Outcomes not yet observed in ongoing groups are predicted as in Appendix D.

Table A.6: Group size and early awards per month, by group start year.

		Early awards		
Start year	Size	Bids	Lotteries	Bid share
Panel A. Cars				
2010–2014	483	4.31	1.10	0.80
2015–2018	635	6.10	1.04	0.85
2019–2022	786	6.11	1.09	0.85
2023–2024	1,194	9.10	1.06	0.90
Panel B. Motorcycles				
2010–2014	529	7.28	0.91	0.89
2015–2018	772	7.64	1.03	0.88
2019–2022	955	9.89	1.27	0.89
2023–2024	978	10.26	1.28	0.89
Panel C. Real estate				
2010–2014	1,109	2.58	1.14	0.69
2015–2018	1,211	2.78	1.19	0.70
2019–2022	2,032	4.80	1.26	0.79
2023–2024	2,807	6.04	1.30	0.82

Notes: This table shows group size and early awards by asset category and the year the group started. Size is the average number of contracts sold by the time a group reaches 10% of its duration. Bids and Lotteries are awards per group per month in the first 25% of the group's duration, and Bid share is the share of these awards made by bid. The sample includes groups started between 2010 and 2024; groups not yet observed through 25% of their duration contribute the months observed.

Table A.7: Correlation between self-reported liquidity and income and savings variables.

	All respondents				Simulated contract ≥ BRL 1,000			
	Income (1)	Coping (2)	Savings (3)	Loan (4)	Income (5)	Coping (6)	Savings (7)	Loan (8)
Household income (reference: up to BRL 2,000)								
BRL 2,001–5,000	-0.036				-0.032			
	(0.054)				(0.053)			
BRL 5,001–10,000	0.033				0.028			
	(0.054)				(0.054)			
BRL 10,001–20,000	0.106*				0.116**			
	(0.058)				(0.057)			
BRL 20,001–100,000	0.159**				0.174***			
	(0.063)				(0.063)			
BRL 100,001 or more	0.285***				0.297***			
	(0.074)				(0.073)			
Emergency-coping capacity (1–5)		0.160***				0.161***		
		(0.010)				(0.010)		
Any savings in the last 12 months			0.213***				0.223***	
			(0.023)				(0.023)	
Access to a personal loan				0.218***				0.226***
				(0.025)				(0.025)
R^2	0.040	0.167	0.056	0.059	0.046	0.174	0.061	0.065
Observations	1296	1296	1296	1296	1262	1262	1262	1262

Notes: This table shows how self-reported liquidity to bid correlates with income and savings measures among Wave 1 consumers. The outcome is an indicator that the respondent could bid about 50% of the simulated contract's face value with their own resources in the first years of the contract. Each column is a separate linear probability model on one measure; household income enters as indicators for income brackets. Coping capacity is a 1–5 rating of the ability to handle a financial emergency. Columns (5)–(8) exclude the 34 respondents whose simulated contract is below BRL 1,000. Robust standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8: Broker sample description.

	Mean (SD)	Median
Panel A. Demographics		
Age	43.8 (11.7)	44.0
Male (%)	75.7	
College complete or higher (%)	64.4	
Panel B. Employer (%)		
Specialized consórcio sales firm	45.9	
Consórcio administrator	31.2	
Financial advisory firm	11.3	
Commercial bank	6.5	
Car dealership	5.1	
Panel C. Business		
Years of experience	6.8 (8.2)	3.0
Sales per month	8.3 (10.5)	5.0
Number of administrators	1.9 (1.3)	1.0
Sells for two or more administrators (%)	40.1	
Sells consórcios exclusively (%)	57.2	
Advertises on social media (%)	78.4	
Observations	292	

Notes: This table describes the brokers who completed the survey. Binary variables report the share in percent; continuous variables report the mean, standard deviation in parentheses, and median. Sales per month is trimmed to 0–100, and the number of administrators codes "5 or more" as 5.

Table A.9: Distributional overlap between the consumer and broker samples.

	Weighted Jaccard	Cosine	Overlap
Segment	0.91	1.00	0.95
Region	0.84	0.99	0.91
Administrator	0.49	0.91	0.66

Notes: This table compares the distributions of consumers and brokers across asset categories, regions, and administrators. For distributions p and q over categories k, the weighted Jaccard index is $\sum_k \min(p_k, q_k) / \sum_k \max(p_k, q_k)$, the cosine similarity is $\sum_k p_k q_k / (\|p\| \, \|q\|)$, and the overlap is $\sum_k \min(p_k, q_k)$; each equals one for identical distributions and zero for disjoint ones. Consumers are classified by their asset of interest and brokers by their sales mix. Regions are located from each response's IP address. Administrators are named in free text: the one a consumer is considering (45% of consumers name one) and up to three that each broker sells for. The sample includes 1,296 Wave 1 consumers and 292 brokers.

Table A.10: Wave 1 and Wave 2 consumer samples.

	Wave 1 Sample		Wave 2 Sample	
	Mean (SD)	Median	Mean (SD)	Median
Panel A. Demographics				
Age	47.2 (12.5)	47.0	47.1 (12.5)	47.0
Male (%)	59.1		57.5	
Employed (%)	87.2		86.2	
College complete or higher (%)	69.7		70.3	
Primary financial decision maker (%)	82.7		81.1	
Panel B. Financial profile				
Financial literacy score (0–4)	3.2 (1.0)	3.0	3.2 (1.0)	3.0
Savings buffer (months)	5.1 (4.0)	4.0	5.1 (4.0)	4.0
Implied monthly installment (BRL)	1,792 (2,176)	1,250	1,792 (2,376)	1,250
Installment as % of monthly income	27.6 (61.3)	14.8	27.4 (64.7)	14.3
Panel C. Consórcio profile				
Past consórcio participation (%)	56.9		56.4	
Currently paying a consórcio (%)	25.5		25.9	
Discussed consórcio with vendor (%)	33.9		33.4	
Likely to join ($\leq$3 mo.) (%)	56.0		55.9	
Would recommend consórcio (%)	69.4		69.2	
Already considering an administrator (%)	44.8		46.6	
Panel D. Asset of interest				
Real estate (%)	47.7		46.0	
Cars / vehicles (%)	40.8		43.3	
Motorcycles (%)	11.5		10.7	
N	1,296		845	

Notes: This table shows the characteristics of the Wave 1 and Wave 2 consumer samples, measured in Wave 1. The Wave 1 sample includes respondents who passed the screening quality filters and completed Wave 1. The Wave 2 sample is the experimental sample, after completion and quality filters; 801 of its 845 respondents are also in the Wave 1 sample, and the remaining 44 passed only the Wave 2 quality filters. Variables are defined as in Table 2.

Table A.11: Balance of Wave 1 covariates across information-treatment arms.

	Control mean (SD)	$D - C$	$DG - C$	N	Test: $D = C$	Test: $DG = C$	Test: $D = DG$
Panel A. All assigned in Wave 2 ($N = 985$)							
Age	47.09 (12.35)	-0.36 (0.96)	0.16 (0.99)	985	0.705	0.875	0.597
Male (%)	54.46 (49.88)	7.54 (3.83)	-1.02 (3.87)	985	0.049	0.792	0.024
Employed (%)	87.69 (32.90)	1.29 (2.53)	-4.44 (2.74)	985	0.611	0.106	0.033
College complete or higher (%)	70.77 (45.55)	-2.00 (3.51)	-1.13 (3.52)	985	0.568	0.748	0.805
Race: White (%)	52.03 (50.06)	-2.68 (4.52)	6.52 (4.49)	732	0.553	0.147	0.042
Race: Brown (%)	35.37 (47.91)	2.83 (4.36)	-3.96 (4.28)	732	0.516	0.355	0.117
Income bracket 1: $\leq$ BRL 5,000 (%)	32.92 (47.07)	-0.77 (3.58)	1.30 (3.60)	985	0.829	0.717	0.565
Income bracket 2: BRL 5,001–20,000 (%)	50.46 (50.07)	-0.72 (3.89)	-5.03 (3.89)	985	0.852	0.196	0.268
Household income floor (BRL 1,000s)	12.32 (23.38)	0.71 (1.83)	1.40 (1.86)	985	0.699	0.454	0.718
Primary financial decision maker (%)	84.31 (36.43)	-0.20 (2.86)	-6.73 (3.02)	985	0.943	0.026	0.032
Financial literacy score (0–4)	3.19 (0.96)	-0.06 (0.08)	0.01 (0.08)	985	0.446	0.872	0.358
Prepared for a financial emergency (%)	18.15 (38.61)	7.38 (2.88)	3.01 (2.92)	985	0.011	0.304	0.154
Access to a personal loan (%)	40.92 (49.25)	3.99 (3.70)	1.52 (3.72)	985	0.281	0.683	0.512
Any savings in the last 12 months (%)	59.08 (49.25)	3.58 (3.63)	1.90 (3.68)	985	0.325	0.605	0.648
Financial risk tolerance (0–3)	1.24 (1.27)	-0.00 (0.11)	-0.03 (0.11)	732	0.986	0.814	0.827
Financial patience (0–5)	1.90 (1.83)	-0.02 (0.16)	0.11 (0.17)	732	0.906	0.499	0.431
Emergency-coping capacity (1–5)	2.52 (1.12)	0.18 (0.08)	0.09 (0.08)	985	0.025	0.245	0.294
Bought a public lottery ticket (%)	30.46 (46.10)	-0.73 (3.60)	-0.61 (3.58)	985	0.841	0.866	0.973
Bets or gambles (%)	9.85 (29.84)	-1.27 (2.27)	2.69 (2.46)	985	0.576	0.273	0.096
Likely to join ($\leq$3 mo.) (1–4)	2.53 (0.96)	0.09 (0.07)	0.07 (0.07)	985	0.239	0.349	0.810
Past consórcio participation (%)	54.46 (49.88)	2.28 (3.87)	1.78 (3.88)	985	0.555	0.646	0.896
Currently paying a consórcio (%)	24.31 (42.96)	4.24 (3.46)	-0.66 (3.34)	985	0.220	0.844	0.152
Family/friends in a consórcio (%)	75.38 (43.14)	-1.44 (3.40)	-2.00 (3.41)	985	0.671	0.557	0.870
Discussed consórcio with vendor (1–4)	2.15 (1.08)	0.13 (0.08)	0.00 (0.08)	985	0.123	0.987	0.120
Lottery vs. auction preference (1–5)	2.07 (1.36)	0.08 (0.11)	0.19 (0.11)	984	0.450	0.087	0.346
Credit-timing preference (1–4)	2.22 (0.89)	-0.06 (0.07)	-0.07 (0.07)	985	0.420	0.282	0.826
Willing to bid $\geq$50% (%)	42.77 (49.55)	-1.19 (3.55)	0.40 (3.60)	985	0.738	0.911	0.661
Unconstrained: can fund a 50% bid (%)†	26.77 (44.34)	-0.39 (3.47)	-0.12 (3.45)	985	0.911	0.972	0.938
Bidding-share belief index (SD)	-0.01 (0.67)	0.09 (0.05)	-0.05 (0.05)	985	0.083	0.371	0.011
Lottery belief index (SD)	0.09 (0.73)	-0.08 (0.06)	-0.20 (0.06)	985	0.207	0.001	0.066
Bidding-belief certainty (1–5)	2.41 (0.94)	0.11 (0.07)	0.14 (0.07)	985	0.124	0.055	0.699
Lottery-belief certainty (1–5)	2.51 (0.98)	0.13 (0.08)	0.16 (0.08)	985	0.113	0.041	0.689
Hypothetical WTP in Wave 1 (%)	64.31 (47.98)	1.63 (3.73)	-1.67 (3.77)	985	0.662	0.658	0.376
Joint orthogonality (p)					0.634	0.392	0.074

Notes: This table shows the balance of Wave 1 covariates across treatment arms. Panel A includes all respondents assigned to an arm in Wave 2, and panel B the analysis sample, after completion and quality filters. Each row regresses the covariate on arm indicators with randomization-strata fixed effects; $D - C$ and $DG - C$ are differences from the control group, with robust standard errors in parentheses, and the test columns report p-values. The first column reports the control mean and standard deviation. Belief indices average standardized Wave 1 beliefs, with higher values more favorable to bidding. Hypothetical WTP is the share choosing consórcio access over a BRL 50 bonus in Wave 1. †Liquidity to bid is a randomization stratum, so it is estimated without strata fixed effects and excluded from the joint test. The joint test is a homoskedastic F-test that the covariates predict assignment, with strata fixed effects. Binary variables are in percent.

Table A.11: Balance of Wave 1 covariates across information-treatment arms (continued).

	Control mean (SD)	$D - C$	$DG - C$	N	Test: $D = C$	Test: $DG = C$	Test: $D = DG$
Panel B. Final analysis sample ($N = 845$)							
Age	47.30 (12.67)	-0.63 (1.04)	-0.13 (1.08)	845	0.546	0.902	0.635
Male (%)	54.84 (49.85)	7.66 (4.14)	-0.14 (4.14)	845	0.065	0.972	0.057
Employed (%)	86.02 (34.74)	3.82 (2.78)	-3.07 (3.03)	845	0.170	0.311	0.017
College complete or higher (%)	70.97 (45.47)	0.31 (3.79)	-1.44 (3.79)	845	0.934	0.704	0.643
Race: White (%)	51.50 (50.10)	-2.17 (5.04)	6.82 (4.93)	595	0.666	0.167	0.072
Race: Brown (%)	36.00 (48.12)	1.64 (4.85)	-4.25 (4.74)	595	0.736	0.369	0.216
Income bracket 1: $\leq$ BRL 5,000 (%)	30.82 (46.26)	1.71 (3.87)	3.09 (3.84)	845	0.659	0.420	0.723
Income bracket 2: BRL 5,001–20,000 (%)	51.25 (50.07)	-3.70 (4.23)	-5.10 (4.17)	845	0.382	0.221	0.737
Household income floor (BRL 1,000s)	13.12 (24.42)	0.32 (2.04)	0.88 (2.06)	845	0.874	0.669	0.789
Primary financial decision maker (%)	83.15 (37.49)	-0.69 (3.22)	-5.45 (3.27)	845	0.831	0.096	0.154
Financial literacy score (0–4)	3.19 (0.98)	-0.02 (0.08)	0.03 (0.08)	845	0.775	0.681	0.478
Prepared for a financial emergency (%)	16.85 (37.49)	9.45 (3.07)	4.48 (3.08)	845	0.002	0.146	0.130
Access to a personal loan (%)	41.94 (49.43)	4.98 (4.03)	-0.70 (4.00)	845	0.216	0.861	0.159
Any savings in the last 12 months (%)	59.50 (49.18)	5.53 (3.94)	2.95 (3.94)	845	0.160	0.455	0.510
Financial risk tolerance (0–3)	1.25 (1.28)	0.02 (0.13)	-0.02 (0.12)	595	0.882	0.885	0.768
Financial patience (0–5)	1.89 (1.80)	0.06 (0.18)	0.13 (0.18)	595	0.739	0.490	0.726
Emergency-coping capacity (1–5)	2.49 (1.10)	0.24 (0.08)	0.12 (0.08)	845	0.003	0.152	0.149
Bought a public lottery ticket (%)	31.90 (46.69)	-0.84 (3.97)	-1.85 (3.87)	845	0.832	0.633	0.796
Bets or gambles (%)	8.60 (28.09)	-0.53 (2.35)	2.96 (2.51)	845	0.822	0.238	0.161
Likely to join ($\leq$3 mo.) (1–4)	2.51 (0.97)	0.10 (0.08)	0.08 (0.08)	845	0.204	0.323	0.762
Past consórcio participation (%)	54.12 (49.92)	3.53 (4.20)	3.16 (4.17)	845	0.401	0.449	0.929
Currently paying a consórcio (%)	24.01 (42.79)	5.74 (3.78)	0.33 (3.60)	845	0.129	0.927	0.148
Family/friends in a consórcio (%)	74.19 (43.84)	-1.89 (3.76)	-2.35 (3.71)	845	0.615	0.527	0.904
Discussed consórcio with vendor (1–4)	2.13 (1.07)	0.19 (0.09)	-0.04 (0.09)	845	0.037	0.658	0.010
Lottery vs. auction preference (1–5)	2.06 (1.34)	0.13 (0.12)	0.17 (0.12)	844	0.287	0.141	0.714
Credit-timing preference (1–4)	2.23 (0.92)	-0.06 (0.08)	-0.06 (0.07)	845	0.444	0.411	0.999
Willing to bid $\geq$50% (%)	40.14 (49.11)	1.01 (3.90)	0.17 (3.87)	845	0.795	0.965	0.830
Unconstrained: can fund a 50% bid (%)†	25.09 (43.43)	-0.18 (3.69)	0.85 (3.65)	845	0.961	0.816	0.779
Bidding-share belief index (SD)	-0.02 (0.68)	0.11 (0.06)	-0.02 (0.06)	845	0.052	0.757	0.027
Lottery belief index (SD)	0.08 (0.73)	-0.03 (0.07)	-0.13 (0.06)	845	0.597	0.035	0.144
Bidding-belief certainty (1–5)	2.41 (0.92)	0.09 (0.08)	0.15 (0.08)	845	0.240	0.057	0.465
Lottery-belief certainty (1–5)	2.49 (0.96)	0.13 (0.08)	0.16 (0.08)	845	0.128	0.040	0.668
Hypothetical WTP in Wave 1 (%)	64.87 (47.82)	0.57 (4.04)	-0.72 (4.01)	845	0.889	0.858	0.749
Joint orthogonality (p)					0.412	0.639	0.217

Notes: This table continues Table A.11; see its notes.

Table A.12: Wave 2 attrition and quality-filter flow, by information-treatment arm.

	Control (C)	Debiasing (D)	Debiasing + Group Size Mechanism (DG)	p-value (C vs. D)	p-value (C vs. DG)
Assigned in Wave 2	325 (33.0%)	326 (33.1%)	334 (33.9%)		
Survey attrition	26 (8.0%)	30 (9.2%)	22 (6.6%)	0.593	0.480
Quality filter	20 (6.2%)	23 (7.1%)	19 (5.7%)	0.605	0.825
Final sample	279 (85.8%)	273 (83.7%)	293 (87.7%)	0.440	0.487

Notes: This table shows Wave 2 completion by treatment arm, among the 985 respondents assigned to an arm. Entries are counts, with percentages in parentheses: the arm's share of all assigned respondents in the first row, and the share of the arm's assigned respondents in the other rows. Survey attrition counts respondents who did not finish Wave 2, and Quality filter those who finished but failed the Wave 2 quality checks. p-values are from regressions of each indicator on arm indicators with randomization-strata fixed effects and robust standard errors.

Table A.13: Treatment effects on beliefs (AIPW).

	Bidding Allocation				Lottery Wait Time				Group Structure
	Qualit.	Quanti.	Probab.	Certainty	Qualit.	Quanti.	Probab.	Certainty	Group Size
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Debiasing (D)	0.852***	0.125***	0.045**	0.218***	0.439***	0.095***	0.102***	0.188***	164.6***
	(0.087)	(0.018)	(0.020)	(0.064)	(0.085)	(0.019)	(0.017)	(0.066)	(59.8)
Debiasing + Group (DG)	0.771***	0.124***	0.069***	0.235***	0.480***	0.076***	0.107***	0.220***	93.9*
	(0.092)	(0.017)	(0.019)	(0.064)	(0.081)	(0.019)	(0.017)	(0.067)	(53.7)
D % Gap Correction		47.4%	20.6%			66.0%	45.2%		23.5%
DG % Gap Correction		47.1%	31.0%			52.7%	47.7%		13.4%
$D = DG$ (p)	0.288	0.974	0.265	0.792	0.612	0.315	0.738	0.621	0.235
Control Mean	3.806	0.533	0.576	2.495	3.774	0.596	0.644	2.613	596.1
N	845	845	845	845	845	845	845	845	845
Measurement	Scale 1–5	Share 0–1	Chance 0–1	Scale 1–5	Scale 1–5	Month 0–1	Chance 0–1	Scale 1–5	Members

Notes: This table shows treatment effects on Wave 2 beliefs. Estimates are augmented inverse-probability-weighted (AIPW) effects from a multi-arm causal forest (Athey et al., 2019) with the Wave 1 covariates and the randomization strata; standard errors in parentheses. Gap correction is the arm's effect as a share of the gap between the true value and the control mean, and is blank for outcomes without a true value. The $D = DG$ row reports the p-value for equal effects in the two arms. Group size is winsorized at the 99th percentile. $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.14: Learning rates from the information treatments.

	Bid share		Lottery wait		Lottery chance
	OLS	ORIV	OLS	ORIV	OLS
	(1)	(2)	(3)	(4)	(5)
Panel A. Pooled treatment					
Treatment	0.198***	-0.080	0.351***	0.333***	0.434***
	(0.076)	(0.191)	(0.073)	(0.090)	(0.092)
Gap	0.503***	0.150	0.496***	0.423***	0.586***
	(0.055)	(0.151)	(0.053)	(0.077)	(0.065)
Treatment × gap	0.246***	0.490***	0.070	0.103	0.085
	(0.066)	(0.173)	(0.069)	(0.101)	(0.078)
Panel B. By treatment arm					
D	0.183**	-0.181	0.339***	0.286***	0.448***
	(0.089)	(0.205)	(0.080)	(0.100)	(0.103)
DG	0.214**	0.041	0.363***	0.381***	0.419***
	(0.087)	(0.224)	(0.082)	(0.103)	(0.104)
D× gap	0.249***	0.571***	0.091	0.169	0.074
	(0.079)	(0.188)	(0.083)	(0.119)	(0.090)
DG× gap	0.241***	0.400**	0.053	0.047	0.095
	(0.075)	(0.202)	(0.081)	(0.118)	(0.090)
Respondents	845	845	845	845	845

Notes: This table shows how much of the initial belief gap treated respondents close. The dependent variable is the change in belief from Wave 1 to Wave 2; each column regresses it on a treatment indicator, the gap between the true value and the Wave 1 belief, and their interaction, with asset category fixed effects. The interaction is the learning rate. Changes and gaps are in Wave 1 standard deviations. OLS columns average the questions on each belief (three for the bid share, two for the lottery wait), with robust standard errors; ORIV columns instrument the gap in each question with the gap in another (Gillen et al., 2019), with standard errors clustered by respondent. Lottery chance is the probability of no award in the first half of the group's duration, so higher values mean a longer wait. For the qualitative questions, the true value is the top category. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.15: Treatment effects on attitudes and recommendation (AIPW).

	Bidding-Advantage Beliefs		Would Recommend to Non-Bidders		
	Only Good if Bidding	Admin. Favors Bidders	Yes	No	Don't Know
	(1)	(2)	(3)	(4)	(5)
Debiasing (D)	0.106**	0.093**	-0.031	0.062**	-0.036
	(0.042)	(0.039)	(0.036)	(0.029)	(0.029)
Debiasing + Group (DG)	0.081**	0.050	-0.016	0.065**	-0.054*
	(0.041)	(0.039)	(0.036)	(0.029)	(0.029)
$D = DG$ (p)	0.546	0.251	0.668	0.923	0.512
Control Mean	0.477	0.649	0.624	0.151	0.226
N	845	845	845	845	845
Measurement	Share 0–1	Share 0–1	Share 0–1	Share 0–1	Share 0–1

Notes: This table shows treatment effects on Wave 2 attitudes toward consórcios, coded 0/1. Estimates are AIPW effects from a multi-arm causal forest, as in Table A.13; standard errors in parentheses. The first two columns are the shares agreeing that consórcios are a good deal only for those who can bid, and that administrators favor bidders. The Would recommend columns are the shares who would, would not, or do not know enough to recommend a consórcio to someone who cannot bid. The $D = DG$ row reports the p-value for equal effects in the two arms. $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.16: Information treatment and hypothetical choices of a loan or saving over a consórcio, by liquidity and optimism.

	Loan over consórcio			Saving over consórcio		
	(1)	(2)	(3)	(4)	(5)	(6)
Debiasing (D)	0.062*	-0.012	-0.052	0.040	-0.024	-0.030
	(0.033)	(0.047)	(0.051)	(0.042)	(0.056)	(0.065)
× optimist		0.165**	0.157**		0.141*	0.120
		(0.066)	(0.073)		(0.084)	(0.098)
× liquidity to bid			0.182			0.028
			(0.122)			(0.130)
× optimist × liquidity			-0.022			0.063
			(0.164)			(0.189)
Debiasing + Group (DG)	-0.019	-0.084**	-0.084*	-0.060	-0.163***	-0.114*
	(0.030)	(0.042)	(0.049)	(0.041)	(0.055)	(0.064)
× optimist		0.145**	0.126*		0.228***	0.131
		(0.059)	(0.067)		(0.082)	(0.096)
× liquidity to bid			0.001			-0.222*
			(0.098)			(0.122)
× optimist × liquidity			0.064			0.385**
			(0.137)			(0.183)
Optimist		-0.150***	-0.153***		-0.243***	-0.200***
		(0.043)	(0.049)		(0.060)	(0.070)
Liquidity to bid			0.033			0.185*
			(0.091)			(0.101)
Optimist × liquidity			0.015			-0.165
			(0.105)			(0.136)
$D = DG$ (p-value)	0.012	0.097	0.483	0.015	0.012	0.175
Control mean	0.18	0.24	0.23	0.48	0.58	0.55
Observations	845	845	845	845	845	845

Notes: This table shows treatment effects on hypothetical choices of a loan (columns 1–3) or saving (columns 4–6) over a consórcio with average market attributes, with the loan rate or the savings return randomized. Estimates are AIPW effects from one multi-arm causal forest per outcome, with the randomization strata and the randomized rate as covariates. Arm rows are effects in the reference group: all respondents in columns 1 and 4, non-optimists in columns 2 and 5, and non-optimists without liquidity to bid in columns 3 and 6; × rows are differences from it. The optimist and liquidity rows are differences within the control arm. The $D = DG$ row and the control mean refer to the reference group. Optimists report a Wave 1 chance of a lottery award in the first half of the group's duration above the median of 30%; liquidity to bid means having own resources to bid 50% of the contract's face value. Standard errors in parentheses. $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

Table A.17: Information treatment and willingness to pay for the lottery-intensive offer.

	Willingness to pay (BRL, incentivized price list)						
		Heterogeneity by liquidity			Heterogeneity by beliefs		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Debiasing (D)	4.77*	5.77*	6.79**	7.05**	2.63	5.18	5.75
	(2.89)	(3.33)	(3.27)	(3.41)	(4.00)	(4.46)	(4.55)
× liquidity to bid 50%		-4.00		-4.28		-12.53	
		(6.76)		(6.73)		(10.09)	
× strategy = bid early			-8.72				-13.05
			(7.01)				(9.52)
× optimist about early award					4.78	1.43	2.28
					(5.79)	(6.70)	(6.54)
× optimist × liquidity						15.14	
						(13.66)	
× optimist × bids early							10.21
							(14.12)
Debiasing + Group (DG)	-0.16	0.24	-1.12	0.01	0.63	2.95	0.29
	(2.71)	(3.14)	(3.10)	(3.24)	(3.75)	(4.22)	(4.26)
× liquidity to bid 50%		-1.58		1.28		-11.49	
		(6.22)		(6.20)		(9.24)	
× strategy = bid early			4.55				1.65
			(6.35)				(9.07)
× optimist about early award					-1.75	-6.54	-3.11
					(5.42)	(6.32)	(6.22)
× optimist × liquidity						19.22	
						(12.57)	
× optimist × bids early							6.39
							(12.71)
$D = DG$ (p-value)	0.08	0.09	0.01	0.05	0.62	0.65	0.24
Sample	Full	Full	Full	WTP-consistent	Full	Full	Full
Control mean	36.6	36.6	36.6	34.8	36.6	36.6	36.6
Observations	845	845	845	810	845	845	845

Notes: This table shows treatment effects on willingness to pay for access to the offer, in BRL, from each respondent's price list (up to BRL 50 or BRL 100, depending on the offer-value arm). Columns, estimator, and tests are as in Table 3. Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.18: Treatment effects on bidding beliefs and intentions.

	Expected winning bid (1)	Likelihood of bidding (2)	Competitiveness of own bid (3)	Reported own bid (4)
Debiasing (D)	5.53***	0.157	-0.046	1.71
	(1.71)	(0.098)	(0.087)	(1.54)
Debiasing + Group (DG)	6.25***	0.120	0.063	2.16
	(1.71)	(0.094)	(0.085)	(1.56)
$D = DG$ (p-value)	0.680	0.695	0.193	0.777
Control mean	44.51	3.018	2.989	30.92
Wave 1 measure included	Yes	–	–	–
Observations	845	845	845	845
Measurement	% of face value	Scale 1–5	Scale 1–5	% of face value

Notes: This table shows treatment effects on Wave 2 bidding beliefs and intentions. Estimates are augmented inverse-probability-weighted (AIPW) effects of each arm relative to the control group, from a multi-arm causal forest that conditions on pre-treatment covariates, randomization strata, and, in column (1), the Wave 1 value of the outcome; standard errors in parentheses. Expected winning bid and reported own bid are percentages of face value; likelihood of bidding and competitiveness of own bid are on 1–5 scales. The $D = DG$ row reports the p-value for equal effects in the two arms. $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

B Institutional context

This appendix describes the rules that govern consórcio contracts, groups, awards, and firms. Consórcios are regulated by Law 11,795/2008 and by Central Bank rules: Circular 3,432/2009, in force for most of our sample period, and Resolution BCB 285/2023, which replaced it in July 2024 and keeps the provisions we describe below.

Contracts. The monthly installment has up to four components: the contribution to the pool, the administration fee, a contribution to a reserve fund, and an insurance premium. Each is set as a percentage of the price of the asset referenced in the contract, so all of them move with that price. Contracts traditionally referenced a specific asset, such as a given car model, so that payments and credit followed that model's list price. This practice is now less common: contracts usually reference a credit value updated by a price index or a price table set in the contract. The administration fee is a fixed percentage spread over the months of the contract, but firms may collect part of it at signing to pay sales representatives and brokers. Moreover, firms may generally anticipate the collection of fees, charging higher fees in the earlier months of a contract. Sales commissions can therefore be paid when a contract is sold, while the rest of the fee accrues over the contract's life.

Contracts are often bundled with credit life insurance. If a participant who has not yet been awarded dies and the insurer's payment covers the remaining balance, the payment counts as a winning bid in the next monthly meeting with enough funds, and any surplus goes to the participant's beneficiaries. Coverage beyond death, such as disability or job loss, varies by contract.

Groups. A group must be formed within 90 days of the first contract sold; otherwise, the firm refunds any amounts collected, with returns. Participants who join a group already under way must complete their payments within the group's remaining duration, so their contract's term is constrained to the duration of the group.

The pool legally belongs to the group's participants, and not to the firm. It must be deposited in a commercial bank and invested only in federal government bonds or in short-term fixed-income funds, excluding funds in which the firm itself invests. Funds waiting to be awarded therefore earn approximately the base interest rate, and any of these returns accrue to the group. In practice, the common pool is usually kept at small levels, so the money accrued is not significant.

Groups may also hold a reserve fund. It covers shortfalls in the pool, insurance against participants' defaults, and collection costs. When it is not needed for these, it can fund additional lottery awards. Within 60 days of the last award meeting, the firm must notify participants that unused credit and outstanding refunds are available in cash, and the group must close within 120 days. Balances left in the pool and the reserve fund are returned to active participants in proportion to

their payments.

Awards. Awards take place at monthly meetings. Only participants who are current on their payments are eligible. Lottery awards are drawn first, and bids are awarded only after the first lottery award, or when the pool is too small for lotteries. A bid must be paid before the award is confirmed. Some contracts allow "embedded bids", which is a mode of payment in which part of the credit being awarded pays the bid, and the winners' effective funds are only the difference. Other contracts allow "fixed bids", where a pre-defined bid amount is set, and a lottery takes place among participants submitting that fixed bid.

Usage of funds. The award can be used with any seller, and the firm pays the seller directly. In vehicle groups, the credit can buy any vehicle, new or used, as well as aircraft, boats, and capital machinery. Often, vehicles that are too old are not accepted. In real-estate groups, it can buy a finished or off-plan property, or land, and, if the contract allows, pay for construction or renovation. In every category, the credit can also pay off an existing loan on an asset of the same category. If the asset costs less than the credit, the difference can cover related costs (transfer taxes, registration, and insurance) up to 10% of the credit, prepay future installments, or be paid out in cash once the participant has settled their obligations to the group. The asset bought serves as collateral, and the firm may require additional guarantees before releasing the credit.

Awarded credit is not necessarily used immediately. The firm must make it available within three business days of the award, and until it is used it remains invested with the group's funds and accrues their returns. A participant who waits for a better purchase therefore earns approximately the Selic rate on the credit in the meantime. Therefore, considering financial returns, an early lottery award is always more profitable even if the asset is not desired to be acquired immediately. After settling all remaining payments and waiting 180 days, the participant can take the award in cash instead of purchasing an asset, so consórcios do not function as a source of liquidity or insurance. Because cashing out requires paying off the remaining balance, the award is in practice a right to buy an asset of the category, not a cash loan.

Cancellations and refunds. A contract is canceled when the participant files a cancellation explicitly or misses three consecutive payments. Canceled contracts do not receive their refund on leaving: they remain in the group's lottery draws, and a canceled contract that is drawn receives a refund instead of credit. The refund equals the share of the asset's price the participant had paid into the pool, valued at the price on the date of the draw, plus the pool's returns, minus contractual fines. Since 2024, fines cannot exceed the administration fee the firm would have collected had the participant stayed. Canceled contracts that are never drawn are refunded when the group closes.

Once the credit has been used, the contract can no longer be canceled, missed payments are a

default, and the firm must move to seize the collateral after more than one late installment.

Secondary markets. Contracts can be transferred to a third party with the firm's consent, and the firm assesses the buyer's ability to pay. An informal market trades contracts that have been awarded but whose credit is unused: the buyer pays the seller and takes over the remaining installments, obtaining credit at once without bidding. This market also lets participants join as investors, buying contracts in order to resell them after award.

Firms. Firms that manage groups must be authorized and supervised by the Central Bank. Their revenue is the administration fee: returns on the pool accrue to the group, and at least half of any fines and late-payment interest must go to the group, with the remainder to the firm. Firms also bear credit risk at award, as they must compensate the group for losses from guarantees they approved that prove insufficient. This liability gives firms a reason to screen participants before releasing credit. In practice, the cost of defaults is passed through the reserve fund fee.

C Central Bank data: extraction and sample construction

This appendix describes how we extract, clean, and filter the administrative consórcio data reported to the Central Bank of Brazil, and how we build the analysis sample.[31] The source is the quarterly contract-level filings that administrators are required to submit. We merge this data at the taxpayer-ID (*CPF*) level with the national credit registry, the federal revenue individual and firm registries, formal labor-market records, and the social-assistance registry. Our paper restricts analysis to the real-estate, automobile, and motorcycle assets, which represent 90% of contracts sold. We therefore exclude heavy vehicles (trucks, farm machinery) and services.

Our two main datasets include a contract-level sample of 8,642,735 contracts in 7,328 groups and a corresponding longitudinal group-level dataset, which covers 6,887 of these groups after filtering.

C.1 Extraction, sampling, and de-duplication

The full filing history is too large to extract at once on Central Bank servers since observations are quarterly. We sample contracts by stratifying and extracting smaller files at the group-year level, ranging from 2010 to 2025.[32]

We aggregate observations to obtain a cross-section of contracts. A contract is also referred to as a "share". In the dataset, a contract is formally identified by an *asset–administrator–group–share–subshare* key. The subshare number exists to allow a canceled contract to be resold and replaced while preserving the same share number. Typically, the subshare number is a chronological sequence from 1, 2, .., and so on up until the current or last owner. In practice, when a canceled contract is replaced, a subshare is first reported as `0` and then switch to a nonzero code, spuriously multiplying observations of what is essentially a single contract. We therefore ignore the subshare dimension and extract data at the *asset–administrator–group–taxpayerID–share* as the unit of analysis, which removes the double-counting artifact while still distinguishing the legitimate case of a number resold to a new holder. One consequence is that public published aggregate totals, which count share–subshare pairs, will not match our contract–CPF counts exactly.

C.2 Contract-level cleaning

Cleaning reduces the filings to one row per contract–CPF. As noted above, filings can carry more than one row per contract–CPF, both from the subshare artifact and from the monthly panel, and these rows need not agree. We therefore keep only contracts whose fields are invariant across all of

[31] The SQL scripts that extract the datasets are available from the authors for replication, pending Central Bank authorization.

[32] Sampling whole groups, rather than individual contracts, preserves the within-group allocation dynamics that the group-level analysis studies.

their rows — the enrollment, award, and cancellation dates, the allocation type, the planned term, and the administration fee — and drop the 19% of contract–CPF units that are not.[33] Monetary values are deflated to constant reais by the enrollment-month price index. Table C.1 defines the variables the analysis relies on.

Table C.1: Key variables in the contract-level data

Variable	Definition
Contract–CPF	Unit of analysis: a contract collapsed across subshare codes and identified with its holder's CPF.
Enrollment, award, cancellation dates	Dates the contract enters the group, is awarded credit, and (if applicable) is canceled.
Allocation type	Whether the award came by lottery (*sorteio*) or bid (*lance*).
Wait for award	Months from enrollment to award, for awarded contracts.
Bid size	Bid amount as a share of the total contract value (the credit value plus the administration fee).
Contract value	Value of the asset at enrollment (pre-correction) and at its final level (post-correction), in constant BRL.
Individual/business dummy	Whether the contract is held by an individual or a firm.

- *Group start.* A group's start is the date of its first award — the earliest *contemplação* observed across the group's contracts.
- *Entry timing.* A contract's entry timing is its enrollment date relative to the group start, scaled by the group's planned duration: near zero for a contract that joins around the first award, near one for one that joins at the group's planned end.
- *Default and repossession.* We flag a default when a judicial collection action is recorded against the contract, and a repossession when the financed good is returned. They therefore represent defaults on obligations, not missed payments.

C.3 Group-panel construction

For the group-level analysis, we collapse the cleaned contracts into a panel with one row per group and award month. A group starts at its first award date and ends at the later of either (i) its planned end (start plus the planned duration) or (ii) its last observed award. The planned duration is defined as the maximum term among its contracts. We index periods in relative time, as a fraction of the group's duration, and measure group size both at fixed fractions of the planned duration (5, 10, and 20 percent) and over the first 5 percent of *observed* periods, the latter so that young groups are not dropped for want of history; active size nets out cancellations. A group is finished when its end date precedes the panel cutoff (31 December 2025).

[33] The share is 15% for cars, 26% for motorcycles, and 10% for real estate, and 35% for groups started in 2010–2014 against 2% for groups started in 2024–2025.

C.4 Group inclusion criteria

We keep groups that behave like standard, well-populated consórcio groups. We drop any that meets one of the following criteria:

1. a cancellation share of 80 percent or more;
2. a group smaller than its planned duration in months (N < T);
3. a finished group that lasted less than 80 percent of its planned duration;
4. groups where a single CPF holding more than 10 percent of the contracts;[34]
5. groups where firms represent 40 percent or more of total contracts;
6. total of non-canceled awards is larger than number of active contracts at any point;
7. very short group durations (below 100 months for real estate, below 30 for cars and motorcycles).

The population census of groups is cross-sectional, so we apply only three rules computable from census columns: (i) cancellation share; (ii) size versus duration; (iii) standard duration. The remaining screens require a monthly panel and cannot be reproduced from the census due to computing constraints.

C.5 Weighting and representative samples

Because the estimation sample is a cluster sample of groups, we reweighted groups according to the population samples. The census fixes the population of groups and their observable margins. From these frames we construct a survey weight, normalized to mean one. A group's base weight is the inverse of its sampling probability, known by design as the number of population groups over the number sampled in its (start-year, asset) stratum. The group-level weight w_g is raked to the group census on start year, group size, asset, and planned duration, and makes the sampled groups representative of the population.

[34] This criterion is also a regulatory requirement from the Central Bank.

D Prediction of censored group histories

This appendix documents the predictions introduced in Section 3 in four steps. First, what is estimated: seven objects, of which four are rates at the group-by-period level, two are prices of an imputed exit, and one is the cumulative bid share that completes panel A of Figure 6. Second, the data and covariates used for estimation: the cut at each group's own first quarter of planned duration, along with a general time profile estimated from all groups of the same asset category. Third, how we estimate: one binomial logit specification for the four rates, prices as cell means or a regression, and then recursions that map rates and prices into a contract's exit and return. Fourth, the validation: in completed groups censored at the cut and predicted from models estimated without them, the lottery wait of a participant who never bids and the pace of awards are matched within a percentage point for cars and motorcycles, and the imputed return distribution of contracts still waiting sits two cents below the realized one. Predictions are made only for groups observed through the cut, and only for the part of each group's history that is not observed.

D.1 Prediction objects

Table D.1 lists the first six objects to be estimated. The first four are rates at the group-by-period level: three hazards giving the probability that: (i) a waiting contract is awarded; (ii) an active contract cancels; and (iii) a waiting contract is drawn in the lottery; and the bid share of a period's awards. The next two are prices: the winning bid amount paid (when the award is won by bid), and the refund value a contract recovers on cancellation.

These six objects are combined to predict the outcomes of a censored contract. The cancellation hazard first sets the exit outcome. Conditional on not canceling, the award hazard predicts the probability of an award in each remaining period. The bid share splits an award between bid and lottery. Then, for each of the three outcomes, the two prices are components of the return calculation: a bid award is priced at the winning bid (a lottery award sets the bid to zero) and a cancellation at the refund value. We call the exit assigned this way an imputed exit; with the contract's own terms, it becomes a return.

Finally, we estimate the cumulative bid share of a group over time.

Everything the paper uses is derived from these objects. The first three rates give each waiting contract a probability of leaving by bid, by lottery, or by cancellation in each remaining period, and the two prices turn those exits into a return. The lottery hazard gives the wait of a participant who never bids. It equals the award hazard times one minus the bid share, but we estimate it directly, because the never-bidder's wait is the object the paper reports and a direct fit can be selected and validated on it; the product of the first two rows is not used.

Table D.1: Predicted objects, definitions, and uses.

Object	Definition	Use in the paper
Panel A. Group-by-period rates and shares		
1. Award hazard	Awards by cash bid or lottery divided by contracts eligible for an award	Award timing of waiting contracts
2. Cash-bid share	Bid awards divided by all awards	Division of awards between cash bid and lottery
3. Cancellation hazard	Cancellations divided by contracts at risk of cancellation	Cancellation timing and probability
4. Lottery hazard	Lottery awards divided by contracts eligible for an award	Waiting time of a participant who never bids
Panel B. Prices assigned to imputed exits		
5. Winning bid	Mean winning bid within asset category and period	Payment at an imputed bid award
6. Cancellation value	Predicted net recovery for a canceled contract	Value of an imputed cancellation

D.2 Data and covariates used for estimation

A group's planned duration is divided into twenty equal periods. For each group and period, the data record lottery awards, bid awards, cancellations, active contracts, and waiting contracts, where waiting contracts are the active contracts not yet awarded. Completed groups contribute all twenty periods while ongoing groups contribute their observed periods.

We construct seven group-level predictors using the first 25% of the groups planned duration: (i) log group size at the cut; (ii) group size at the cut divided by planned duration in months; (iii) log planned duration; (ivvi) the average monthly numbers of lottery awards, bid awards, and cancellations during the first quarter; and (vii) the log ratio of the waiting-contract pool (at the start of the period) to group size at the cut.

Of the 6,887 groups observed through the cut, 43 have lottery awards and no bid award within it. We treat them as pure lotteries by definition, therefore requiring no fit, and they enter neither the estimation nor the validation.

D.3 Estimation procedure

The four rates share one specification. For rate h_{gt} of group g in period t, in asset category s,

$$\log \frac{h_{gt}}{1 - h_{gt}} = \alpha_t + \beta_t \ln N_g + \gamma_s + \delta_s \, \mathbf{1}[t = 20] + \theta' X_g + \sum_k \phi_k(t) \, x_{gk}, \tag{D.1}$$

where α_t are period effects, β_t is a period-specific loading on log group size N_g, γ_s is a category effect, δ_s is a category-specific effect for the final period, and X_g holds the remaining predictors. The three pace predictors x_{gk} (lottery awards, bid awards, and cancellations per month) are multiplied by natural cubic splines $\phi_k(t)$, so that a group's early pace can change over its life. Each rate is a binomial logit with the numerator and denominator of Table D.1 as successes and trials. It is estimated on all groups, completed and ongoing, with periods weighted by the group's sampling weight.

The winning bid is the mean winning bid in each category and period. The cancellation value is a linear regression, on canceled contracts, of the net value recovered on the timing of the cancellation and the shares paid in; predictions are clamped to the range observed among canceled contracts. The cumulative bid share is a binomial logit of cumulative bid awards over cumulative awards on a natural cubic spline in relative time with three degrees of freedom, log group size, and the bid share within the first period, estimated by category on completed groups only, since its outcome is defined at completion.

We combine the fitted rates in two steps. In both, a group's observed periods keep their realized values, and only its unobserved periods take fitted values. First, for a contract that never bids, the probability of a lottery award in period t is $p_{gt} = h^L_{gt} \prod_{\tau<t}(1 - h^L_{g\tau})$: the lottery hazard in t times the probability of not having been drawn earlier. Summing p_{gt} over periods gives the probability of a lottery award by any point in the group. Second, for each contract still waiting when its group's history ends, we follow the remaining periods one at a time. In each period, the contract may first cancel; if it does not, it may be awarded, and the bid share splits that award between bid and lottery. This gives, for every remaining period, the probability that the contract exits by bid, by lottery, or by cancellation. We value each exit as in the main text, by the net present value of the contract's cash flows per unit of face value: a bid exit pays the category's winning bid for that period, a lottery exit carries a zero bid, and a cancellation exit recovers the predicted refund. A contract's expected return averages these values, weighted by their probabilities. For the return distribution, however, we keep every combination of exit and period of every contract as a separate observation, weighted by its probability. Using one expected return per contract would compress the tails of the distribution.

We compared alternative time profiles, links, predictor blocks, and weighting schemes. The comparison emphasized the objects used in the paper: the cumulative probability of a lottery award at the midpoint and at completion, and expected waiting time conditional on an award. We also examined held-out deviance, giving equal weight to relative-time bins and separately evaluating the second half of the term. These criteria assess both the accuracy of the reported objects and errors that an exposure-weighted score can obscure. The chosen specification had lower pooled deviance than the by-category alternative under all three aggregations, although the differences between these finalists were small.

D.4 Results and validation

For cars and motorcycles, the lottery wait of a participant who never bids and the pace of awards are predicted within a percentage point of their realized values; the imputed exits of contracts still waiting reproduce the realized return distribution to within two cents, on the pessimistic side. The design is the one used for selection: completed groups are censored at the cut and predicted from models estimated without them, under five-fold cross-validation by group and under a vintage split that estimates on groups started through 2013 and predicts later ones. Cars and motorcycles have 2,299 and 2,179 completed groups. Real estate has 59, too few to validate at completion; its lottery path is checked at the midpoint on the 201 groups observed through at least half of their term, 142 of which are ongoing and enter the estimation, so this is an in-sample check of the midpoint and not a validation.

Table D.2 reports the lottery path and Figure D.1 overlays predicted and realized cumulative lottery-award curves. For cars, 19.4% of never-bidding contracts are predicted drawn by the midpoint against 19.8% realized, 98.1% by the end of the term against 98.2%, and the expected wait given a draw is 69.4% of the term against 68.4%, with a group-level mean absolute error of 3.6% of the term, under three months. For motorcycles the same figures are 12.0% against 11.9%, 94.1% against 95.9%, 75.2% against 74.4%, and 2.2 months. The vintage split gives the same accuracy. Relative to a model with only the category time profile, the group predictors lower held-out deviance by 21% in cars and 20% in motorcycles (column 7).

The pace of awards is predicted at least as closely (Table D.2, panel C): 58.3% of waiting car contracts are predicted awarded by the midpoint against 57.1% realized, and 60.5% of motorcycle contracts, predicted and realized alike; the expected award time is 45% and 43% of the term, within half a point of the realized values.

The one systematic miss is in the timing of lottery awards late in a group's life. Lottery awards come in a burst after three quarters of the term, and the model under-predicts it: by three quarters of the term, 33.3% of never-bidding motorcycle contracts are predicted drawn against 37.5% realized, and 45.9% against 47.8% for cars. The error rises with a group's bid share after the cut, which the cut does not forecast: groups that bid more than their first quarter suggested leave more of their lottery awards for the end.

Table D.2: Predicted against realized lottery path and award pace in held-out completed groups censored at the cut.

Sample	Groups	Share awarded by			Mean time to award	MAE	Held-out deviance
		Midpoint	Three quarters	End			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Lottery path, five-fold cross-validation							
Cars	2,299	0.194 / 0.198	0.459 / 0.478	0.981 / 0.982	0.694 / 0.684	0.036	−21%
Motorcycles	2,179	0.120 / 0.119	0.333 / 0.375	0.941 / 0.959	0.752 / 0.744	0.036	−20%
Real estate†	201	0.191 / 0.216					−8%
Panel B. Lottery path, vintage split							
Cars	1,064	0.193 / 0.189	0.493 / 0.462	0.981 / 0.979	0.684 / 0.692	0.040	
Motorcycles	1,020	0.108 / 0.112	0.343 / 0.356	0.950 / 0.961	0.756 / 0.748	0.029	
Panel C. Award pace, five-fold cross-validation							
Cars	2,299	0.583 / 0.571	0.843 / 0.835	0.995 / 0.996	0.446 / 0.450	0.022	
Motorcycles	2,179	0.605 / 0.605	0.866 / 0.866	0.994 / 0.995	0.432 / 0.433	0.014	
Real estate	59	0.583 / 0.599	0.851 / 0.879	0.987 / 0.995	0.440 / 0.431	0.025	

Notes. Every cell in columns (2)–(5) is *predicted / realized*. Groups are completed groups held out of estimation, censored at the cut and predicted from models fit without them. Columns (2)–(4) are the cumulative share awarded by the midpoint, three quarters and the end of the planned term: of contracts that never bid in Panels A and B, of all waiting contracts in panel C. Column (5) is the mean time to award, a fraction of the planned term — the never-bidder's wait in panels A and B, the award time of any waiting contract in panel C. Column (6) is the group-level mean absolute error of that mean time, also as a fraction of the term. Column (7) is the change in held-out deviance against a model carrying only the category time profile, so a negative number is an improvement from the group predictors; it is reported for the cross-validated lottery fits only. †Real estate is an in-sample check at the midpoint, on groups observed through at least half of their term, so its later columns are undefined.

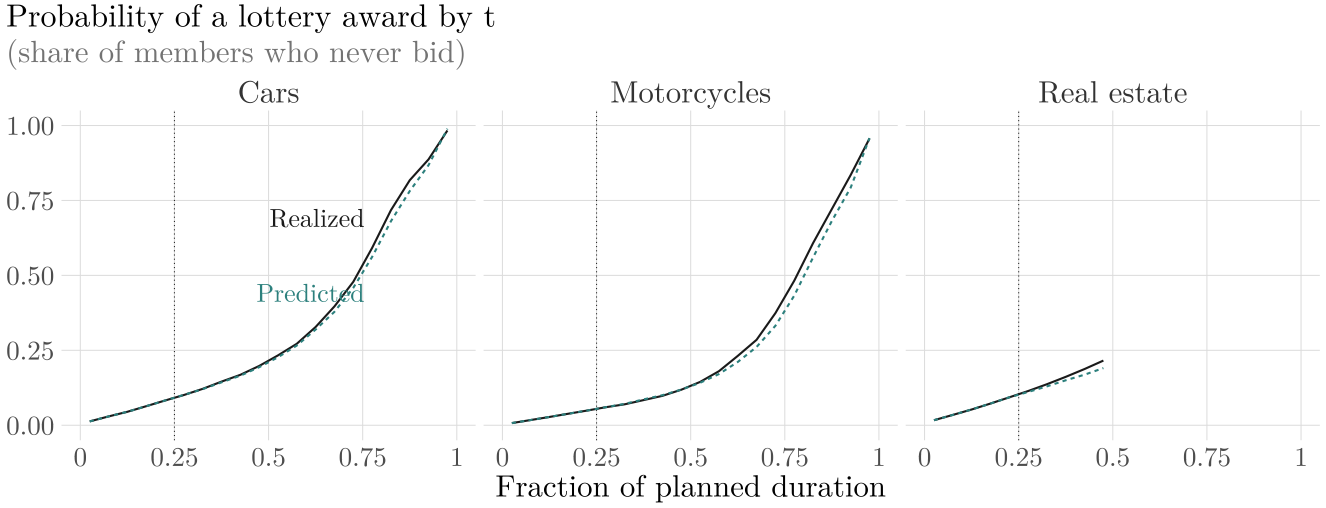


Figure D.1: Cumulative probability of a lottery award, realized and predicted from the cut, in held-out completed groups (cars, motorcycles) and in groups observed through at least half of their term (real estate).

Table D.3 turns to the contracts still waiting at the cut and compares their imputed exits and returns with the realized ones. Pooled over categories, the imputed median return is −0.087 per unit of face value against −0.069 realized, the mean −0.075 against −0.071, and 83% of imputed returns are negative against 79% realized: the imputed median sits about two cents below the realized one. The exit mix is close for bids, 48% imputed against 47% realized pooled and within two points in cars and motorcycles, 6 points low in real estate; cancellations are 38% against 33% pooled, within one point for motorcycles but 8 points high in cars and real estate; lotteries are low, 15% against 20% pooled, by 8 points in cars, 4 in motorcycles, and 2 in real estate. The errors by exit type offset. Imputed bid awards return 4 cents less than realized ones pooled, and 3 to 4 cents less in cars and motorcycles, because the winning bid of the category and period is pooled over all vintages and exceeds what these older completed groups paid; imputed lottery awards return 4 cents more pooled, and 3 to 7 cents more in cars and motorcycles, because exits are placed 1.3 months too early on average.

Table D.3: Imputed against realized exits and returns of contracts still waiting when their group is censored at the cut, completed groups.

	Pooled (1)	Cars (2)	Motorcycles (3)	Real estate (4)
Panel A. Return of waiting contracts				
Median	−0.087 / −0.069			
Mean	−0.075 / −0.071			
Share negative	0.83 / 0.79			
Panel B. Exit shares				
Cash-bid award	0.48 / 0.47	0.47 / 0.47	0.49 / 0.47	0.38 / 0.44
Lottery award	0.15 / 0.20	0.19 / 0.27	0.11 / 0.15	0.18 / 0.20
Cancellation	0.38 / 0.33	0.34 / 0.26	0.40 / 0.39	0.44 / 0.36
Panel C. Mean return by exit				
Cash-bid award	−0.072 / −0.035	−0.050 / −0.008	−0.086 / −0.054	−0.121 / −0.101
Lottery award	−0.072 / −0.115	−0.061 / −0.089	−0.085 / −0.151	−0.122 / −0.154
Cancellation	−0.081 / −0.100	−0.080 / −0.099	−0.082 / −0.100	−0.076 / −0.095

Notes. Every cell is *imputed / realized*. The sample is contracts still waiting when their group is censored at the cut, in completed groups. Exit shares are shares of those waiting contracts; returns are per unit of face value. Column (1) pools the categories, weighting each by its number of waiting contracts. Panel A is reported pooled only.

Three limits remain. First, the imputed winning bid is too high for older groups, which is why imputed bid returns fall short of realized ones; a winning bid by vintage is the natural fix. The exposure of the paper's headline is bounded by Table D.3 and the imputed share of Section 3: imputed contracts are about 15% of the contracts in ongoing groups, and their median is two cents too low. Second, the late-life miss above has a cohort behind it: motorcycle groups started in 2019 or later run a lower mid-life lottery hazard than earlier cohorts, and the cut does not reveal it, so

no completed-group validation can measure this error either way. Third, real estate rests on 59 completed groups, so its end-of-term lottery mass is not validated.

E Simulated bidding strategies

This appendix describes the simulation behind Table 1. We simulate a contract through each group in our dataset under different bidding strategies, assuming bid amounts are constant every month until awarded, and record when it is awarded and the returns. Other participants' bids and the group's lottery awards stay at their observed or predicted paths.

First, the contract joins at the group's start, assuming its term is the group's planned duration and the asset category's average administration fee. We rule out canceling or defaulting. Each award is valued with Equation (1) at the benchmark rates, per BRL of payments to the pool.

Each month, the contract can be awarded by lottery using the group's estimated lottery probability (Appendix D). The model gives one probability per twentieth of the group's duration, spread evenly over its months, with any remainder in the term's last month, when every remaining contract receives its funds.

A bid of b wins the award in a given month if it exceeds that month's threshold: the lowest winning bid above 30% of face value, excluding fixed bids (a value that accounts for more than 20% of a group's bid awards). Each contract faces its own group's thresholds: for each twentieth of duration, we record whether any month's threshold is below b, and the bid wins in the first such twentieth, spread evenly over its months. In a completed group whose thresholds never fall below b, the contract can only be awarded by lottery. In an ongoing group, thresholds after the last observed period are unknown; there, a Kaplan-Meier estimator by asset category, weighting each group by its sampling weight times its size and censoring ongoing groups at their last observed period, gives the probability that b wins in each later twentieth, given that it has not won yet. This applies to 3% of group and bid combinations.

With h_t^L and h_t^B the probabilities of a lottery and a bid award in month t given no earlier award, the contract is awarded in month t with probability

$$a_t = \prod_{s<t} \left(1 - h_s^L\right) \left(1 - h_s^B\right) \left[1 - \left(1 - h_t^L\right) \left(1 - h_t^B\right)\right], \tag{E.1}$$

of which a share $\theta_t = h_t^B / (h_t^L + h_t^B)$ is a bid award. With $V_t(b)$ the value of an award in month t with bid b paid, the expected return is $\sum_t a_t \left[\theta_t V_t(b) + (1 - \theta_t) V_t(0)\right]$. Never bidding sets $h_t^B = 0$.

The statistics in Table 1 are computed from a_t group by group and averaged over each asset category's groups, weighting each group by its sampling weight times its size, so that they describe the average contract.

F The consórcio value

This appendix derives a closed form for the consórcio value V_i of Equation (1).

F.1 Setup

Consider a contract with credit P, a term of T months, and administration fee τ, awarded in month $t^A \leq T$, and let $n := T - t^A$. We drop the contract subscript i until it is needed. The participant pays $c^{pre} := (1+\tau)P/T$ in each month up to the award. A bid of $b(1+\tau)P$, with $0 \leq b \leq 1 - t^A/T$, is paid at the award, and the remaining balance, $(1+\tau)P\,(1 - t^A/T - b)$, is paid in n equal installments c^{post}. The payment stream is therefore

$$c_t = \begin{cases} c^{pre}, & t < t^A, \\ c^{pre} + b(1+\tau)P, & t = t^A, \\ c^{post}, & t > t^A, \end{cases}$$

and the credit P is received at t^A. Let r^S and r^B be the annual savings and borrowing rates, with $0 < r^S \leq r^B$, and $\rho^x := (1+r^x)^{1/12} - 1$ their monthly equivalents. For $x \in \{S, B\}$, $A^x(k) := [1 - (1+\rho^x)^{-k}]/\rho^x$ is the present value of BRL 1 a month for k months at ρ^x. Finally, $y^+ := \max\{y, 0\}$.

F.2 The value

The market replication deposits the contract's own payments up to the award, bid included, in a savings account. At the award, the fund holds

$$F := \sum_{t=1}^{t^A} c_t\,(1+\rho^S)^{t^A - t},$$

and the shortfall is $L := P - F$. If $L > 0$, the replication borrows L at r^B and repays it in n level payments $\mathrm{PMT} := L/A^B(n)$.[35] If $L \leq 0$, it keeps the excess $-L$, and $\mathrm{PMT} := 0$. Discounting both options at ρ^S,

$$\mathrm{NPV}^{\mathrm{con}} := P(1+\rho^S)^{-t^A} - \sum_{t=1}^{T} c_t (1+\rho^S)^{-t},$$

$$\mathrm{NPV}^{\mathrm{mkt}} := \big[P + (F-P)^+\big](1+\rho^S)^{-t^A} - \sum_{t=1}^{t^A} c_t (1+\rho^S)^{-t} - \sum_{t=t^A+1}^{T} \mathrm{PMT}\,(1+\rho^S)^{-t},$$

[35] $L > 0$ requires $n \geq 1$: at $t^A = T$ the fund holds at least the total payments, $(1+\tau)P \geq P$.

and $V := \text{NPV}^{\text{con}} - \text{NPV}^{\text{mkt}}$. The deposits are worth $F(1+\rho^S)^{-t^A}$, and the loan payments are worth PMT $A^S(n)$ at the award. Since $P + (F-P)^+ - F = L^+$, the market replication has NPV

$$\text{NPV}^{\text{mkt}} = \underbrace{-\left(\frac{A^S(n)}{A^B(n)} - 1\right)}_{\text{Additional borrowing cost per BRL borrowed}} L^+(1+\rho^S)^{-t^A},$$

and hence

$$V = \text{NPV}^{\text{con}} + \left(\frac{A^S(n)}{A^B(n)} - 1\right) L^+(1+\rho^S)^{-t^A}. \tag{F.1}$$

Therefore, V is the net present value of consórcios, plus any financial benefits of borrowing through a consórcio instead of the market. When $r^B = r^S$, there is no benefit of borrowing from the market at the opportunity cost r^S, and V is the NPV at r^S. Because r^B enters Equation (F.1) only through $A^B(n)$, which increases with r^B, V is strictly increasing in r^B when $L > 0$ and does not depend on r^B when $L \leq 0$. A break-even borrowing rate is therefore unique when it exists, and undefined for contracts whose savings cover the credit.

F.3 Implementation

For each contract, we compute V_i from Equation (F.1). It depends only on the contract's own terms (P_i, T_i, τ_i, b_i) and award date, not on other members, so it can be measured for early recipients in ongoing groups, whose awards are already observed. It is homogeneous of degree one in P_i, so we report it per BRL of credit.

G Treatment video and survey questions

G.1 Treatment videos

The information treatments are short videos with a Portuguese voice-over produced by the research team. Arms *D* and *DG* share a common set of nine scenes (approximately 90 seconds). The *DG* arm video adds four scenes (approximately 30 seconds), for a total runtime of about two minutes. Each arm is tailored for asset categories (real estate, cars, and motorcycles) with identical scenes and narration except for the asset icon and the asset-specific statistics, all of which are computed from the Central Bank administrative data. Videos were rendered in a vertical format optimized for mobile screens and embedded directly in the endline survey.

Table G.1 lists the scenes. Below, we describe each scene's visuals and reproduce the narrated script verbatim (in Portuguese, with English translations). The narration quoted is from the motorcycle cut; where the cuts differ, the car and real-estate values are indicated.

Table G.1: Scenes of the treatment videos

Scene	Content	Arms
2	The common fund	D, DG
3	Two award mechanisms: lottery and bid	D, DG
4	Bid share of awards in the first half	D, DG
5	Bidding crowds the lottery	D, DG
6	The bid–lottery trade-off	D, DG
7	Timing of lottery awards	D, DG
8	Recap: think about your goal	D, DG
G1	Groups come in different sizes	DG
G2	One lottery per month: group size and bid share	DG
G3	Lottery chances by group size	DG
G4	Group sizes over time	DG

Base module (arms *D* and *DG*)

Scene 1. Disclaimer. An opening card shows a data icon next to the asset icon (house, car, or motorcycle). On-screen text only: *"As informações deste vídeo usam dados reais do Banco Central do Brasil."* ("The information in this video uses real data from the Central Bank of Brazil.")

Scene 2. The common pool. Twenty participant figures each send a coin, representing the monthly installment, into a money bag labeled *Fundo Comum* (common pool), under the caption *"Como funcionam as contemplações de crédito?"* ("How do awards work?").

"No consórcio, todo mês as parcelas dos participantes formam o fundo comum. Todas

as contemplações de crédito usam o dinheiro desse fundo."
("In a consórcio, every month the participants' installments form the common fund. All credit awards use the money from this fund.")

Scene 3. Two award mechanisms. The pool divides into two: a teal path leading to a *Sorteio* (lottery) badge and a gold path leading to a *Lance* (bid) badge. One figure is awarded through lottery and four through bids, reflecting true data.

"Tem dois jeitos de ser contemplado: por sorteio, ou por lance. A administradora decide a regra de contemplação. Sempre tem pelo menos um sorteio. E, normalmente, o restante é por lance."
("There are two ways to be awarded: by lottery or by bid. The administrator decides the award rule. There is always at least one lottery. And, normally, the rest is by bid.")

Scene 4. Bid share of awards in the first half. The five awarded figures regroup into a donut chart titled *"Lance ou sorteio: qual acontece mais?"* ("Bid or lottery: which happens more?") for the first half of the group, splitting awards into a gold bid slice and a teal lottery slice with asset-specific shares: 86%/14% (motorcycles), 82%/18% (cars), 76%/24% (real estate).

"Costuma ter muito mais contemplações por lance do que sorteio. Na primeira metade de um grupo de motos típico, quase 86% das contemplações são por lance, e só 14% são por sorteio."
("There are usually many more awards by bid than by lottery. In the first half of a typical motorcycle group, almost 86% of awards are by bid, and only 14% are by lottery.")

Scene 5. Bidding crowds the lottery. Two conveyor belts of participants move toward a *Crédito* (credit) badge: a gold bid lane designed to move four times faster than a teal lottery lane, illustrating that the queue for bid awards is faster. Figures from the bid lane periodically also move to the lottery award, showing that intending bidders compete in the lottery as well.

"Já que tem muito lance, o sorteio acaba demorando. Até porque quem pretende dar lance também compete no sorteio enquanto não for contemplado."
("Since there is a lot of bidding, the lottery ends up taking long — not least because those who intend to bid also compete in the lottery until they are awarded.")

Scene 6. The bid–lottery trade-off. Monthly award slots rebalance on screen from four bid slots and one lottery slot to two of each: with more lottery awards, the lottery queue absorbs waiting participants faster, but there is less room for bid awards.

> *"A proporção de contemplações por sorteio e por lance é importante. Se diminuem as contemplações por lance, e aumentam as por sorteio, a fila do sorteio anda mais rápido. Mas ao mesmo tempo, fica mais difícil contemplar por lance."*
> ("The proportion of awards by lottery and by bid matters. If bid awards fall and lottery awards increase, the lottery queue moves faster. But at the same time, it becomes harder to be awarded by bid.")

Scene 7. Timing of lottery awards. Twenty figures first spread out evenly along a timeline under the card *"Exemplo: 100% sorteios"* ("Example: 100% lotteries"). They are then pushed toward the end of the timeline into a pyramid concentrated near the final periods, under the card *"Média do mercado hoje"* ("Today's market average"). Two captions quantify the message with segment-specific values: the chance of an early lottery award (*"Chance de contemplar — 2 a 3 em cada 20"*; "Chance of being awarded — 2 to 3 in every 20"; 3 in 20 in the car and real-estate cuts) and, with a shaded region beyond the segment-specific median, *"Metade só perto do final."* ("Half only near the end.").

> *"Imagine o seguinte: se não tivesse lance, os sorteios seriam todos distribuídos igualmente ao longo do tempo. Agora, como os grupos contemplam muito por lance no começo, então a maioria das contemplações por sorteio acontecem na parte final do grupo. Por causa disso, contemplar por sorteio no começo do grupo é bem raro: a cada 20 pessoas, só 2 a 3 vão ser contempladas por sorteio na primeira metade do grupo. E quem depende de sorteio costuma esperar bastante: mais da metade das pessoas que dependem de sorteio só vão ganhar bem tarde."*
> ("Imagine the following: if there were no bids, the lotteries would all be distributed evenly over time. Now, since groups award a lot by bid at the beginning, most lottery awards happen in the final part of the group. Because of this, being awarded by lottery at the beginning of the group is quite rare: out of every 20 people, only 2 to 3 will be awarded by lottery in the first half of the group. And those who depend on the lottery usually wait a long time: more than half of the people who depend on the lottery will only win quite late.")

Scene 8. Recap: think about your goal. Two summary cards appear under the title *"Antes de começar o consórcio, pense no seu perfil."* ("Before starting a consórcio, think about your profile."): a *Lance* card ("more awards"; "brings the credit forward, but you need to pay") and a *Sorteio* card ("few awards at the beginning"; "can take quite long").

"Antes de entrar em um consórcio, pense no seu objetivo. O lance pode antecipar o crédito, mas precisa de dinheiro na mão ou usar parte do próprio crédito como lance. Quem depende só do sorteio não paga lance, mas pode esperar bastante."
("Before joining a consórcio, think about your goal. A bid can bring the credit forward, but it requires money on hand or using part of the credit itself as the bid. Those who depend only on the lottery pay no bid, but may wait a long time.")

Scene 9. Attribution and closing disclaimer. On-screen text only, no narration. Five attribution lines appear in sequence: *"Dados obtidos junto ao Banco Central do Brasil. Estimativas preparadas pela equipe de pesquisa. Estimativas consideram sorteios apenas de cotas ativas. As estimativas exibidas podem variar por grupo e administradora. O Banco Central não é associado a essa pesquisa, apenas é a fonte dos dados."* ("Data obtained from the Central Bank of Brazil. Estimates prepared by the research team. Estimates consider lotteries of active contracts only. The estimates shown may vary by group and administrator. The Central Bank is not associated with this research; it is only the source of the data.") A final warning card closes the video: *"Sempre se informe e pense por conta própria antes de tomar uma decisão financeira."* ("Always inform yourself and think for yourself before making a financial decision.")

Group-size module (arm *DG* only)

Scene G1. Groups come in different sizes. Under the title *"O tamanho do grupo é importante."* ("Group size matters."), two groups appear side by side: a small row of participants labeled *Grupo Menor* (smaller group) and a large grid of participants labeled *Grupo Maior* (larger group).

"Para escolher melhor, é bom pensar no tamanho do grupo. Tem grupos de consórcios menores e grupos maiores."
("To choose better, it is good to think about group size. There are smaller consórcio groups and larger ones.")

Scene G2. One lottery per month: group size and bid share. Month-level award slots are shown for each group: the small group has one lottery slot and two bid slots, while the large group has one lottery slot and nine bid slots — the single monthly lottery is diluted in larger groups.

"Quanto maior o grupo, maior costuma ser a proporção de lances, especialmente no começo. Isso porque os grupos costumam ter só um sorteio por mês."
("The larger the group, the larger the share of bids tends to be, especially at the beginning. That is because groups usually have only one lottery per month.")

Scene G3. Lottery chances by group size. In each group, one participant is highlighted as the monthly lottery winner. Because both groups draw one winner, the smaller group's members face better odds, summarized on screen as *"3 vezes mais chance"* ("3 times higher chance") under the smaller group.

> *"Por causa disso, em grupos menores, a chance de ganhar por sorteio costuma ser maior."*
> ("Because of this, in smaller groups the chance of winning by lottery tends to be higher.")

Scene G4. Group sizes over time. A line chart titled *"Os grupos estão cada vez maiores"* ("Groups are getting bigger and bigger") traces the average group size of newly started groups in the viewer's segment from 2010 to 2024, computed from the Central Bank data. Value labels pop in at 2012, 2016, 2020, and 2024, showing that the average group size roughly doubled over the period.

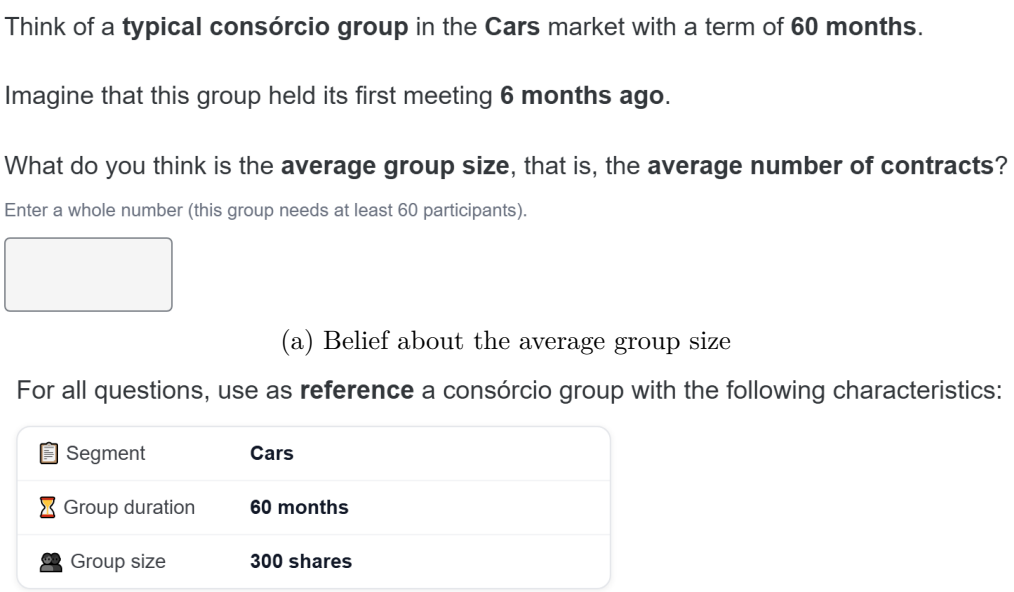
> *"Nos últimos anos o tamanho médio dos grupos vem aumentando. Então, é cada vez mais importante pensar nisso."*
> ("In recent years, the average group size has been increasing. So it is increasingly important to think about this.")

Active control (arm C)

The control arm watched an existing informational video produced by the Central Bank of Brazil, in which a presenter explains basic institutional features of consórcios that we expect consumers to already be familiar with. The video is of comparable length to the treatment videos. It contains no information about award probabilities, bidding shares, lottery wait times, or group sizes, and we verify that it does not move beliefs by comparing Wave 1 and Wave 2 beliefs within the control group.

G.2 Belief questions

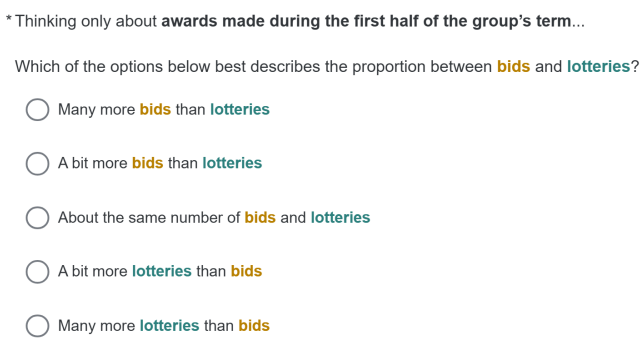
Figures G.1–G.3 reproduce the Wave 1 belief questions as they appear in the survey. The version below is a translated English version, as respondents answered in Portuguese. Each question refers to a scenario built from the respondent's earlier answers: the asset category and the contract duration are those the respondent chose, and the group size in the reference group is randomized across respondents as a multiple of the duration. The screenshots show one such scenario, a 60-month car contract in a group of 300.



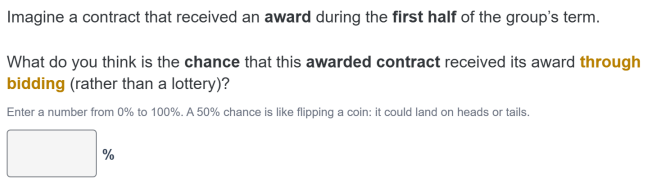
(a) Belief about the average group size

(b) Reference group for the bid-share questions

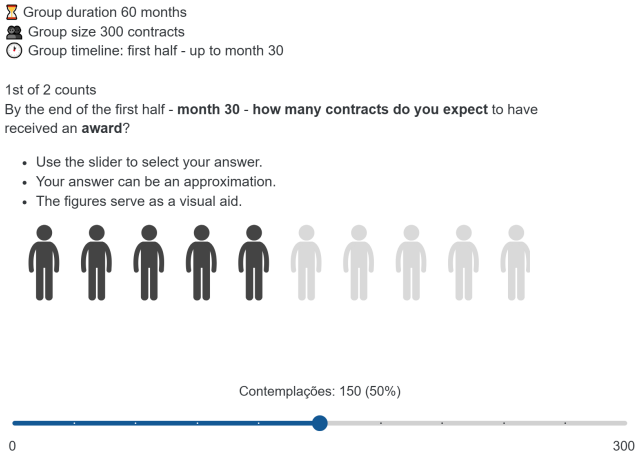
Figure G.1: Group-size question and reference group.



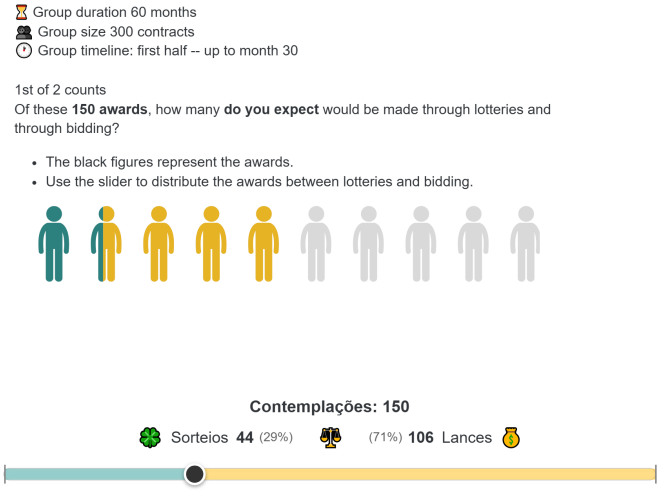
(a) Qualitative



(b) Probabilistic



(c) Quantitative, step 1: number of awards



(d) Quantitative, step 2: lotteries versus bids

Figure G.2: Bid-share questions. In the quantitative question, respondents first choose how many contracts in the reference group are awarded by the midpoint of the contract, then use a slider to split those awards between lotteries and bids.

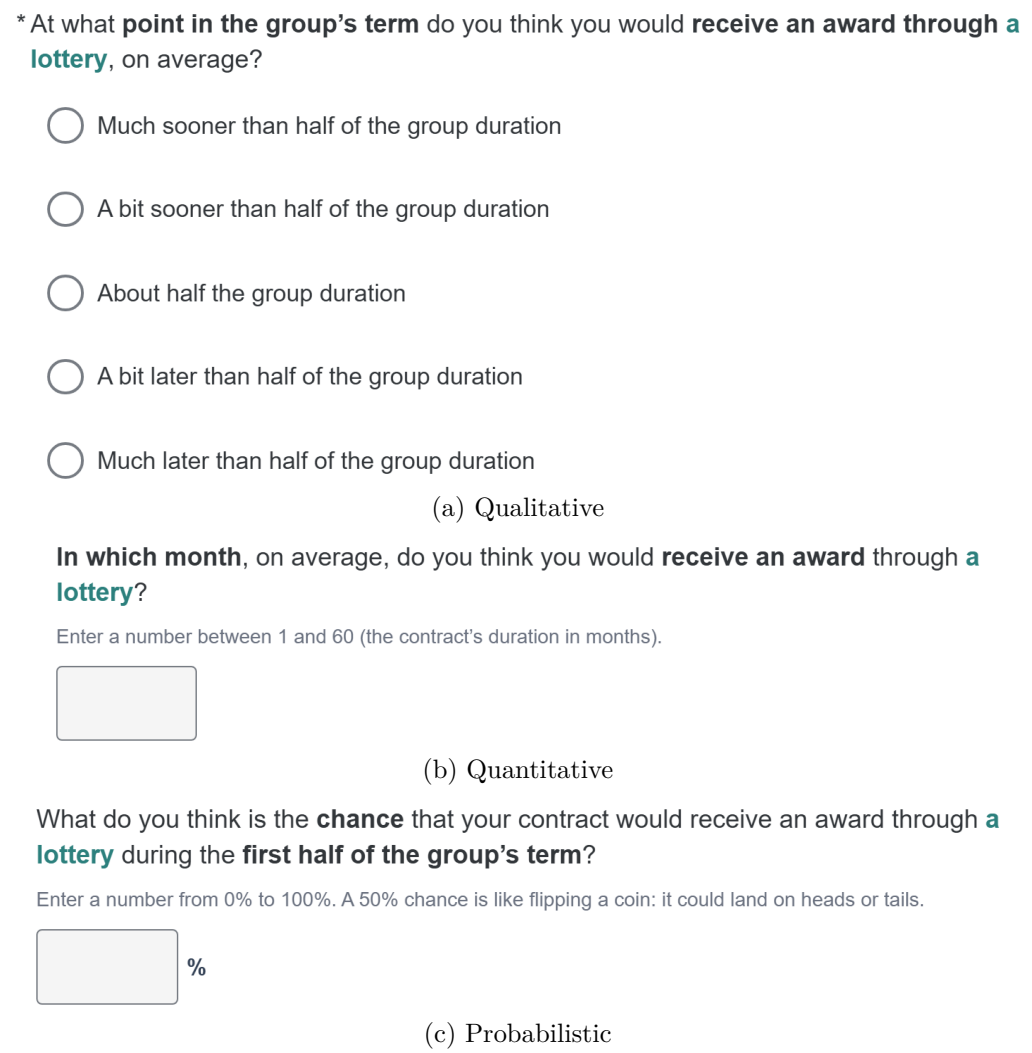


Figure G.3: Lottery-wait questions. Respondents answer assuming they never bid and wait for a lottery award.

H A model of a competitive consórcio market

Section 4 examines one consumer facing one contract. In this appendix we close the model: we add bidders, describe how a group turns installments and bids into awards, and let competitive firms choose the contract. The purpose is to show which contracts survive competition, and in particular when the lottery share $\ell < 1$ that Section 3 documents can be an equilibrium. Two periods of consumption are enough for that question, so the market model has two rather than three. The third period in Section 4 serves to discuss the commitment result.

H.1 Setup

Consider a unit mass of consumers who want to acquire a lumpy indivisible good with normalized price 1, and $N \geq 2$ firms indexed by j, each offering one consórcio group. There are three periods. Period 0 is a decision period; income and consumption occur at $t = 1, 2$. Each consumer joins at most one group or takes an outside option.

Consumers differ in liquidity. There are two types, $\theta \in \{H, L\}$, with masses μ_H and μ_L and per-period income y_θ, with $\frac{1}{2} < y_L < y_H < 1$, so that no consumer can buy the good outright at $t = 1$ but either can pay for it over two periods. Owning the good at $t = 1$ yields a flow $\xi > 0$, and owning it by $t = 2$ yields a terminal value Δ, the value of the good after the model ends. Let $v(\cdot)$ denote utility from nondurable consumption c_t, strictly increasing and strictly concave, with $v''' \geq 0$, $v(0)$ finite and $v'(0) = \infty$. Let $d_t \in \{0, 1\}$ denote ownership at t, and let period utility be $u_1 = v(c_1) + \xi d_1$ and $u_2 = v(c_2) + \Delta d_2$. Consumers are quasi-hyperbolic discounters: the self at date t maximizes $u_t + \beta \sum_{s>t} u_s$, $\beta \in (0, 1]$, taking her later selves' choices as given. The self at $t = 0$ has no consumption and weights $t = 1$ and $t = 2$ equally.

A consumer can save from $t = 1$ to $t = 2$ at rate $r^S = 0$, and can borrow at $t = 1$ at rate $r^B_\theta \geq r$, where r is the market rate; $r^B_\theta = \infty$ is credit exclusion. Outside the group, the $t = 1$ self can borrow and buy at once, if credit is available, save and buy at $t = 2$, or not buy. Let $\bar{U}_\theta$ be the $t = 0$ self's value of the plan her $t = 1$ self carries out, in $t = 1$ plus $t = 2$ utility.

The contract. Each firm chooses a contract (τ, ℓ): a proportional fee $\tau \geq 0$ and a lottery share $\ell \in [0, 1]$, the fraction of period-1 awards allocated by lottery. A member pays an installment $q(\tau) = (1 + \tau)/2$ in each period, of which $1/2$ enters the group's fund and $\tau/2$ is the fee. We write $c_\theta \equiv y_\theta - q(\tau)$ for consumption net of the installment, and assume $y_L > q(\tau)$ for every contract considered, so that installments are affordable. A member who does not pay the first installment leaves at no cost.

A member may also submit a bid $b \geq 0$, measured in principal. A winning bid prepays principal b into the fund in period 1 together with the fee on it, τb, so that the member's period-2 installment

falls by $(1+\tau)b$. Submitting a bid is costless and does not exclude the bidder from the lottery. Consumers cannot borrow to bid, so $(1+\tau)b \le c_\theta$.

Group mechanics. Awards in period 1 are allocated in two steps. First, a share ℓ goes by lottery, with equal odds for every member. Then the remaining $1-\ell$ go by a uniform-price auction: the highest bids win and every winner pays the highest losing bid, a cut-off b^*. We call a member whose bid clears, $b \ge b^*$, a *bidder* and any other member a *non-bidder*.

The fund in period 1 holds $s/2$ from the installments of a group of mass s, plus the prepaid bids, and each award costs 1, so the mass of period-1 awards solves $W_1 = s/2 + (1-\ell)W_1 b^*$. Hence

$$W_1 = \frac{s/2}{1-(1-\ell)b^*}, \qquad p(\ell) = \frac{\ell W_1}{s} = \frac{\ell/2}{1-(1-\ell)b^*}, \qquad B = (1-\ell)\, b^* W_1, \tag{H.1}$$

where $p(\ell)$ is a member's probability of a lottery award and B the total volume of winning bids. These equations capture the fact that larger bids add more resources to the fund. Because ℓ is a share, the bid amount also moves the number of period-1 lotteries; in practice the number of lotteries is fixed as an integer. In period 2 the remaining members receive the award, and the fund balances exactly.

Firms. Firms compete à la Bertrand in contracts: they post simultaneously, any firm may post any contract, and consumers then choose. Administering a member costs $c < 1$, incurred as $c/2$ with each installment. In period 1 the firm collects fees on the first installment, $s\tau/2$, and the fee on prepaid principal, τB, and pays $cs/2$. In period 2 it collects the remaining fees, $s\tau/2 - \tau B$, and pays $cs/2$. Discounting period 2 at $1/(1+r)$, profits in period-1 units are

$$\Pi(\tau,\ell) = \underbrace{\left[s\tfrac{\tau}{2} + \tau B - s\tfrac{c}{2}\right]}_{\text{period 1}} + \frac{1}{1+r}\underbrace{\left[s\tfrac{\tau}{2} - \tau B - s\tfrac{c}{2}\right]}_{\text{period 2}} = \underbrace{s\,\tfrac{\tau-c}{2}\,\tfrac{2+r}{1+r}}_{\text{fee revenue net of cost}} + \underbrace{\tau\,\tfrac{r}{1+r}\,B}_{\text{extra bid revenue}}. \tag{H.2}$$

Bids do not change total fees. They move τB of fee revenue from period 2 to period 1, worth $\tau\frac{r}{1+r}B$ to the firm. Profits are linear in s, so groups operate under constant returns.

Beliefs. As in Section 4, consumers do not observe the odds or ℓ. A consumer evaluating a contract anchors on the odds of a pure-lottery group, here $p(1) = 1/2$, and adjusts a fraction $\lambda \in [0,1]$ for the bid awards,

$$\hat{p}(\ell) = \tfrac{1}{2} + \lambda\big[p(\ell) - \tfrac{1}{2}\big]. \tag{H.3}$$

Beliefs are correct at $\lambda = 1$ and, for any λ, at $\ell = 1$. Both selves hold the same beliefs. Firms know the true odds, and beliefs do not affect what firms collect.

Timing. First, each firm stays out or posts one contract (τ_j, ℓ_j). Second, at $t = 0$, consumers observe the menu, form beliefs by (H.3), and each chooses a firm or the outside option, randomizing with strictly positive probabilities when indifferent. Third, at $t = 1$, each member's self decides whether to pay the first installment, and (H.1) allocates period-1 awards. Fourth, at $t = 2$ the remaining members pay and receive the good. We study subgame-perfect equilibria with sophisticated consumers.

Two assumptions are of note. The first assumption assumes the indivisible good is worth buying. The second assumption sorts the two types into bidders and non-bidders.

Assumption H1 (Durable worth its price)**.** *The terminal value is the flow capitalized at the market rate, $\Delta = \xi\,(1+r)/r$, and the flow covers the interest on the price at every contract considered,*

$$\xi \;\geq\; r\,v'(c_\theta) \qquad \textit{for both types and every } \tau \textit{ considered.} \tag{H.4}$$

The second part says the good is worth its price at the market rate: $\Delta \geq (1+r)\,v'(c_\theta)$, so that a member buys it at the margin out of her consumption net of the installment. The results below depend on this assumption.

Two observations follow. First, in this model, consumers join a consórcio because they want a lumpy indivisible good they cannot buy at once. If the flow utility from owning the good were not worth its price, the consumer would rather save at r, removing the reason to enter a consórcio.

Second, without Assumption H1, Δ is free, and a good whose flow falls short of the interest on its price could still be bought by assumption.

Assumption H2 (Sorting by credit)**.** *For an H consumer, at every break-even contract with $\ell < 1$, a seat beats her outside option and her outside option beats the lottery alone,*

$$2v(c_H) + \Delta + \hat{p}(\ell)\,\xi \;<\; \bar{U}_H \;\leq\; 2v(c_H) + \Delta + \hat{p}(\ell)\,\xi + \big(1 - \hat{p}(\ell)\big)\,\rho_H, \tag{H.5}$$

where $\rho_H > 0$ is the $t = 0$ value of clearing at the L bid rather than waiting, and her $t = 1$ self pays when seated.

Since $\bar{U}_H$ falls in r_H^B, Assumption H2 is a band on the H type's borrowing rate: she needs credit cheap enough that she will not join only for the lottery odds, but expensive enough that a zero-interest loan from others in the group is worth paying the fee. It sorts the group. By Lemma H2 below, $\bar{b}_H > \bar{b}_L$, so an H earns a rent only when the bid slots cover the H lottery losers, $(1-\ell)W_1 \geq s_H(1-p(\ell))$; if they do not, the auction clears at $\bar{b}_H$, the rent is zero, and every H prefers to leave.

The direct consequence is that, in every active group with $\ell < 1$, the cut-off is the L bid, $b^* = \bar{b}_L(\tau)$, every H member is a bidder and every L member is a non-bidder. A group without

non-bidders cannot seat anyone, since it would need $(1-\ell)b^* \geq \frac{1}{2}$ and $b^* < \frac{1}{2}$: this is why bidders follow non-bidders. At $\ell = 1$ there is no seat, so by the lower inequality no H joins a pure lottery. That group is all-L, and H borrows. The pure-lottery result in Proposition H1 does not use this assumption, but the equilibrium's bidders' rent does.

H.2 Discussion

We now discuss specific elements of the model. Some are motivated by observed characteristics of the consórcio market, while others simplify the environment.

- *Uniform fees.* Fees are the same for every member of a group, as is usual in consórcios. A single group cannot discriminate between types ex ante.

- *Concave utility and present bias.* With concave utility the bid trades nondurable consumption today against early access, and present bias adds a second cost: the $t = 1$ self discounts what the prepayment returns tomorrow. Prudence, $v''' \geq 0$, makes the bid increase in income; concavity alone does not.

- *Durable with constant utility.* The constant flow implies bidding is increasing in income. As in Besley et al. (1993), this can be relaxed to $\xi(c)$ with $\xi'(c) > 0$, so that the good and nondurable consumption are complements. If they are sufficiently substitutable, optimal bidding and income can be negatively related, and bidding differences between L and H would be given solely by the cash constraint. The terminal value Δ at $t = 2$ can also be relaxed, in which case the good provides utility ξ in each period: what matters is that the good is indivisible, not that it is durable; in Basu (2011) the good is not a durable.

- *Competitive firms, free entry, constant marginal cost.* There are at least two firms, any firm may post any contract, and a group of any mass operates at constant cost. By a standard Bertrand argument, with constant marginal costs any contract accepted with positive probability earns zero profit, and competition for members forces at least two firms to offer the contract that maximizes the perceived utility of the members the group needs. With convex costs, firms can earn positive profits and equilibria need not be unique. However, convex costs push firms to offer smaller groups, which does not change our main result.

- *Uniform-price auction.* Every winner pays the highest losing bid. The assumption removes strategic bidding: with a uniform price each member's best bid is her reservation value, and the results extend to a first-price rule whenever shading preserves the ranking of bids. In practice auctions are first-price. The uniform price also creates rents from bidding: every H member pays L's reservation bid and keeps the difference to her own.

- *Reduced-form beliefs.* The value of λ can stand for myopia, low salience of lottery levels, low observability of lottery levels (e.g., shrouding), probability weighting, or the outcome of a disclosure game between firms and consumers.

- *No borrowing to bid.* With borrowing, the bid would be set by arbitrage instead of liquidity. In practice, borrowing to bid would be more expensive than r^B, since it would not be collateralized debt. In our survey, among bid winners, only 6% report borrowing from a bank to bid.

H.3 Consumer choices

Autarky. Without credit, the $t = 1$ self who wants the good at $t = 2$ saves $s \in [1 - y, y]$ and consumes $y - s$ now and $y + s - 1$ then.

Lemma H1 (Saving under present bias)**.** *Let $r^B = \infty$. If the $t = 1$ self saves for the good, she saves $s^*(\beta) = \max\{\tilde{s}(\beta), 1 - y\}$, where $\tilde{s}$ solves $v'(y - s) = \beta\, v'(y + s - 1)$; s^* is nondecreasing in β and $s^*(1) = 1/2$. There is a threshold $\underline{\beta}(y) \in (0, 1)$ such that she saves for the good iff $\beta \geq \underline{\beta}(y)$ and does not buy it otherwise. $\underline{\beta}$ falls in y and in Δ.*

A present-biased self saves less than her $t = 0$ self would like, and below $\underline{\beta}$ she gives up the good. This is the saving friction the contract can substitute for: the installment is forced saving, and Lemma H3 gives the band of β for which the $t = 1$ self accepts it.

Bidding. A bid matters only when the member has lost the lottery and her bid clears, and submitting a bid is costless. Thus, the bid compares a payment of $(1 + \tau)b$ today to obtain the good now, or wait and get the good tomorrow for sure. Her reservation bid $\bar{b}_\theta$ is the price at which she is indifferent, or the cash cap $c_\theta/(1 + \tau)$.

Lemma H2 (Optimal bid)**.** *The optimal bid $\bar{b}_\theta$ is unique, and bidding it is weakly dominant in the uniform-price auction. It is weakly increasing in ξ, β and y_θ, strictly in y_θ when $\beta < 1$ or $v''' > 0$, and weakly decreasing in τ.*

Under the assumptions above, $\bar{b}_L \leq \bar{b}_H$, strictly when $\beta < 1$ or $v''' > 0$; we take the strict ranking throughout, which fails only for quadratic utility without present bias.[36] Knowing her type, a consumer knows whether her bid will clear: under Assumption H2 the cut-off is $b^* = \bar{b}_L$, H members clear with a strict rent, and L members are indifferent.

[36]The ranking $\bar{b}_L \leq \bar{b}_H$ uses a common ξ. If the flow gains from owning the durable differed by type, ξ_θ, then if $\xi_L > \xi_H$, even an illiquid consumer could have the higher reservation bid, since $\bar{b}_\theta$ rises in ξ. When the cap binds, the bid is defined by liquidity, not valuation. Bidding then sorts by liquidity even when the illiquid type wants the good more.

The $t = 0$ self weights the prepayment's return in full and prefers a larger bid. A present-biased member may then face two commitment problems inside the contract: the $t = 1$ self is tempted to cancel, but also she may bid less than her $t = 0$ self wanted.

Entry. Fix a contract (τ, ℓ). Entry needs two conditions: the $t = 1$ self must prefer paying the installment to her best alternative at $t = 1$, and the $t = 0$ self, anticipating this, must prefer the contract to the outside option. For a non-bidder,

$$\underbrace{\hat{p}(\ell)\,\xi}_{\substack{\text{perceived chance}\\ \text{of early access}}} \;\geq\; \underbrace{c_{1,\theta}(\tau) - \beta\, g_{2,\theta}}_{\text{cost today net of discounted gain}}, \tag{H.6}$$

$$\underbrace{2v(c_\theta) + \Delta + \hat{p}(\ell)\,\xi}_{\text{contract at } t=0} \;\geq\; \bar{U}_\theta, \tag{H.7}$$

where $c_{1,\theta}$ is the installment's utility cost today relative to her alternative and $g_{2,\theta}$ the future gain of staying.[37] Condition (H.6) is the commitment problem: at $\beta = 1$ it is implied by (H.7), and at $\beta < 1$ the $t = 1$ self is tempted to leave. Condition (H.7) nests the usual accounts of rotating groups. Without credit and at $\tau = 0$ the odds of early access are worth joining for, as in Besley et al. (1993); with $\beta < \underline{\beta}(y)$ the installment is forced saving the $t = 0$ self wants, as in Basu (2011); under misperceptions, $\hat{p}(\ell) > p(\ell)$ makes a consumer join when saving or borrowing was better.

A bidder obtains early access for sure and, when the lottery does not select her, prepays the clearing bid. Her conditions are (H.6) and (H.7) with her expected rent from clearing at b^* added on the left: the probability of losing the lottery, $1 - \hat{p}(\ell)$, times the rent as the $t = 1$ self values it in the first and as the $t = 0$ self values it in the second. Both are positive for $\bar{b}_\theta > b^*$.

Lemma H3 (Participation band)**.** *Let $r^B_\theta = \infty$ and $\beta < \underline{\beta}(y_\theta)$, so that the consumer would not save alone. At a contract (τ, ℓ) the $t = 1$ self pays iff $\beta \geq \hat{\beta}_\theta(\tau, \ell)$, where*

$$\hat{\beta}_\theta(\tau, \ell) \equiv \frac{v(y_\theta) - v(y_\theta - q(\tau)) - \hat{p}(\ell)\,\xi}{\Delta - v(y_\theta) + v(y_\theta - q(\tau))}. \tag{H.8}$$

Where $\hat{\beta}_\theta > 0$, the threshold rises in τ and, for $\ell < 1$, in λ, and falls in ℓ, ξ, y_θ and Δ.

This is the two-period counterpart of Proposition 1. The lottery helps the $t = 1$ self to keep paying: each bid slot lowers her perceived odds and raises $\hat{\beta}_\theta$, so the most present-biased members are the first to leave when bidding rises, and the fee works the same way. Misperception lowers $\hat{\beta}_\theta$ by raising $\hat{p}$ above p, so it sustains at $t = 1$ the bidding contracts it makes attractive at $t = 0$.

[37] For a consumer who would not save alone, $c_1 = v(y) - v(y - q)$ and $g_2 = \Delta - c_1$. For one who would save s^*, $c_1 = v(y - s^*) - v(y - q)$ and $g_2 = v(y - q) - v(y + s^* - 1)$. For one who would borrow at $t = 1$, the alternative is the borrowing plan.

H.4 Firms and the break-even fee

Free entry and constant returns force every active contract to break even: any contract with positive profit is undercut at the same ℓ and a lower fee. Setting $\Pi = 0$ in (H.2) with B from (H.1),

$$\tau(\ell) = \frac{c}{D(\ell)}, \qquad D(\ell) \equiv 1 + \frac{r}{2+r}\,\frac{(1-\ell)\,b^*}{1-(1-\ell)\,b^*}, \tag{H.9}$$

with $D(\ell) \geq 1$ and $D(1) = 1$, so $\tau(1) = c$: a pure lottery is priced at cost. D falls in ℓ and rises in b^*, so $\tau'(\ell) > 0$: bidding collects fees earlier and supports a fee below cost, the only compensation a firm can offer non-bidders since fees are uniform. A firm poaches when it posts another break-even contract that some members of an existing one strictly prefer.

H.5 Equilibrium

An equilibrium is a vector of contracts offered by each firm, a consumer acceptance decision at $t = 0$, and a payment decision at $t = 1$ with the following properties: (i) each firm's contract maximizes its expected profit given the other contracts; (ii) each consumer chooses the contract that maximizes her perceived utility, the left side of (H.7) or of its bidder version, randomizing with strictly positive probabilities when several contracts give the same perceived utility, or takes the outside option if none dominates it; (iii) each member's $t = 1$ self chooses whether to pay to maximize her continuation utility, so (H.6) holds on every active contract. Within each contract b_j^* clears the auction given the members it attracts. When more H consumers demand a contract's seats than it has, seats are rationed at random and a rationed consumer takes her next option. As usual, equilibrium pins down the contract accepted but not which firm serves whom: with more than two firms there are equilibria in which some firms offer contracts that are never accepted, but the set of contracts accepted with positive probability is unique, and we refer to it as the equilibrium contract. Non-bidders are necessary to hold the group together: their installments fund the early awards, and they are the side an entrant can peel off with a better offer. Bidders obtain early access with a rent at any break-even contract.

H.6 Results

The cut-off. By Lemma H2 and Assumption H2, the cut-off in a group with fee τ is the non-bidder's reservation bid, $b^*(\tau) = \bar{b}_L(\tau)$, a function of the posted contract. Along the break-even locus it is $b^*(\tau(\ell))$, and (H.9) is then a fixed point in τ at each ℓ, $\tau = c/D(\ell, \bar{b}_L(\tau))$. The right side rises in τ, since $\bar{b}_L$ falls in τ and D rises in b^*, so uniqueness needs its slope below one. The slope is at most $c\,\frac{r}{2+r}\,|\bar{b}_L'(\tau)|/(1-b^*)^2$ and $|\bar{b}_L'| < 1$ (Remark 1), so $c\,\frac{r}{2+r} < (1-b^*)^2$ suffices: with $b^* < \frac{1}{2}$ it holds for every $c \leq 1$ when $r \leq 2/3$ per period, and at $r = 1$ for $c < 3/4$. We take the break-even

fee to be affordable, $\tau(1) < 2y_L - 1$. Write $c_L(\ell) \equiv y_L - q(\tau(\ell))$ for the non-bidder's consumption along the locus, and $p(\ell)$, $\tau(\ell)$ for the odds and fee along it, with derivatives taken along the locus, cut-off included. The non-bidder's value of the contract is $V_L(\ell) = 2v(c_L(\ell)) + p(\ell)\,\xi$.

Seats. Let s_H and s_L be the masses of each type in a group, $s = s_H + s_L$. Under Assumption H2 every H member receives early access and every L member who loses the lottery waits, so membership satisfies

$$s_H\,(1 - p(\ell)) = (1 - \ell)\,W_1, \qquad \text{so} \qquad h(\ell) \equiv \frac{s_H}{s} = \frac{(1-\ell)\,W_1}{s\,(1-p(\ell))} = \frac{1-\ell}{1 + (1-\ell)(1-2b^*)}\,: \tag{H.10}$$

the H members who lose the lottery exactly fill the bid slots. The share $h(\ell)$ of bidders in the group falls from $1/(2(1-b^*))$ at a pure auction to zero at a pure lottery.

Spillovers. Bid slots change a non-bidder's odds. We call the *spillover* of bidding the flow she loses relative to a pure lottery,

$$S(\ell) \equiv \big[p(1) - p(\ell)\big]\,\xi, \qquad S'(\ell) = -p'(\ell)\,\xi < 0 \;\text{ at a fixed bid.} \tag{H.11}$$

By (H.1) the spillover nets two effects: each slot removes a lottery award, and the prepayment it brings in funds part of another. A consumer with beliefs (H.3) perceives only $\lambda S(\ell)$.

Single crossing. Along the break-even locus, with the cut-off $b^*(\tau(\ell))$, let

$$R(\ell) \equiv \frac{\xi\,p'(\ell)}{\tau'(\ell)\,v'\big(c_L(\ell)\big)} \tag{H.12}$$

be the ratio of the marginal spillover to the marginal fee compensation.

Assumption H3 (Single crossing)**.** *R is strictly decreasing in ℓ.*

Since $V_L'(\ell) = \tau'(\ell)\,v'(c_L(\ell))\,[R(\ell) - 1]$ with $\tau' > 0$, Assumption H3 says the non-bidder's gain from one more unit of lottery share crosses zero at most once, from above: V_L is single-peaked. It holds at a fixed cut-off, where D falls and $v'(c_L)$ rises in ℓ; with the cut-off $\bar{b}_L(\tau(\ell))$ a higher fee lowers the bid and so the slots' value, and the assumption bounds that fee-sensitivity. The same bound keeps $p' > 0$ along the locus, so the spillover is decreasing there as well.

Lemma H4 (Optimal contract for the non-bidder)**.** *Let $\lambda = 1$, let Assumption H3 hold, and fix every parameter other than ξ. There are thresholds $0 < \underline{\xi} < \bar{\xi}$, independent of ξ when the cut-off is held fixed, such that:*

(i) if $\underline{\xi} < \xi < \bar{\xi}$*, the maximizer* $\ell^* \in (0,1)$ *solves*

$$S'(\ell) = -\tau'(\ell)\, v'\big(c_L(\ell)\big), \tag{H.13}$$

the marginal spillover equal to the compensation from fees, and at a fixed cut-off ℓ^* *increases in* ξ*;*

(ii) if $\xi \geq \bar{\xi}$*, then* $\ell^* = 1$*: a pure lottery, since no fee cut compensates the spillover;*

(iii) if $\xi \leq \underline{\xi}$*, then* $\ell^* = 0$*: pure bidding.*

Assumption H4 (Strict participation)**.** *Condition* (H.6) *holds strictly for* L *at* $(\tau(\ell^*), \ell^*)$*, and entry* (H.7) *holds there.*

Early acquisition of the good is valuable, but bidding increases firms' revenues, which under competition are passed through to fees. An L type may therefore find this trade-off worth it. At intermediate values of ξ, the optimal contract exactly offsets the trade-off between a lower lottery chance, the spillover from bidding, and the lower fee and the higher nondurable consumption it allows. If ξ is high enough, however, the fee reduction through the pass-through is not sufficient to compensate the lower lottery chances. Assumption H4 rules out a non-bidder who is exactly indifferent at $t = 1$, for whom a fee ϵ higher would break the contract. At the equilibrium of this game, a group must serve the L types' optimal contract because they are the poachable margin.

Proposition H1 (Benchmark)**.** *Let* $\lambda = 1$.[38] *An equilibrium exists and is unique up to which firm serves whom: bidders' participation constraints are slack, and non-bidders are served by the break-even contract* $(\tau(\ell^*), \ell^*)$ *of Lemma H4. Under Assumption H1,* $\xi > \bar{\xi}$ *and every equilibrium contract is a pure lottery.*

At a fixed cut-off the last claim does not use Assumption H3: under Assumption H1, $R(\ell) > 1$ at every ℓ, so V_L rises on all of $[0,1]$. Single crossing is needed only when the cut-off moves with the fee.

Proposition H1 shows that the full-lottery contract is the equilibrium of the market with accurate beliefs, and that this follows directly from Assumption H1. The reasoning is as follows. L types find the contract worth it because they want to acquire a lumpy, indivisible good. Bidding only increases revenues through the anticipation of bidders' fees, which is priced at the firms' interest rate; under perfect pass-through, the gain in fees is therefore at most interest-rate sized. The same force that drives the demand for the consórcio contract in the first place is the force that makes fee rebates through bidding unable to compensate for the lower probability of acquiring the good. Through this intuition, the result is robust to different formulations of the value of the good.

[38] Assumptions H1, H2, H3 and H4 are maintained throughout: H1 makes the good worth buying, H2 fixes the cut-off and the bidders' rent, H3 enters through Lemma H4, and H4 is used in the poaching step of the proof.

Proposition H2 (Misperceptions)**.** *Hold fixed every other parameter of the model, and let Assumption H3 hold. There are thresholds* $0 < \underline{\lambda} < \bar{\lambda} < 1$ *such that:*

(i) if $\underline{\lambda} < \lambda < \bar{\lambda}$*, the equilibrium contract has* $\ell^* \in (0,1)$*, which solves (H.13) at* $\lambda\xi$ *and increases in* λ*;*

(ii) if $\lambda \geq \bar{\lambda}$*, then* $\ell^* = 1$*;*

(iii) if $\lambda \leq \underline{\lambda}$*, then* $\ell^* = 0$*.*

A consumer with $\lambda < 1$ perceives only a fraction λ of the spillover, so the fee cut needed to retain her is smaller, and a low enough λ sustains bidding where the full-information contract is the pure-lottery corner.

The contract has two prices, in the sense of Gabaix and Laibson (2006). The fee is the "base" or upfront price, and everyone "sees" it because it is reflected in the installment. The odds a member gives up to bid slots, the spillover $S(\ell)$, is a hidden price to non-bidders, and they perceive only λ of it. Competition can push the upfront price below cost, $\tau(\ell) < c$, but require bidding, so the hidden price is paying for the cut. The "sophisticated" action does not necessarily rely on information about lotteries, but rather on liquidity available to bid. An H member who bids obtains early access, and the diluted lottery matters less to her. This is the cross-subsidy from L to H. In Gabaix and Laibson (2006) the sophisticate avoids the add-on because she knows its price, and pays an effort cost to do so. Here the bidder can avoid the hidden price without knowing it. Bidding can be a better deal at any beliefs, since her rent is strict by Lemma H2. What this consumer must know is the cut-off, not the odds. Sophistication about bidding and sophistication about odds are different things, and H can have the first without the second.

Two consequences follow. First, there is no curse of debiasing on the L side: an L member who learns the odds cannot earn the rent by bidding, so she demands more lottery instead, and the equilibrium share rises with λ by Proposition H2. Second, for the mechanism to be sustained, it is important that the bidding side be public and the odds side not. The cut-off must be easy to learn, because bidders act on it. The probability of a lottery award can stay hidden. That is a prediction about what the supply side prefers to disclose. As we have seen in Section 5.3.2, this is exactly what meeting minutes do: winning bids are revealed while odds are not.

H.7 Extensions

H.7.1 Monopoly

Suppose a single firm serves the market. It still faces the entry conditions, the fund identity and the seat structure, but no rival pins the fee. The polar case shows what competition was doing.

Reservation fee. For a lottery share ℓ, the highest fee at which non-bidders enter is the $\tau_r(\ell)$ at which (H.7) binds,

$$2v\big(y_L - q(\tau_r)\big) + \Delta + \hat{p}(\ell)\,\xi = \bar{U}_L, \tag{H.14}$$

provided (H.6) holds there; otherwise the binding constraint is (H.6) and the fee is the one at which the $t = 1$ self is indifferent. Along (H.14), $\tau_r'(\ell) = \lambda\, p'(\ell)\,\xi / v'(c_L)$: the monopolist sells odds to the non-bidder at her marginal valuation in fee units. Write $M \equiv \tau_r(1) - c$ for the markup at the pure lottery.

Profit. With μ_L non-bidders and H members filling the seats by (H.10), the group has mass $s(\ell) = \mu_L\, m(\ell)$, where $m(\ell) \equiv 1/(1 - h(\ell))$. Per-member profit is $\pi(\tau, \ell) \equiv \Pi / s$ from (H.2), and the monopolist solves

$$\max_{\ell \in [0,1]} \; \mu_L\, m(\ell)\, \pi\big(\tau_r(\ell), \ell\big). \tag{H.15}$$

Proposition H3 (Monopoly)**.** *Let the monopolist price at* (H.14) *and hold b^* at its value at the pure lottery. The monopolist does not post a pure lottery if*

$$\underbrace{\frac{\lambda\,\xi\,(1 - b^*)}{2\,v'\big(c_L(\tau_r(1))\big)}}_{\text{reservation fee lost per slot}} \;<\; \underbrace{M}_{\text{profit on the seated bidder}} \;+\; \underbrace{\tau_r(1)\,\frac{r\,b^*}{2 + r}}_{\text{fee anticipation}}. \tag{H.16}$$

If (H.15) *is single-peaked in ℓ, the three cases of Proposition H2 hold with thresholds that are increasing in M. At $\lambda = 1$ the monopolist posts bidding iff M exceeds $\xi(1 - b^*)/(2v'(c_L)) - \tau_r(1) r b^*/(2 + r)$: market power alone sustains bidding under correct beliefs when the markup is large, and misperception lowers the markup needed one for one in fee units.*

The left side of (H.16) is what a bid slot costs the monopolist: the non-bidder values the odds it removes, and the reservation fee falls by that amount in fee units. The right side is what the slot earns: an extra seated bidder who pays the full fee and costs c, which is the markup, and the fee on her prepayment collected early. Under competition the markup is zero and only the anticipation term remains, which Proposition H1 shows is too small; that is why competition needs misperception to produce bidding and a monopolist does not.

Present bias enters the monopoly case through the reservation fee. When (H.6) binds before (H.7), the fee is capped by the $t = 1$ self rather than the $t = 0$ self, τ_r is lower, and so is the markup; the lottery is then what lets the monopolist charge more, and the same corner condition applies with the $t = 1$ self's valuation of odds in place of the $t = 0$ self's.

H.7.2 Bidding with interest

In a bidding RoSCA (Besley et al., 1993) the winner's bid is paid to the members who wait. In a consórcio the bid amortizes the bidder's own installments, and the members who wait receive nothing. This extension adds the first feature as a contract term. A contract is (τ, ℓ, α) with $\alpha \geq 0$. A winning bidder prepays b as before and, in period 2, pays interest αb into a pool divided equally among the members who wait, who receive it as a reduction of their period-2 installment. The consórcio is $\alpha = 0$.

Mechanics. The prepayment still enters the fund, so W_1, $p(\ell)$, B and the break-even locus $\tau(\ell)$ are as in (H.1) and (H.9): interest is a transfer among members and never reaches the firm. The pool is αB and the members who wait have mass $s - W_1$, so each receives

$$\iota(\ell) \equiv \frac{\alpha B}{s - W_1} = \frac{\alpha\,(1-\ell)\,b^*}{1 - 2(1-\ell)\,b^*}, \qquad \iota(1) = 0. \tag{H.17}$$

A non-bidder waits when she loses the lottery, so her value of the contract is

$$V_L(\ell;\alpha) = 2v\big(c_L(\ell)\big) + \hat{p}(\ell)\,\xi + \big(1-\hat{p}(\ell)\big)\Big[v\big(c_L(\ell)+\iota(\ell)\big) - v\big(c_L(\ell)\big)\Big]. \tag{H.18}$$

Interest raises the cost of clearing: a winner's period-2 installment falls by $(1+\tau-\alpha)b$ instead of $(1+\tau)b$, so $\bar{b}_\theta$, the cut-off and the bidder's rent fall in α. Let $\bar{\alpha}$ be the largest interest at which the bidder conditions of Assumption H2 hold; it is positive since $\rho_H > 0$.

Proposition H4 (Bidding with interest)**.** *Hold b^* at its value at the pure lottery.*

(i) If firms choose $\alpha \in [0, \bar{\alpha}]$, no equilibrium contract has bidding without interest: every active contract with $\ell < 1$ has $\alpha > 0$.

(ii) The non-bidder's value rises as ℓ falls from 1 iff the interest exceeds

$$\alpha^*(\lambda) \;\equiv\; \lambda\,\frac{\xi}{v'(c_L(1))}\,\frac{1-b^*}{b^*} \;-\; \frac{2rc}{2+r}. \tag{H.19}$$

Such an interest is feasible iff $\alpha^*(\lambda) < \bar{\alpha}$, and $\alpha^*(\lambda) \leq 0$ iff $\lambda \leq \bar{\lambda} \equiv \bar{\xi}/\xi$.

(iii) α^* increases in λ and in ξ: the more of the spillover the non-bidder perceives, and the more the flow is worth, the more interest a slot must carry. Write $\alpha_1 \equiv \alpha^*(1)$ for the interest that sustains bidding under correct beliefs.

(iv) Equivalently, for a given α, the pure lottery loses to bidding iff $\lambda < \bar{\lambda}^{\text{int}}(\alpha)$, where

$$\bar{\lambda}^{\text{int}}(\alpha) \;\equiv\; \bar{\lambda}\left[1 + \alpha\,\frac{2+r}{2rc}\right] \;=\; \bar{\lambda} + \alpha\,\frac{v'(c_L(1))}{\xi}\,\frac{b^*}{1-b^*} \;>\; \bar{\lambda} \quad \textit{for } \alpha > 0. \tag{H.20}$$

At $\alpha = 0$ it is the threshold of Proposition H2, $\bar{\lambda}^{\text{int}}(0) = \bar{\lambda}$.

If $V_L(\cdot;\alpha)$ is single-peaked, the equilibrium contract is a pure lottery iff $\alpha \leq \alpha^*(\lambda)$, that is iff $\lambda \geq \bar{\lambda}^{\text{int}}(\alpha)$.

The expression compares two interest rates. Without interest a slot compensates the non-bidder, per unit bid, by $2rc/(2+r)$: the firm's return on the fee it collects early, which is below the interest on the cost and far below r. With interest the compensation is $2rc/(2+r) + \alpha$. The factor $(2+r)/(2rc)$ is large, about $1/(rc)$: at $r = 0.10$ and $c = 0.15$ one point of interest raises the threshold by seventy percent.

Under Assumption H1, $\xi/v'(c_L(1)) \geq r$, so $\alpha_1 = \alpha^*(1) \geq r\,[(1-b^*)/b^* - 2c/(2+r)]$, above the market rate unless b^* is near its cap: a slot dilutes the odds on the whole price while the bid prepays only b^* of it. Whether bidders pay that is whether $\alpha_1 \leq \bar{\alpha}$, the heterogeneity condition of a bidding RoSCA: the gap between y_H and y_L must be wide enough for the bidder's rent to cover the interest.

Part (i) says that under competition $\alpha = 0$ is never chosen. Interest is cash the non-bidder perceives in full, and bidders' constraints are slack, so a rival posting $\alpha > 0$ at the same (τ, ℓ) poaches non-bidders, and at a fixed cut-off competition pushes α to $\bar{\alpha}$. In consórcios $\alpha = 0$ is an institutional restriction, as in random RoSCAs and unlike bidding RoSCAs. Under that restriction bidding needs $\lambda < \bar{\lambda}^{\text{int}}(0) = \bar{\lambda}$, the threshold of Proposition H2. With interest it would need only $\lambda < \bar{\lambda}^{\text{int}}(\alpha)$, and no misperception at all if $\alpha \geq \alpha_1$. The misperception that sustains bidding in a consórcio is worse, by the factor $1 + \alpha(2+r)/(2rc)$, than what a bidding RoSCA paying α would need. Interest and misperception are substitutes, as markup and misperception are in Proposition H3.

Present bias enters only through timing. Interest paid in period 2 enters (H.6) discounted by β; paying it in period 1 removes the discount and changes nothing in (H.20), since the $t = 0$ self weights both periods equally and the non-bidder consumes c_L in both.

H.8 Proofs for Section 4

Section 4 has three periods and one consumer. Its propositions use a few things the text does not spell out, which we collect here.

The decision is the $t = 1$ self's: she joins by paying the first installment. There is no $t = 0$ self. A member not yet awarded is drawn with probability $p(\ell)$ at $t = 1$ and again at $t = 2$, and is awarded for sure at $t = 3$. We take the two draws to have the same odds; this is what lies behind the single anchor $\frac{1}{3}$ in the text. The odds rise and are concave in ℓ, as the fund identity (H.1) implies. Installments are due at $t = 1, 2, 3$ whether or not she has been awarded. A member who cancels at $t = 2$ forfeits what she paid. She has no credit. She also holds no savings: with $r^S = 0$

and $\beta \leq 1$, saving a unit out of flat consumption changes a self's utility by $-v'(c) + \beta v'(c) \leq 0$, inside or outside the contract.

Write $\omega(\tau) \equiv v(y) - v(y - q(\tau))$ for the utility cost of one installment. We assume $\Delta > 2\omega$, so that the good is worth its last two installments to the selves who pay them, and $\omega(0) > p(1)\xi$, so that one lottery chance does not pay for an installment on its own. Let $\bar{\beta}(y)$ be the smallest β at which she buys the good without any contract; it plays the role of $\underline{\beta}(y)$ in Lemma H1. The bands in Proposition 1 are for consumers below $\bar{\beta}(y)$, which is therefore both $\bar{\beta}^R$ and $\bar{\beta}^C$. The lower bounds $\underline{\beta}^R$ and $\underline{\beta}^C$ correspond to $\hat{\beta}_\theta$ in Lemma H3.

Lemma H5 (Thresholds)**.** *Let $\beta < \bar{\beta}(y)$ and $p = p(\ell)$. The consumer joins (τ, ℓ) and completes it iff $\beta \geq \hat{\beta}(\tau, \ell) \equiv \max\{\hat{\beta}_1, \hat{\beta}_2\}$, where*

$$\hat{\beta}_1 = \frac{\omega - p\,\xi}{\Delta - 2\omega + \xi\, p\,(2 - p)}, \qquad \hat{\beta}_2 = \frac{\omega - p\,\xi}{\Delta - \omega}. \tag{H.21}$$

Both are positive, rise in τ and fall in ℓ.

Proof. Start with the $t = 2$ self. If she has not been awarded and cancels, she never buys the good, since $2y < 1$, and she consumes her income: this is worth $(1 + \beta)v(y)$. If she pays, she gets $v(y - q) + p\xi + \beta[v(y - q) + \Delta]$. So she pays iff

$$p\,\xi \;\geq\; \omega - \beta\,(\Delta - \omega), \tag{H.22}$$

which is $\beta \geq \hat{\beta}_2$.

Now the $t = 1$ self. Not joining is worth $(1 + 2\beta)v(y)$. If she joins and her $t = 2$ self completes, she consumes $y - q$ in every period, holds the good at $t = 3$, holds it at $t = 1$ with probability p and by $t = 2$ with probability $p(2 - p)$. This is worth $v(y - q) + p\xi + \beta[2v(y - q) + \Delta + p(2 - p)\xi]$. So she joins iff

$$p\,\xi\,[1 + \beta(2 - p)] \;\geq\; \omega - \beta\,(\Delta - 2\omega), \tag{H.23}$$

which is $\beta \geq \hat{\beta}_1$. There is one more case: she could join knowing that her $t = 2$ self will cancel, hoping for an award at $t = 1$. Relative to not joining this is worth $-\omega + p[\xi + \beta(\xi + \Delta - 2\omega)]$. It is never positive when (H.22) fails. Failure of (H.22) needs $\omega > p\xi$ and gives $\beta < (\omega - p\xi)/(\Delta - \omega)$; positivity gives $\beta \geq (\omega - p\xi)/[p(\xi + \Delta - 2\omega)]$; together, $p(\xi + \Delta - 2\omega) > \Delta - \omega$. With $p \leq 1$ this needs $\xi > \omega$, and with $p < \omega/\xi$ it needs $\omega > \xi$. So she joins iff both (H.22) and (H.23) hold.

The numerators in (H.21) are at least $\omega(0) - p(1)\xi > 0$, and the denominators are positive because $\Delta > 2\omega$. Both thresholds rise with ω, hence with τ, and fall with p, hence with ℓ. ■

Proof of Proposition 1. Parts (i) and (ii) are Lemma H5 with $\underline{\beta}^R = \hat{\beta}(0, 1)$, $\underline{\beta}^C(\tau, \ell) = \hat{\beta}(\tau, \ell)$ and $\bar{\beta}^R = \bar{\beta}^C = \bar{\beta}(y)$. The containment says $\hat{\beta}(\tau, \ell) \geq \hat{\beta}(0, 1)$, which holds because $\tau \geq 0$ and $p(\ell) \leq p(1)$.

For (iii), $\underline{\beta}^R$ does not depend on ℓ, so we need $\hat{\beta}(\tau,\ell)$ to be decreasing and convex in ℓ. It is the larger of $\hat{\beta}_1$ and $\hat{\beta}_2$, and the maximum of two convex functions is convex, so we take them in turn. $\hat{\beta}_2$ is linear in p. For $\hat{\beta}_1$, write $m(p) \equiv p(2-p)$, so that

$$\hat{\beta}_1 = \frac{\omega - p\,\xi}{\Delta - 2\omega + \xi\, m(p)}.$$

The numerator falls linearly in p. The denominator rises and is concave in p, so its reciprocal falls and is convex. The product of two positive, decreasing, convex functions is convex. Finally $p(\ell)$ rises and is concave, and a decreasing convex function of such a p is decreasing and convex in ℓ. ■

Proof of Proposition 2. With misperceived odds, $\hat{p}(\ell) = p(1) + \lambda[p(\ell) - p(1)]$ replaces $p(\ell)$ in (H.22) and (H.23). Nothing else changes, because the odds enter only through the flow terms and both selves hold the same beliefs. For $\ell < 1$, $\hat{p}$ falls as λ rises, and the thresholds fall in the odds. So the lower bound $\hat{\beta}(\tau,\ell;\lambda)$ rises with λ, from $\hat{\beta}(\tau,1)$ at $\lambda = 0$ to $\hat{\beta}(\tau,\ell)$ at $\lambda = 1$.

(i) At $\lambda = 0$ she perceives (τ,ℓ) as $(\tau,1)$: the fee is the same and $\hat{p}(\ell) = p(1)$ at both draws. Her outside option, saving alone or not buying, does not involve the odds, and the perceived odds enter the value of the contract continuously, including through the $t = 2$ self's decision to pay. So if she strictly prefers $(\tau,1)$ to her outside option, she takes up (τ,ℓ) for all λ up to some $\bar{\lambda}(\tau,\ell) > 0$, at any β, including $\beta = 1$. For a consumer below $\bar{\beta}(y)$ the condition reads $\beta > \hat{\beta}(\tau,1) = \hat{\beta}(\tau,\ell;0)$, and a higher β stays above the threshold for longer, so $\bar{\lambda}$ rises with β.

(ii) The lower bound rises with λ and the upper bound $\bar{\beta}(y)$ does not move. The band widens as λ falls.

(iii) She pays fee τ iff $\beta \geq \hat{\beta}(\tau,\ell;\lambda)$, and the threshold rises with τ and with λ. As λ rises, the set of fees she accepts shrinks, so the highest one, $\bar{\tau}(\ell;\lambda)$, falls. ■

H.9 Proofs for the market model

Proof of Lemma H1. The first-order condition $v'(y-s) = \beta v'(y+s-1)$ has a unique solution $\tilde{s}(\beta)$, since the left side rises and the right side falls in s. Raising β raises the right side, so $\tilde{s}$ rises in β; at $\beta = 1$ it is $s = \frac{1}{2}$, feasible because $y > \frac{1}{2}$. Hence $s^* = \max\{\tilde{s}, 1-y\}$ is nondecreasing, with $s^*(1) = \frac{1}{2}$.

For the threshold, let $g(\beta) \equiv \max_{s \geq 1-y}\{v(y-s) + \beta[v(y+s-1)+\Delta]\} - (1+\beta)v(y)$ be her gain from saving over not buying. By the envelope theorem $g'(\beta) = v(y+s^*-1) + \Delta - v(y) > 0$. As $\beta \to 0$, $g \to v(2y-1) - v(y) < 0$. At $\beta = 1$, $g = 2v(y-\frac{1}{2}) + \Delta - 2v(y) \geq 0$: by concavity $2[v(y) - v(y-\frac{1}{2})] \leq v'(y-\frac{1}{2})$, and $v'(y-\frac{1}{2}) \leq v'(c_\theta) \leq \Delta/(1+r)$ because $q \geq \frac{1}{2}$ and Assumption H1. So g crosses zero once, at $\underline{\beta}(y) \in (0,1)$. (The $t = 2$ self then buys: her loss $v(y+s) - v(y+s-1) \leq$

$v'(2y-1) \le v'(c_\theta) < \Delta$ by the same assumption, since $2y-1 \ge c_\theta$.) Finally, g rises in y, since $v'(y-s^*)$ and $v'(y+s^*-1)$ both exceed $v'(y)$, and in Δ; so $\underline{\beta}$ falls in both. ■

Proof of Lemma H2. The $t=1$ self of type θ gains from clearing at price b

$$G_\theta(b) \equiv \xi - \big[v(c_\theta) - v(c_\theta - P)\big] + \beta\big[v(c_\theta + P) - v(c_\theta)\big], \qquad P \equiv (1+\tau)b. \tag{H.24}$$

Write $X \equiv [v'(c_\theta - P) - v'(c_\theta)] - \beta[v'(c_\theta) - v'(c_\theta + P)]$ and $Y \equiv v'(c_\theta - P) - \beta v'(c_\theta + P)$. Then $0 \le X \le Y$: $X \ge 0$ because v' is convex and $\beta \le 1$, strictly if $\beta < 1$ or $v''' > 0$, and $Y - X = (1+\beta)v'(c_\theta) - 2\beta v'(c_\theta + P) \ge 0$.

Uniqueness and dominance. $G'_\theta = -(1+\tau)Y < 0$ and $G_\theta(0) = \xi > 0$, so the root $\bar{b}_\theta$ is unique. A bid affects the member's payoff only through whether it clears, and under the uniform price b^* is set by others' bids, so she wants to clear iff $G_\theta(b^*) \ge 0$ iff $b^* \le \bar{b}_\theta$: bidding $\bar{b}_\theta$ is weakly dominant.

Comparative statics. At an interior root $d\bar{b}_\theta/d\omega = -G_\omega/G_b$ with $G_b < 0$, so $\bar{b}_\theta$ moves with the sign of G_ω:

$$G_\xi = 1, \qquad G_\beta = v(c_\theta + P) - v(c_\theta) > 0, \qquad G_y = X \ge 0, \qquad G_\tau = -\tfrac{1}{2}X - bY \le 0.$$

At the cap, $\bar{b}_\theta = c_\theta/(1+\tau)$ rises in y_θ and falls in τ, so the statics hold weakly there. Finally $\bar{b}_L \le \bar{b}_H$ because $y_L < y_H$. ■

Remark 1. *The fee-sensitivity of the bid is bounded:* $|\bar{b}'_\theta(\tau)| = [X/(2Y)+b]/(1+\tau) \le (\frac{1}{2}+\bar{b}_\theta)/(1+\tau) < 1$ *at an interior root, using* $X \le Y$ *and* $\bar{b}_\theta < \frac{1}{2}$*, and* $|\bar{b}'_\theta| = y_\theta/(1+\tau)^2 < 1$ *at the cap. This is what the uniqueness of the break-even fixed point in the Results uses.*

Proof of Lemma H3. Let $\beta < \underline{\beta}$, so without the contract she does not buy, and her $t=1$ alternative is consuming her income: $c_1 = v(y) - v(y-q)$ and $g_2 = \Delta - c_1$. Write $n \equiv c_1 - \hat{p}\xi$ and $d \equiv \Delta - c_1$. Since $c_1 \le q\, v'(c_\theta) < \Delta$ under Assumption H1, $d > 0$, and (H.6) reads $\beta d \ge n$, that is $\beta \ge n/d = \hat{\beta}_\theta$. For the statics: n falls in ℓ and ξ, and for $\ell < 1$ rises in λ; $\partial\hat{\beta}_\theta/\partial\tau = v'(y-q)(\Delta - \hat{p}\xi)/(2d^2) > 0$ (with $\Delta > \hat{p}\xi$ because $\hat{p} \le \frac{1}{2}$ and $\Delta = \xi(1+r)/r$); c_1 falls in y by concavity, so n falls and d rises; and d rises in Δ. ■

Proof of Lemma H4. Along the break-even locus, differentiating $V_L(\ell) = 2v(c_L(\ell)) + p(\ell)\xi$ with $dc_L/d\ell = -\tau'/2$ gives

$$V'_L(\ell) = -v'(c_L(\ell))\,\tau'(\ell) + p'(\ell)\,\xi = \tau'(\ell)\,v'(c_L(\ell))\,[R(\ell) - 1], \tag{H.25}$$

with $\tau' > 0$. Under Assumption H3, R is strictly decreasing, so V'_L changes sign at most once,

from positive to negative: V_L is single-peaked. Define

$$\bar{\xi} \equiv \frac{\tau'(1)\, v'(c_L(1))}{p'(1)}, \qquad \underline{\xi} \equiv \frac{\tau'(0)\, v'(c_L(0))}{p'(0)}, \tag{H.26}$$

so that $R(1) = \xi/\bar{\xi}$ and $R(0) = \xi/\underline{\xi}$, with $\underline{\xi} < \bar{\xi}$ by Assumption H3. The maximizer is $\ell^* = 1$ if $\xi \geq \bar{\xi}$, $\ell^* = 0$ if $\xi \leq \underline{\xi}$, and otherwise the unique root of (H.13). At a fixed cut-off, τ', p' and c_L do not depend on ξ, so the thresholds are independent of ξ, R is proportional to ξ, and the root rises in ξ.

At a fixed cut-off the objects have closed forms. Write $A(\ell) \equiv 1 - (1-\ell)b^*$; from (H.1) and (H.9),

$$p'(\ell) = \frac{1-b^*}{2A(\ell)^2}, \qquad \tau'(\ell) = \frac{r}{2+r}\,\frac{c\,b^*}{A(\ell)^2 D(\ell)^2}, \qquad R(\ell) = \frac{\xi\,(1-b^*)\,(2+r)\,D(\ell)^2}{2\,c\,r\,b^*\,v'(c_L(\ell))}, \tag{H.27}$$

so R falls in ℓ, because D falls and $v'(c_L)$ rises, and Assumption H3 holds there. Evaluating at $\ell = 1$ and $\ell = 0$, with $D(1) = 1$ and $D(0) = 1 + \frac{r}{2+r}\frac{b^*}{1-b^*}$,

$$\bar{\xi} = v'(c_L(1)) \cdot c \cdot \frac{2r}{2+r} \cdot \frac{b^*}{1-b^*}, \qquad \underline{\xi} = \bar{\xi} \cdot \frac{v'(c_L(0))}{v'(c_L(1))} \cdot \left[1 + \frac{r}{2+r}\,\frac{b^*}{1-b^*}\right]^{-2}. \tag{H.28}$$

■

Remark 2. *With the cut-off $\bar{b}_L(\tau(\ell))$, a higher ξ also raises the bid, by Lemma H2, and with it the fee cut each slot buys; the sign of $d\ell^*/d\xi$ is then not determined.*

Proof of Proposition H1. We check the deviations a firm can make.

Zero profit. Let (τ, ℓ) attract a positive mass of non-bidders. If $\pi < 0$ the firm prefers to stay out. If $\pi > 0$, a rival posts $(\tau - \epsilon, \ell)$. The left side of (H.7) falls in τ and (H.6) is relaxed by a lower fee, so every non-bidder at (τ, ℓ) moves, and by Assumption H2 the H members needed to fill its slots follow, since no other group can seat them and a seat beats their outside option. For small ϵ the rival captures every member of the contract at a per-member profit near $\pi > 0$, more than its share. Hence every active contract is break-even, $\tau = \tau(\ell)$.

Non-bidder optimality. Suppose an active contract has $\ell \neq \ell^*$. It is break-even, so it gives non-bidders $V_L(\ell) < V_L(\ell^*)$ by Lemma H4. A firm posting $(\tau(\ell^*) + \epsilon, \ell^*)$ attracts all non-bidders for small ϵ: (H.7) holds strictly at ℓ^*, since it holds at the incumbent's ℓ and $V_L(\ell^*) > V_L(\ell)$, and (H.6) holds strictly there by Assumption H4. Bidders follow by Assumption H2. The deviator earns $\epsilon\,\frac{2+r}{2(1+r)}D(\ell^*) > 0$ per member against zero. Hence every active contract is $(\tau(\ell^*), \ell^*)$; how non-bidders split across the firms posting it is indeterminate and payoff-irrelevant.

Existence. Let two firms post $(\tau(\ell^*), \ell^*)$ and split the non-bidders, with H members allocated to fill both groups' slots. A deviation (τ', ℓ') that non-bidders strictly prefer must give them more

than $V_L(\ell')$ at the break-even fee, since ℓ^* maximizes V_L along the locus; so $\tau' < \tau(\ell')$ and it loses money. One they weakly prefer earns at most zero. One they do not prefer attracts no non-bidder, hence by Assumption H2 no bidder either, since without non-bidders it cannot seat the H members it would need. Bidders' constraints are slack by Assumption H2, with the strict rent $G_H(\bar{b}_L) > 0$ since $\bar{b}_H > \bar{b}_L$.

Pure lottery. At a fixed cut-off, the cash cap gives $b^* = \bar{b}_L < c_L/(1+\tau) < \frac{1}{2}$, so $b^*/(1-b^*) < 1$, and $2r/(2+r) < r$; by (H.28),

$$\bar{\xi} = v'(c_L(1))\, c\, \frac{2r}{2+r}\, \frac{b^*}{1-b^*} \;<\; v'(c_L(1))\, c\, r \;<\; r\, v'(c_L(1)) \;\leq\; \xi,$$

using $c < 1$ and then Assumption H1. So $\ell^* = 1$ by Lemma H4(ii). With the cut-off $b^*(\tau(1))$ the same bound holds, since at $\ell = 1$ the slots' value does not depend on the cut-off's fee-sensitivity: $p'(1) = (1-b^*)/2$ and $D'(1) = -\frac{r}{2+r} b^*$ as at a fixed cut-off.

This establishes existence, uniqueness up to which firm serves whom, and the pure lottery. ■

Remark 3. *At a fixed cut-off the pure-lottery claim does not need Assumption H3: by* (H.27) *and Assumption H1,* $R(\ell) \geq (1-b^*)(2+r)D(\ell)^2/(2cb^*) > 1$ *for every* ℓ*, so* $V_L' > 0$ *on all of* $[0,1]$.

Proof of Proposition H2. By (H.3), $\hat{p}(\ell)\xi = \lambda\xi\, p(\ell) + (1-\lambda)\xi/2$, so the non-bidder's value of a contract is V_L with $\lambda\xi$ in place of ξ, plus a constant. The bid is set at $t = 1$ from the true flow, so b^*, τ', p' and c_L do not depend on λ. Hence Lemma H4 and Proposition H1 apply with $\lambda\xi$ in place of ξ, with thresholds $\bar{\lambda} \equiv \bar{\xi}/\xi$ and $\underline{\lambda} \equiv \underline{\xi}/\xi$ from (H.26), which are independent of λ even with the cut-off $\bar{b}_L(\tau(\ell))$; R is proportional to λ, so the root of (H.13) rises in λ. Finally $\underline{\lambda} < \bar{\lambda}$ by Assumption H3 and $\bar{\lambda} < 1$ by Proposition H1. ■

Proof of Proposition H3. Differentiating (H.14) in ℓ gives $\tau_r' = \lambda p'\xi/v'(c_L)$. Per-member profit is $\pi(\tau,\ell) = (\tau - c)\frac{2+r}{2(1+r)} + \tau\frac{r}{1+r}B/s$ with $B/s = (1-\ell)b^*/(2A)$ by (H.1) and A as in (H.27), so $\pi_\tau = \frac{2+r}{2(1+r)} + \frac{r}{1+r}B/s$ and, at a fixed cut-off, $\pi_\ell = -\tau\frac{r}{1+r}\, b^*/(2A^2)$. The derivative of (H.15) has the sign of $m\,[\pi_\tau\tau_r' + \pi_\ell] + m'\pi$. At $\ell = 1$: $B = 0$, $A = 1$, $p'(1) = (1-b^*)/2$, $h(1) = 0$ and $h'(1) = -1$ by (H.10), so $m'/m = -1$ and $\pi(\tau_r(1),1) = M\,\frac{2+r}{2(1+r)}$. Dividing by $m\,\frac{2+r}{2(1+r)} > 0$, the derivative at $\ell = 1$ is negative iff

$$\frac{\lambda\,\xi\,(1-b^*)}{2\,v'(c_L)} \;<\; \tau_r(1)\,\frac{r\,b^*}{2+r} + M,$$

which is (H.16); profit then rises as ℓ falls from 1. The same computation at $\ell = 0$, with $A(0) = 1-b^*$, $B > 0$, and $h(0)$ and $h'(0)$ from (H.10), gives the lower threshold, and both thresholds rise in M, since the markup enters the right side with a positive coefficient. Under single-peakedness the two corners determine the three cases, and the claims at $\lambda = 1$ read off (H.16). ■

Proof of Proposition H4. (i). Let $(\tau, \ell, 0)$ with $\ell < 1$ attract non-bidders, and let a rival post (τ, ℓ, α) with $\alpha \in (0, \bar{\alpha}]$. By (H.17), $\iota(\ell) > 0$, so the bracket in (H.18) is positive and the non-bidder's value rises, while (H.6) is relaxed, since the interest raises her period-2 continuation; profit is unchanged by (H.2), since interest never reaches the firm; and bidders join by the definition of $\bar{\alpha}$. The rest is the non-bidder-optimality step of Proposition H1: every non-bidder moves, bidders follow by Assumption H2, and the rival earns a positive profit after raising τ by a small ϵ. Hence no active contract has $\ell < 1$ and $\alpha = 0$.

(ii) to (iv). Interest leaves (H.2) and the locus (H.9) unchanged. Differentiate (H.18) at $\ell = 1$, where $\iota(1) = 0$, so the bracket vanishes and its derivative is $(1 - \hat{p}(1))\, v'(c_L(1))\, \iota'(1)$ with $\iota'(1) = -\alpha b^*$. Using $p'(1) = (1 - b^*)/2$ and $\tau'(1) = rcb^*/(2 + r)$ from (H.27),

$$V_L'(1; \alpha) = -\, v'(c_L(1))\, b^* \left[\frac{rc}{2 + r} + \frac{\alpha}{2} \right] + \lambda\, \xi\, \frac{1 - b^*}{2}. \tag{H.29}$$

This is negative iff $\alpha > \alpha^*(\lambda)$ as in (H.19), which is (ii); and $\alpha^*(\lambda) \leq 0$ iff $\lambda\xi(1-b^*)/(v'(c_L(1))b^*) \leq 2rc/(2 + r)$, which by (H.28) is $\lambda \leq \bar{\xi}/\xi$. Part (iii) reads off (H.19). Part (iv) solves the same inequality for λ: $\lambda < v'(c_L(1))\, b^* \left[\frac{2rc}{2+r} + \alpha \right] / [\xi(1-b^*)]$, and by (H.28), $\bar{\lambda} = v'(c_L(1))\, b^* \frac{2rc}{2+r} / [\xi(1-b^*)]$, which gives (H.20), equal to $\bar{\lambda}$ at $\alpha = 0$. Under single-peakedness the sign of (H.29) decides the corner, and the zero-profit and poaching steps of Proposition H1 apply with α fixed. ■

I How present-biased must consumers be? A calibration

Following the calibration of Anagol et al. (2017) for whole life insurance in India, we ask how present-biased a sophisticated consumer must be for commitment (Section 4) to justify joining a consórcio. We assume full sophistication, which plays in favor of joining the consórcio. We first use their model, in which the contract enters only through its return, and then a monthly version of the model in Section 4, in which the indivisible good and the timing of awards enter explicitly. In both, consumers cannot borrow and face no income risk.

I.1 A simple model

A consumer consumes her wealth from year $t = 0$ to year $t = N$ with utility $\log c_t$. The self at t maximizes $\log c_t + \beta \sum_{s>t} \delta^{s-t} \log c_s$ and is fully sophisticated, correctly anticipating the choices of later selves. In a liquid account, wealth earns $r^S = 4\%$ a year and each self chooses its own consumption. With log utility, the equilibrium rule is $c_t = x_t/(1 + \beta \sum_{k=1}^{N-t} \delta^k)$ for wealth x_t. In an illiquid product, wealth earns $r^W < r^S$, and the path chosen at $t = 0$ cannot be revised. The product trades returns for commitment. The self at $t = 0$ prefers it when β is below a threshold. We set $\delta = 1/(1 + r^S)$.

Section 3 shows that the median never-bidding contract is equivalent to saving at -4.9% for cars, -8.5% for motorcycles, and -2.7% for real estate. Over the average term of each contract, the thresholds give 0.39 for cars (6 years), 0.33 for motorcycles (5 years), and 0.37 for real estate (15 years). They rise to 0.44, 0.36, and 0.44 if the liquid account pays 1.5% instead (Table I.1). Because β discounts every future period relative to the current one, its value depends on the length of a period: with monthly periods, as in the next subsection, the thresholds are 0.51, 0.48, and 0.43. With CRRA utility, the thresholds fall as curvature rises (0.23, 0.18, and 0.19 with a coefficient of 2), because a lower elasticity of intertemporal substitution reduces the distortion that commitment corrects. Only an elasticity above one raises them (0.53, 0.47, and 0.55 with a coefficient of 0.5).

Table I.1: Present bias required to prefer an illiquid product to saving at 4%.

	Real annual return of the illiquid product								
Horizon	0%	−1%	−2%	−3%	−4%	−5%	−6%	−7%	−8%
6 years	0.53	0.49	0.46	0.43	0.41	0.38	0.36	0.35	0.33
8 years	0.51	0.47	0.44	0.41	0.39	0.37	0.35	0.33	0.32
10 years	0.49	0.46	0.42	0.40	0.37	0.35	0.33	0.32	0.30
12 years	0.48	0.44	0.41	0.38	0.36	0.34	0.32	0.30	0.29
15 years	0.46	0.42	0.39	0.36	0.34	0.32	0.30	0.29	0.27
20 years	0.43	0.39	0.36	0.34	0.32	0.30	0.28	0.26	0.25

Notes: This table shows the value of β below which a consumer prefers to lock in her consumption path at the column's real return rather than save at 4% in a liquid account, over the row's horizon. Periods are years; $\delta = 1/1.04$.

I.2 Extension to an indivisible good

We now calibrate the model of Section 4 at a monthly frequency. A consumer earns y each month, cannot borrow, and has utility $\log c_t$; a good with price 1 adds ξ to utility in every month she owns it. Saving alone at $r^S = 4\%$, each self chooses how much to save and whether to buy, and we solve for the sophisticated equilibrium backward from a purchase deadline of 45 to 60 years. In a consórcio, she pays $q = (1+\tau)/T$ for T months, where τ is the administration fee, and has no other savings. She receives the good at a date drawn from the distribution of lottery awards for never-bidders in her asset category (Figure 6), or at T at the latest, and keeps paying after the award. As in Section 4, the self who joins pays the first installment, and a member not yet awarded may cancel in any month, forfeiting what she paid. We set τ and T to category averages and installments to 15, 12, and 14% of median reported income for cars, motorcycles, and real estate. As a benchmark for the unobserved ξ, a consumer with $\beta = 1$ saving alone would save exactly the installment, buying after 58, 43, and 123 months.

Figure I.1 reports the gain from joining. The random RoSCA of Proposition 1, with no fee and uniformly distributed award dates, is joined for any β above 0.05. Below it, the joining self will not pay the first installment. The consórcio is joined only for β between about 0.08 and 0.25 for cars and between 0.10 and 0.25 for motorcycles, and never for real estate, where an 18% fee and a median award in month 157 of 179 make the contract worth less than not buying. Consistent with Proposition 1, the consórcio's band lies inside the RoSCA. The fee and the late awards shrink it mostly from above. Delivering the good only at maturity barely changes the band (0.10 to about 0.25 for cars, 0.15 to 0.25 for motorcycles), so at observed lottery shares nearly all of the contract's appeal would have to be commitment, rather than early access. For values of the good between half and three times the benchmark, and with a 1.5% saving rate, the upper end never exceeds 0.33. At low values of the good nobody joins.

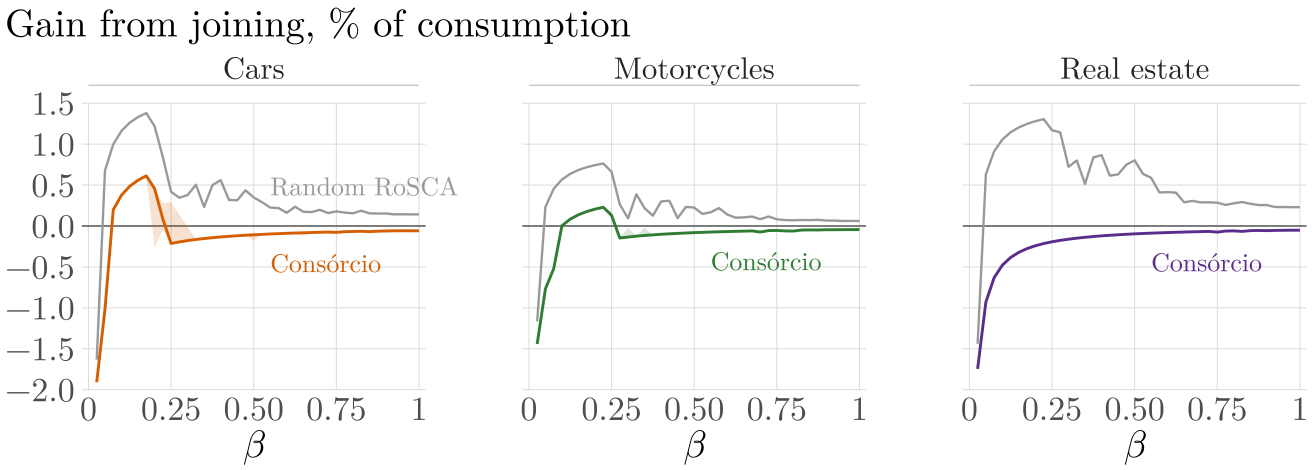


Figure I.1: Gain from joining a consórcio and a random RoSCA, by present bias.

Notes: This figure shows the joining self's gain from joining instead of saving alone, as a permanent percentage change in consumption, against the monthly β. Lines use a 50-year purchase deadline; bands span deadlines of 45 to 60 years.